MATHEMATICS

BINOMIAL THEOREM

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BINOMIAL THEOREM

JEE EXERCISES

Exercise J-I

BASED ON IMPORTANT TERMS

1. If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a+b)^{n+3}$ are equal, then n is equal to :

यदि $(a+b)^n$ के प्रसार में $\frac{T_2}{T_3}$ तथा $(a+b)^{n+3}$ के प्रसार में $\frac{T_3}{T_4}$ बराबर है तब n का मान होगा:

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. $\frac{T_2}{T_3}(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r$

$$\Rightarrow \frac{T_2}{T_3} = \frac{{}^nC_1 a^{n-1} b^1}{{}^nC_2 a^{n-2} b^2} = \frac{\lfloor n \rfloor 2 \lfloor n-2}{\lfloor n-1 \rfloor n} \cdot \frac{a}{b} = \frac{2a}{(n-1)b}$$

$$(a+b)^{n+3} = \sum_{r=0}^{n+3} {}^{n+3}C_r a^{n+3-r} b^r$$

$$\Rightarrow \frac{T_3}{T_4} = \frac{{}^{n+3}C_2 a^{n+3-2} b^2}{{}^{n+3}C_3 a^{n+3-3} b^3} = \frac{\lfloor n+3 \rfloor n \rfloor 3}{\lfloor n+1 \rfloor 2 \rfloor n+3} \cdot \frac{a^{n+1} \cdot b^2}{a^n b^3}$$

$$= \frac{3}{n+1} \cdot \frac{a}{b}$$

Given $\frac{T_2}{T_3} = \frac{T_3}{T_4}$

$$\Rightarrow \frac{2a'}{(n-1)b'} = \frac{3a'}{(n+1)b'}$$

$\Rightarrow 2n + 2 = 3n - 3$

$\Rightarrow n = 5$

2. If number of terms in the expansion of $(x - 2y + 3z)^n$ are 45, then n is equal to :

(A) 7 (B) 8 (C) 9 (D) None of these

यदि $(x - 2y + 3z)^n$ के प्रसार में कुल पदों की संख्या 45 है तब n का मान होगा :

(A) 7 (B) 8 (C) 9 (D) इनमें से कोई नहीं

Ans. B

Sol. No. of terms in the expansion of $(x - 2y + 3z)^n$ are 45

$$(x-2y+3z)^n = \sum_{r_1+r_2+r_3=n} \frac{\lfloor n}{\lfloor r_1 \lfloor r_2 \lfloor r_3} \cdot x^{r_1} \cdot (-2)^{r_2} \cdot y^{r_2} \cdot 3^{r_3} \cdot z^{r_3}$$

$$\Rightarrow n + {}^{3-1}C_{3-1} = 45 \Rightarrow {}^{n+2}C_2 = 45 \Rightarrow n^2 + 3n + 2 = 90$$

$$\Rightarrow n = -11 \text{ or } n = 8$$

3. The greatest coefficient in the expansion of $(1 + x)^{2n+2}$ is :

$(1 + x)^{2n+2}$ के प्रसार में महत्तम पद का गुणांक होगा :

(A) $\frac{(2n)!}{(n!)}$ (B) $\frac{(2n+2)!}{[(n+1)!]^2}$ (C) $\frac{(2n+2)!}{n!(n+1)!}$ (D) $\frac{(2n)!}{n!(n+1)!}$

Ans. B

Sol. $(1+x)^{2n+2} = \sum_{r=0}^{2n+2} {}^{2n+2}C_r x^r$

${}^{2n+2}C_r$ is maximum if $r = n + 1$

$\Rightarrow$ Greatest coefficient is ${}^{2n+2}C_{n+1} = \frac{\lfloor 2n+2}{(\lfloor n+1)^2}$

4. The expression $\frac{1}{\sqrt{4x+1}}\left[\left[\frac{1+\sqrt{4x+1}}{2}\right]^7 - \left[\frac{1-\sqrt{4x+1}}{2}\right]^7\right]$ is a polynomial in x of degree :

व्यंजक $\frac{1}{\sqrt{4x+1}}\left[\left[\frac{1+\sqrt{4x+1}}{2}\right]^7 - \left[\frac{1-\sqrt{4x+1}}{2}\right]^7\right]$, x में किस घात का बहुपदीय व्यंजक है :

(A) 7 (B) 5 (C) 4 (D) 3

Ans. D

Sol. Given expression can be rewritten as

$$\frac{2}{2^7\sqrt{4x+1}}\left[{}^7C_1(\sqrt{4x+1})+{}^7C_3(\sqrt{4x+1})^3+\ldots+{}^7C_7(\sqrt{4x+1})^7\right]$$

$$\frac{1}{2^6}\left[{}^7C_1+{}^7C_3(\sqrt{4x+1})^2+\ldots..+{}^7C_7(\sqrt{4x+1})^6\right]$$

$\therefore$ Last term becomes $(4x+1)^3$

Hence degree is 3

5. If the coefficient of $(r+2)^{th}$ and $(3r)^{th}$ term in the exp. of $(1+x)^{2n}$ are equal then :

(A) $n = 2r+1$ (B) $n = 2r-1$ (C) $n = 2r$ (D) None of these

$(1+x)^{2n}$ के प्रसार में $(r+2)$वें तथा $(3r)$ वें पद के गुणांक बराबर हो तो :

(A) $n = 2r+1$ (B) $n = 2r-1$ (C) $n = 2r$ (D) इनमें से कोई नहीं

Ans. C

Sol. $(r+2)^{th} = (3r)^{th}$

${}^{2n}C_{r+2} = {}^{2n}C_{3r}$

$r+2 = 3r$ or $r+2+2r = 2n$

$r = 1$ or $n = 2r+1$

6. The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and of $(1-\alpha x)^6$ is the same if α equals :

$(1+\alpha x)^4$ तथा $(1-\alpha x)^6$ के द्विपद प्रसार, जो कि x की घातों के रूप में है, के मध्य पदों के गुणांक समान होंगे यदि α का मान है :

(A) $-\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $-\frac{3}{10}$ (D) $\frac{3}{5}$

Ans. C

Sol. Coefficient of middle term in $(1+\alpha x)^4 = {}^4C_2\,\alpha^2$

coefficient of middle term in $(1-\alpha x)^6 = {}^6C_3\,(-\alpha)^3$

${}^4C_2\,\alpha^2 = -{}^6C_3\,\alpha^3$

$-\frac{6}{20} = \alpha$

$\alpha = \frac{-3}{10}$

7. If the coefficients of rth, $(r+1)$th and $(r+2)$th terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation :

एक द्विपद $(1+y)^m$ के प्रसार में यदि r वें, $(r+1)$ वें तथा $(r+2)$ वें पदों के गुणांक समान्तर श्रेणी में हैं, तो m तथा r निम्न समीकरण को सन्तुष्ट करते हैं :

(A) $m^2 - m(4r - 1) + 4r^2 - 2 = 0$ (B) $m^2 - m(4r + 1) + 4r^2 + 2 = 0$

(C) $m^2 - m(4r + 1) + 4r^2 - 2 = 0$ (D) $m^2 - m(4r - 1) + 4r^2 + 2 = 0$

Ans. C

Sol. Coefficient of rth, $(r + 1)^{th}$ and $(r + 2)^{th}$ terms in expansion of $(1 + y)^n$ are ${}^mC_{r-1}, {}^mC_r, {}^mC_{r+1}$

Then 2. ${}^mC_r = {}^mC_{r-1} + {}^mC_{r+1}$

$\Rightarrow m^2 - m(4r + 1) + 4r^2 - 2 = 0$

Short Method: Let r = 1, hence ${}^mC_0$, ${}^mC_1$ and ${}^mC_2$ are in A.P.

$\Rightarrow 2.{}^mC_1 = {}^mC_0 + {}^mC_2 \Rightarrow 2m = 1 + \frac{m(m-1)}{2}$

$\Rightarrow 4m = 2 + m^2 - m \Rightarrow m^2 - 5m + 2 = 0$

Which is given by (C)

8. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation :

यदि $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ के प्रसार में x^7 का गुणांक $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$ के प्रसार में x^{-7} के गुणांक के बराबर है, तो a और b में सम्बन्ध है :

(A) $a - b = 1$ (B) $a + b = 1$ (C) $\frac{a}{b} = 1$ (D) $ab = 1$

Ans. D

Sol. In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r a^{11-r} \frac{1}{b^r} x^{22-3r}$$

For x^7, we must have $22 - 3r = 7 \Rightarrow r = 5$, and the coefficient of $x^7 = {}^{11}C_5 . a^{11-5} \frac{1}{b^5} = {}^{11}C_5 \frac{a^6}{b^5}$.

Similary, in the expansion of $\left(ax^2 - \frac{1}{bx}\right)^{11}$, the general term is $T_{r+1} = {}^{11}C_r (-1)^r \frac{a^{11-r}}{b^r} x^{11-3r}$

For x^{-7} we must have $11 - 3r = -7 \Rightarrow r = 6$, and the coefficient of x^{-7} is ${}^{11}C_6 \frac{a^5}{b^6} = {}^{11}C_5 \frac{a^5}{b^6}$

9. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3}+\frac{1}{4^{1/3}}\right)^n$. If y = 12x then the value of n is :

माना $\left(3^{1/3}+\frac{1}{4^{1/3}}\right)^n$ के प्रसार में शुरू से 7वाँ पद x तथा अंत से 7वाँ पद y है। यदि y = 12x है, तो n बराबर होगा:

(A) 9 (B) 8 (C) 10 (D) 11

Ans. A

Sol. $\left(3^{\frac{1}{3}}+\frac{1}{4^{\frac{1}{3}}}\right)^n=\sum_{r=0}^{n}{}^nC_r\,3^{\frac{n-r}{3}}\cdot4^{\frac{-r}{3}}$

$x={}^nC_6\,3^{\frac{n-6}{3}}\cdot4^{\frac{-6}{3}}$ & $y={}^nC_{n-6}\,3^{\frac{n-(n-6)}{3}}\cdot4^{-\frac{(n-6)}{3}}$

Given y = 12x

$\Rightarrow {}^nC_6\,3^2\cdot4^{-\frac{(n-6)}{3}}={}^nC_6\cdot3^{\frac{n-6}{3}}\cdot4^{-2}\cdot4\cdot3$

$\Rightarrow 4^{-\frac{(n-6)}{3}+2}=3^{\frac{n-6}{3}-2}\cdot4\cdot3$

$\Rightarrow 4^{\frac{-n+6+6}{3}}=3^{\frac{n-12}{3}}\cdot4\cdot3$

$\Rightarrow 4^{\frac{-n+12-3}{3}}=3^{\frac{n-12+3}{3}}$

This is possible if n = 9

10. In the binomial expansion of $(a-b)^n$, $n\geq5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals :

$(a-b)^n$, $n\geq5$ के द्विपद प्रसार में, 5वें तथा 6वें पदों का योग 0 है, तब $\frac{a}{b}$ बराबर है:

(A) $\frac{5}{n-4}$ (B) $\frac{6}{n-5}$ (C) $\frac{n-5}{6}$ (D) $\frac{n-4}{5}$

Ans. D

Sol. $(a-b)^n \Rightarrow$ general term $T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r(-1)^r$

${}^nC_4 \cdot a^{n-4} \cdot b^4 - {}^nC_5 \cdot a^{n-5} \cdot b^5 = 0$

$${}^nC_4\left(\frac{a}{b}\right) = {}^nC_5 \Rightarrow \frac{n!}{4!(n-4)!}\left(\frac{a}{b}\right) = \frac{n!}{5!(n-5)!}$$

$$\left(\frac{a}{b}\right) = \frac{(n-4)}{5}$$

BASED ON FINDING COEFFICIENTS

11. If $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots\ldots\ldots$then the value of a and n is :

यदि $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots$ तब a तथा n का मान होगा:

(A) 2, 4 (B) 2, 3 (C) 3, 6 (D) 1, 2

Ans. A

Sol. $(1 + ax)^n = 1 + 8x + 24x^2 + \ldots\ldots$

$\Rightarrow 1 + {}^nC_1 (ax) + {}^nC_2(ax)^2 + \ldots. = 1 + 8x + 24x^2 + \ldots..$

$$\Rightarrow na = 8 \;\&\; \frac{\lfloor n}{\lfloor n-2 \rfloor 2} a^2 = 24$$

$\Rightarrow na = 8 \;\&\; n(n-1)a^2 = 48$

$\Rightarrow na = 8 \;\&\; na\,(na - a) = 48$

$8\,(8 - a) = 48$

$\Rightarrow 8 - a = 6$

$\Rightarrow a = 2$

$\Rightarrow n = 4$

12. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^n$ is :

$(1 + x + x^2 + x^3)^n$ के प्रसार में x^4 को गुणांक होगा :

(A) ${}^nC_4$ (B) ${}^nC_4 + {}^nC_2$ (C) ${}^nC_4 + {}^nC_2 + {}^nC_4 \cdot {}^nC_2$ (D) ${}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

Ans. D

Sol. $(1 + x + x^2 + x^3)^n = ((1 + x^2)\,(1 + x))^n$

$$=\left(1+x^2\right)^n\left(1+x\right)^n=\left[\sum_{r=0}^{n}{}^nC_r x^{2r}\right]\left[\sum_{r=0}^{n}{}^nC_r x^r\right]$$

Coefficient of x^4

$= {}^nC_0\ {}^nC_4 + {}^nC_1\ {}^nC_2 + {}^nC_2 \cdot {}^nC_0$

$= {}^nC_4 + {}^nC_2 + {}^nC_1 \cdot {}^nC_2$

13. The co-efficient of x^9 in the polynomial given by $(x + 1)(x + 2) \ldots\ldots (x + 10) + (x + 2)(x + 3) \ldots\ldots (x + 11) + \ldots\ldots\ldots + (x + 11)(x + 12) \ldots\ldots (x + 20)$ is :

व्यंजक $(x + 1)(x + 2) \ldots\ldots (x + 10) + (x + 2)(x + 3) \ldots\ldots (x + 11) + \ldots\ldots\ldots + (x + 11)(x + 12) \ldots\ldots (x + 20)$ में x^9 का गुणांक है:

(A) 5511 (B) 5151 (C) 1515 (D) 1155

Ans. D

Sol. Coefficient of x^9 in $(x + 1)(x + 2) \ldots\ldots (x + 10) + (x + 2) \ldots\ldots (x + 11) + \ldots\ldots\ldots + (x + 11)(x + 12) \ldots\ldots (x + 20)$

Let $P(x) = (x + 1)(x + 2) \ldots\ldots (x + 10)$

$\Rightarrow x = -1, \ldots\ldots, -10$ are roots of $P(x)$

Coefficient of $x^9 = -(-1 - 2 \ldots\ldots - 10) = 1 + \ldots\ldots + 10$

$$=\frac{10\times 11}{2}=55$$

Coefficient of x^9

$= 55 + 65 + 75 + 85 + 95 + 105 + 115 + 125 + 135 + 145 + 155 \rightarrow$ A.P. with $d = 10$

$$=\frac{11}{2}\left[110+10\times 10\right]=\frac{210}{2}\times 11$$

$= 105 \times 11 = 1155$

14. The coefficient of x^{39} in the expansion of $\left(x^4-\frac{1}{x^3}\right)^{15}$ is :

$\left(x^4-\frac{1}{x^3}\right)^{15}$ के प्रसार में x^{39} का गुणांक है:

(A) – 455 (B) – 105 (C) + 455 (D) + 105

Ans. A

Sol. $\left(x^4 - \frac{1}{x^3}\right)^{15} = \sum_{r=0}^{15} {}^{15}C_r \left(x^4\right)^{15-r} \cdot \left(\frac{-1}{x^3}\right)^r$

$= \sum_{r=0}^{15} {}^{15}C_r (-1)^r x^{60-7r}$

For x^{39}; $60 - 7r = 39$

$\Rightarrow 7r = 60 - 39$

$\Rightarrow r = 3$

Coefficient of x^{39} is $(-1)^3 \cdot {}^{15}C_3$

$= -455$

15. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is :

$(1 + x)(1-x)^n$ के प्रसार में x^n का गुणांक होगा :

(A) $(n-1)$ (B) $(-1)^n(1-n)$ (C) $(-1)^{n-1}(n-1)^2$ (D) $(-1)^{n-1} n$

Ans. B

Sol. Coefficient of x^n in expansion of $(1 + x)(1 - x)^n$ i.e., coefficient of x^n in expansion of $(1 - x)^n$ + coefficient of x^{n-1} in expansion of $(1 - x)^n$]

Now, $(-1)^n {}^nC_n + (-1)^{n-1} {}^nC_{n-1}$

$(-1)^n [{}^nC_n - {}^nC_{n-1}] = (-1)^n[1-n]$

16. For natural numbers m, n if $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + ...$, and $a_1 = a_2 = 10$, then (m, n) is :

प्राकृत संख्याओं m, n के लिए यदि $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + ...$, तथा $a_1 = a_2 = 10$, तो (m, n) है :

(A) (35, 20) (B) (45, 35) (C) (35, 45) (D) (20, 45)

Ans. C

Sol. $(1 - y)^m (1 + y)^n = 1 + a_1 y + a_2 y^2 + ...$

$(1 - my + {}^mC_2 y^2 .....)(1 + ny + {}^nC_2 y^2 .....) = 1 + a_1 y + a_2 y^2 + ...$

$a_1 = n - m = 10$ ..... (1)

$a_2 = {}^mC_2 + {}^nC_2 - mn = 10$ ..... (2)

solving (1) & (2), we get $(m, n) \equiv (35, 45)$

17. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is : **[AIEEE 2011]**

$(1-x-x^2+x^3)^6$ के प्रसार में x^7 का गुणांक है :

(A) – 144 (B) 132 (C) 144 (D) –132

Ans. A

Sol. $(1-x-x^2+x^3)^6 \Rightarrow [(1-x)-x^2(1-x)]$

$\Rightarrow (1-x)^6(1-x^2)^6$

$\Rightarrow (1-{}^6C_1\,x+{}^6C_2x^2-{}^6C_3x^3+{}^6C_4\,x^4-{}^6C_5+{}^6C_6\,x^6)\times(1-{}^6C_1x^2+{}^6C_2x^4-{}^6C_3x^6+....)$

$\Rightarrow$ cofficient of x^7 is $\rightarrow$

${}^6C_1.\,{}^6C_3-{}^6C_3.\,{}^6C_2+{}^6C_5\,.\,{}^6C_1$

$\Rightarrow \frac{6.5.4}{3.2}-\frac{6.5.4}{3.2.1}+6.6=120-300+36$

$\Rightarrow -144$

18. If the coffcicnet of x^3 and x^4 in the expansion of $(1+ax+bx^2)\ (1-2x)^{18}$ in powers of x are both zero then (a, b) is equal to : **[JEE(Main) 2014]**

यदि $(1+ax+bx^2)\ (1-2x)^{18}$ के x की घातों में प्रसार में x^3 तथा x^4 दोनों के गुणांक शून्य हैं, तो (a, b) बराबर है :

(A) $\left(14,\frac{272}{3}\right)$ (B) $\left(16,\frac{272}{3}\right)$ (C) $\left(16,\frac{251}{3}\right)$ (D) $\left(14,\frac{251}{3}\right)$

Ans. B

Sol. $(1+ax+bx^2)(1-2x)^{18}$

coeff of $x^3={}^{18}C_3(-2)^3+a.(-2)^2\,.\,{}^{18}C_2+b.(-2).\,{}^{18}C_1=0$

coeff of $x^4\equiv{}^{18}C_4(-2)^4+a(-2)^3\,.\,{}^{18}C_3+b(-2)^2\,.\,{}^{18}C_2=0$

$\Rightarrow$ $51a-3b=544$ and $32a-3b=240$

Subtracting we get $a=16$

$\Rightarrow$ $b=\frac{272}{3}$

REMAINDER BASED PROBLEMS

19. The number 7^{1995} when divided by 100 leaves the remainder :

(A) 43 (B) 45 (C) 50 (D) None of these

जब संख्या 7^{1995} को 100 से विभाजित किया जाता है तब शेषफल होगा

(A) 43 (B) 45 (C) 50 (D) इनमें से कोई नहीं

Ans. A

Sol. $7^{1995} = 7[(7)^2]^{997} = 7(50-1)^{997}$

using expansion

$7\{^{997}C_0.50^{997} + {}^{997}C_1.50^{996} + ...+{}^{997}C_{996}.50-1\}$

$= 100K + 7(997 \times 50 - 1)$

$= 100K + 7(900 \times 50 + 7 \times 50 - 1)$

$= 100K_1 + 49 \times 50 - 7 = 100K_1 + 2450 - 7$

$= 100K_1 + 2400 + 43$

$\therefore$ Remainder is 43

20. The last digit of (3^P+2), where $P = 3^{4n+2}$ is :

(3^P+2) का अन्तिम अंक क्या होगा जबकि $P = 3^{4n+2}$:

(A) 1 (B) 9 (C) 4 (D) 5

Ans. D

Sol. $(3^P + 2)$; where $P = 3^{4n+2}$

$P = 3^{4n}\cdot 9 = 81^n\cdot 9$

$P = 9\cdot(80+1)^n$

$$= 9\sum_{r=0}^{n} {}^nC_r (80)^{n-r}$$

$P = 10\lambda + 9\cdot{}^nC_n$

$\Rightarrow$ Remainder = 9 last digit of $\left(3^{3^{4n+2}} + 2\right)$ = last digit of $3^9 + 2$

$= 9^4\cdot 3 + 2$

$= (10-1)^4\cdot 3 + 2$

$= 3 + 2 + 10\lambda'$

$\therefore$ Remainder = 5

21. The remainder left out when $8^{2n}-(62)^{2n+1}$ is divided by 9 is : **[AIEEE-2009]**

$8^{2n}-(62)^{2n+1}$ को 9 से भाग देने पर शेषफल बचता है :

(A) 0 (B) 2 (C) 7 (D) 8

Ans. B

Sol. $8^{2n} - (62)^{2n+1}$

$= (9-1)^{2n} - (63-1)^{2n+1}$

$= \sum_{r=0}^{2n} {}^{2n}C_r 9^{2n-r}\cdot(-1)^r - \sum_{r=0}^{2n+1} {}^{2n+1}C_0 (63)^{2n+1-r}(-1)^r$

$= 9\lambda + 1 + 1$

$= 9\lambda + 2$

$\therefore$ Remainder $= 2$

22. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N\}$, where N is the set of natural numbers, then

$X \cup Y$ is equal to : **[JEE(Main) 2014]**

यदि $X = \{4^n - 3n - 1 : n \in N\}$ तथा $Y = \{9(n-1) : n \in N\}$, हैं, जहाँ N प्राकृत संख्याओं का समुच्चय है, तो

$X \cup Y$ बराबर है :

(A) X (B) Y (C) N (D) $Y - X$

Ans. B

Sol. $4^n - 3n - 1$

$= (1+3)^n - 3n - 1$

$= {}^nC_2\, 3^2 + {}^nC_3\, 3^3 + ...$

$= 9[{}^nC_2 + 3 \times {}^nC_3 + ...]$

Set X contains numbers which are multiples of 9 but not all multiples of 9

$\therefore X \cup Y = Y$

NUMERICALLY GREATEST TERM

23. Only 4th term is numerically greatest term in $\left(2+\frac{3x}{8}\right)^{10}$. Then the range of x is :

$\left(2+\frac{3x}{8}\right)^{10}$ के प्रसार में केवल 4th पद महत्तम पद हो तो x का परिसर है :

(A) $\left(\frac{-64}{21},-2\right)\cup\left(1,\frac{64}{25}\right)$ (B) $\left(\frac{-64}{21},-2\right)\cup\left(2,\frac{64}{21}\right)$

(C) $\left(-2,\frac{-64}{21}\right)\cup\left(2,\frac{64}{21}\right)$ (D) $\left(\frac{-64}{21},-2\right)\cup\left(1,\frac{64}{21}\right)$

Ans. B

Sol. $\left(2+\frac{3x}{8}\right)^{10}$

T_4 is only NGT

$\Rightarrow [P] = 3 \;\&\; P \notin Z$

$\Rightarrow 3 < P < 4$

$\Rightarrow 3 < \frac{11}{1+\left|\frac{16}{3x}\right|} < 4$

$\Rightarrow 3\left(1+\left|\frac{16}{3x}\right|\right)-11<0$ & $11<4\left(1+\left|\frac{16}{3x}\right|\right)$

$\Rightarrow \frac{16}{|x|}<8$ $\quad 7<\frac{64}{3|x|}$

$\Rightarrow |x|>2\{\because |x|>0\}$ $\quad \Rightarrow |x|<\frac{64}{21}$

$\Rightarrow x\in(-\infty,-2)\cup(2,\infty)$ $\quad x\in\left(-\frac{64}{21},0\right)\cup\left(0,\frac{64}{21}\right)$

$\Rightarrow x\in\left(-\frac{64}{21},-2\right)\cup\left(2,\frac{64}{21}\right)$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

24. If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is : **[AIEEE-2012]**

(A) A rational number other than positive integers

(B) An irrational number

(C) An odd positive integer

(D) An even positive integer

यदि n एक धनपूर्णांक है, तो $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$: **[AIEEE-2012]**

(A) धनपूर्णांकों को छोड़कर एक परिमेय संख्या हैं (B) एक अपरिमेय संख्या है

(C) एक विषम धनपूर्णांक है (D) एक सम धनपूर्णांक है।

Ans. B

Sol. $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$

$= 2[T_2 + T_2 + T_6 + ... + T_{2n}]$

$= 2[{}^{2n}C_1(\sqrt{3})^{2n-1} + {}^{2n}C_3(\sqrt{3})^{2n-3} + ...]$

= An Irrational Number

SUMMATION OF SERIES

25. $\sum_{r=1}^{n} r.\ {}^{2n}C_r$ is equal to :

$\sum_{r=1}^{n} r.\ {}^{2n}C_r$ का मान है :

(A) $n.2^{n-1}$ (B) 2^{2n-1} (C) $2^{n-1}+1$ (D) $n.2^{2n-1}$

Ans. D

Sol. $\sum_{r=1}^{n} r\cdot{}^{2n}C_r = \sum_{r=1}^{n} r\cdot\frac{2n}{r}\cdot{}^{2n-1}C_{r-1}$

$= 2n\sum_{r=1}^{n} {}^{2n-1}C_{r-1}$

$= 2n\cdot\left({}^{2n-1}C_0 + {}^{2n-1}C_1 + ... + {}^{2n-1}C_{n-1}\right)$

$= 2\cdot n$

$= 2n\cdot\frac{2^{2n-1}}{2} = n\cdot 2^{2n-1}$

26. If C_0, C_1, C_2, ...C_n denote the coefficients in the expansion of $(1 + x)^n$, then the value of $\sum_{r=0}^{n}(r+1)C_r$ is :

यदि $(1 + x)^n$ के प्रसार में $C_0, C_1, C_2, ...C_n$ द्विपद गुणांक है तब $\sum_{r=0}^{n}(r+1)C_r$ है :

(A) $n.2^n$ (B) $(n+1)2^{n-1}$ (C) $(n+2)2^{n-1}$ (D) $(n+2)2^{n-2}$

Ans. C

Sol. $\sum_{r=0}^{n}(r+1)C_r = \sum_{r=0}^{n} r\cdot C_r + \sum_{r=0}^{n} C_r$

$= n\sum_{r=1}^{n} {}^{n-1}C_{r-1} + 2^n$

$= n\cdot 2^{n-1} + 2^n$

$= 2^{n-1}(n+2)$

27. If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots + C_n x^n$, then $C_0 + 5C_1 + 9C_2 + \ldots\ldots\ldots + (4n+1)C_n$ is equal to :

यदि $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots + C_n x^n$, तब $C_0 + 5C_1 + 9C_2 + \ldots\ldots\ldots + (4n+1)C_n$ बराबर होगा :

(A) $n.2^n$ (B) $(n+1)2^n$ (C) $(2n+1)2^n$ (D) $(4n+1)2^n$

Ans. C

Sol. $C_0 + {}^5C_1 + {}^9C_2 + \ldots\ldots + (4n+1)\,C_n$

$= \sum_{r=0}^{n}(4r+1)^n\, C_r$

$= 4\sum_{r=0}^{n} r\cdot {}^nC_r + \sum_{r=0}^{n} {}^nC_r$

$= 4\sum_{r=1}^{n} r\cdot\frac{n}{r}\,{}^{n-1}C_{r-1} + \sum_{r=0}^{n} {}^nC_r$

$= 4n\cdot 2^{n-1} + 2^n$

$= (4n+2)2^{n-1}$

$= (2n+1)\,2^n$

28. If $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \ldots\ldots + a_{2n}$ equals :

यदि $(1 - x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, तब $a_0 + a_2 + a_4 + \ldots\ldots + a_{2n}$ बराबर होगा:

(A) $\frac{3^n+1}{2}$ (B) $\frac{3^n-1}{2}$ (C) $\frac{1-3^n}{2}$ (D) $3^n + \frac{1}{2}$

Ans. A

Sol. $(1 - x + x^2)^n = a_0 + a_1x + ... + a_{2n}x^{2n}$

Put $x = 1$

$1 = a_0 + a_1 + ... + a_{2n}$ ...(i)

Put $x = -1$

$3^n = a_0 - a_1 + ... + a_{2n}$ ...(ii)

(i) + (ii)

$2\,[a_0 + a_2 + ... + a_{2n}] = 3^n + 1$

$$\Rightarrow a_0 + a_2 + ... + a_{2n} = \frac{3^n + 1}{2}$$

29. If $(1+ x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{nC_n}{C_{n-1}} =$

यदि $(1+ x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, तो $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{nC_n}{C_{n-1}} =$

(A) $\frac{n}{2}$ (B) $n(n+1)$ (C) $\frac{n(n+1)}{12}$ (D) $\frac{n(n+1)}{2}$

Ans. D

Sol. $$\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n \cdot C_n}{C_{n-1}}$$

$$= \sum_{r=1}^{n} r \cdot \frac{C_r}{C_{r-1}}$$

$$\left\{ \because \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \right\}$$

$$= \sum_{r=1}^{n} r \cdot \left(\frac{n-r+1}{r} \right) = \sum_{r=1}^{n} (n-r+1)$$

$$= (n+1)\sum_{r=1}^{n} 1 - \sum_{r=1}^{n} r$$

$$= n(n+1) - \frac{n(n+1)}{2}$$

$= \frac{n(n+1)}{2}$

30. If $S_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, then $\frac{t_n}{s_n}$ is equal to :

यदि $S_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ एवं $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$, तो $\frac{t_n}{s_n}$ बराबर है :

(A) $\frac{1}{2}n$ (B) $\frac{1}{2}n - 1$ (C) n–1 (D) $\frac{2n-1}{2}$

Ans. A

Sol. $s_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$

$t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$

$t_n = \sum_{r=0}^{n} \frac{n-r}{{}^nC_r}$ $\therefore$ $2t_n = n\sum_{r=0}^{n} \frac{1}{{}^nC_r} = ns_n$ $\Rightarrow$ $\frac{t_n}{s_n} = \frac{n}{2}$

31. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots.. + {}^{20}C_{10}$ is:

श्रेणी, ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots.. + {}^{20}C_{10}$ का योग है :

(A) $-{}^{20}C_{10}$ (B) $\frac{1}{2}\,{}^{20}C_{10}$ (C) 0 (D) ${}^{20}C_{10}$

Ans. B

Sol. $s = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 \ldots\ldots + {}^{20}C_{10}$

$s = {}^{20}C_{10} - {}^{20}C_{11} + \ldots\ldots\ldots + {}^{20}C_{20}$

$2s = 0 + {}^{20}C_{10}$

$s = \frac{1}{2}({}^{20}C_{10})$

32. The sum of coefficients of integral powers of x in the binomal expansion of $\left(1-2\sqrt{x}\right)^{50}$ is :

$\left(1-2\sqrt{x}\right)^{50}$ के द्विपद प्रसार में x को पूर्णांकीय घातों के गुणांको का योग है : **[JEE(Main) 2015]**

(A) $\frac{1}{2}(3^{50}+1)$ (B) $\frac{1}{2}(3^{50})$ (C) $\frac{1}{2}(3^{50}-1)$ (D) $\frac{1}{2}(2^{50}+1)$

Ans. A

Sol. $(1-2\sqrt{x})^{50} = 1 - {}^{50}C_1(2\sqrt{x}) + {}^{50}C_2(x^2)(\sqrt{x})^2 + ...$

Put $\sqrt{x} = 1$ & $\sqrt{x} = -1$ & then add

Ans, $\frac{3^{50}+1}{2}$

33. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + ... + ({}^{21}C_{10} - {}^{10}C_{10})$ is :

$({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + ... + ({}^{21}C_{10} - {}^{10}C_{10})$ का मान है –

[JEE(Main) 2017]

(A) $2^{20} - 2^9$ (B) $2^{20} - 2^{10}$ (C) $2^{21} - 2^{11}$ (D) $2^{21} - 2^{10}$

Ans. B

Sol. $\left({}^{21}C_0 + {}^{21}C_1 + {}^{21}C_2 + ... + {}^{21}C_{10}\right) - \left({}^{10}C_0 + {}^{10}C_1 + ... + {}^{10}C_{10}\right) - \left({}^{21}C_0 - {}^{10}C_0\right)$

$\frac{2^{21}}{2} - 2^{10} = 2^{20} - 2^{10}$

34. The sum of the co-efficients of all odd degree terms in the expansion of $\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$,

$(x > 1)$ is : **[JEE(Main) 2018]**

$\left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$, $(x > 1)$ के प्रसार में सभी विषम घातों वाले पदों के गुणांको का योग है –

(A) 1 (B) 2 (C) –1 (D) 0

Ans. B

Sol. $f(x) = \left(x+\sqrt{x^3-1}\right)^5 + \left(x-\sqrt{x^3-1}\right)^5$

$= 2(x^5 + 10x^3(x^3-1) + 5x(x^3-1)^2)$

Sum of coefficients of odd degree $= \frac{f(1)-f(-1)}{2} = 2$

35. For $r = 0, 1, ... 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r\left(B_{10}\ B_r - C_{10}\ A_r\right)$ is equal to : **[JEE 2010]**

$(1+x)^{10}$, $(1+x)^{20}$ और $(1+x)^{30}$ के प्रसार में माना x^r के गुणांक क्रमशः A_r, B_r और C_r है जहां r = 0, 1, ... 10, तब $\sum_{r=1}^{10} A_r \left(B_{10}\ B_r - C_{10}\ A_r\right)$ बराबरः

(A) $B_{10} - C_{10}$ (B) $A_{10}\left(B_{10}^2 - C_{10}\,A_{10}\right)$

(C) 0 (D) $C_{10} - B_{10}$

Ans. D

Sol. $A_r = {}^{10}C_r$, $B_r = {}^{20}C_r$, $C_r = {}^{30}C_r$

$$\sum_{r=1}^{10}\left({}^{20}C_{10}\ {}^{10}C_r\ {}^{20}C_r - {}^{30}C_{10}\left({}^{10}C_r\right)^2\right)$$

$$= {}^{20}C_{10}\left({}^{10}C_1\ {}^{20}C_1 + {}^{10}C_2\ {}^{20}C_2 + ... + {}^{10}C_{10}\ {}^{20}C_{10}\right)$$

$$-{}^{30}C_{10}\left({}^{10}C_1^2 + {}^{10}C_2^2 + ... + {}^{10}C_{10}^2\right)$$

$$= {}^{20}C_{10}\left({}^{30}C_{10} - 1\right) - {}^{30}C_{10}\left({}^{20}C_{10} - 1\right)$$

$$= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}$$

BINOMIAL THEOREM FOR ANY INDEX

36. Number of terms in the expansion of $(1 - 4x)^{-30}$ are :

(A) 29 (B) 30 (C) 31 (D) infinitely many

$(1 - 4x)^{-30}$ के प्रसार में पदों की संख्या होगी :

(A) 29 (B) 30 (C) 31 (D) अनन्त

Ans. D

Sol. No. of terms in the expansion of $(1 - 4x)^{-30}$ are infinite.

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON IMPORTANT TERMS

1. (a)If the coefficients of $(2r+4)^{th}$, $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, find r.

(b) If the coefficients of the rth, $(r+1)^{th}$ & $(r+2)^{th}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r.

(c) If the coefficients of 2nd, 3rd & 4th terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.

(a) यदि $(1+x)^{18}$ के प्रसार में $(2r+4)^{th}$, $(r-2)^{th}$ पदों के गुणांक समान है तो r का मान ज्ञात कीजिए।

(b) यदि $(1+x)^{14}$ के प्रसार में r^{th}, $(r+1)^{th}$ तथा $(r+2)^{th}$ पद समान्तर श्रेणी में है तो r का मान ज्ञात कीजिए।

(c) $(1+x)^{2n}$ के प्रसार में 2nd, 3rd तथा 4th पद समान्तर श्रेणी में है तो सिद्ध कीजिए $2n^2 - 9n + 7 = 0$.

Ans. (a) $r = 6$ (b) $r = 5$ or 9

Sol. (a) Coefficient of T_{2r+4} term = Coefficient of T_{r-2} in $(1+x)^{18}$

$\Rightarrow {}^{18}C_{2r+3} = {}^{18}C_{r-3}$

$\Rightarrow 2r + 3 = r - 3$ or $2r + r + 3 - 3 = 18$

$\Rightarrow r = 6$ or $r = 6$

$\Rightarrow r = 6$

(b) $T_r, T_{r+1}, T_{r+2} \rightarrow$ A.P. $(1+x)^{14}$

$\Rightarrow {}^{14}C_{r-1}, {}^{14}C_r, {}^{14}C_{r+1} \rightarrow$ A.P.

$2 \cdot {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$

$\Rightarrow 4 \cdot {}^{14}C_r = {}^{14}C_r + {}^{14}C_{r-1} + {}^{14}C_{r+1} + {}^{14}C_r$

$= {}^{15}C_r + {}^{15}C_{r+1}$

$4 \cdot {}^{14}C_r = {}^{16}C_{r+1}$

$$\Rightarrow 4 \cdot \frac{\lfloor 14}{\lfloor 14-r \rfloor r} = \frac{\lfloor 16}{\lfloor 15-r \rfloor r+1} \Rightarrow 4 = \frac{16 \times 15}{(15-r)(r+1)}$$

$\Rightarrow (r+1)(r-15) = -60 \Rightarrow r^2 - 14r + 45 = 0 \Rightarrow r = 9 \text{ or } 5$

(c) $T_2, T_3, T_4 \rightarrow$ A.P. $(1+x)^{2n}$

$\Rightarrow 2 \cdot {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$

$\Rightarrow 4 \cdot {}^{2n}C_2 = {}^{2n}C_2 + {}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_2$

$\Rightarrow \frac{4 \cdot \lfloor 2n}{\lfloor 2n-2 \rfloor 2} = {}^{2n+1}C_2 + {}^{2n+1}C_3 = {}^{2n+2}C_3$

$\Rightarrow \frac{4 \cdot \lfloor 2n}{\lfloor 2n-2 \rfloor 2} = \frac{(2n+2)(2n+1)\lfloor 2n}{(2n-1)\lfloor 2n-2 \rfloor 3}$

$\Rightarrow 12(2n-1) = 4n^2 + 6n + 2$

$\Rightarrow 4n^2 + 6n + 2 - 24n + 12 = 0$

$\Rightarrow 4n^2 - 18n + 14 = 0$

$\Rightarrow 2n^2 - 9n + 7 = 0$

2. (a) Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

(b) Find the value of x for which the fourth term in the expansion, $\left(5^{\frac{2}{5}\log_5\sqrt{4^x+44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1}+7}}}\right)^8$ is 336.

(a) निम्न के प्रसार में x रहित पद ज्ञात कीजिए। **(i)** $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ **(ii)** $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

(b) x का मान ज्ञात कीजिए जिसके लिए $\left(5^{\frac{2}{5}\log_5\sqrt{4^x+44}} + \frac{1}{5^{\log_5\sqrt[3]{2^{x-1}+7}}}\right)^8$ के प्रसार में 4th पद 336 है।

Ans. (a) (i) $\frac{5}{12}$ (ii) $T_6 = 7$, (b) $x = 0$ or 1

Sol. (a) (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$

$$= \sum_{r=0}^{10} {}^{10}C_r \left(\frac{x}{3}\right)^{\frac{10-r}{2}} \left(\frac{\sqrt{3}}{2x^2}\right)^r$$

Term independent of x if

$$5 - \frac{r}{2} - 2r = 0 \Rightarrow 10 - 5r = 2$$

$$\Rightarrow r = 2$$

$$\text{Term} = {}^{10}C_2 \cdot \left(\frac{1}{3}\right)^4 \cdot \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{45}{27} \cdot \frac{1}{4} = \frac{5}{12}$$

$$\text{(ii)} \left[\frac{1}{2}x^{\frac{1}{3}} + x^{-\frac{1}{5}}\right]^8 = \sum_{r=0}^{8} {}^8C_r \left(\frac{1}{2}\right)^{8-r} (x)^{\frac{8-r}{3}} \cdot x^{-\frac{r}{5}}$$

Term independent of x if $\frac{8-r}{3} - \frac{r}{5} = 0$

$$\Rightarrow 40 - 5r - 3r = 0$$

$$\Rightarrow r = 5$$

$$\text{Term } T_6 = {}^8C_5 \left(\frac{1}{2}\right)^3 = \frac{\lfloor 8}{\lfloor 5 \lfloor 3} \cdot \frac{1}{8} = 7$$

$$\text{(b)} \left(5^{\frac{2}{5}\log_5 \sqrt{4^x+44}} + \frac{1}{5^{\log_5 \sqrt[3]{2^{x-1}+7}}}\right)^8$$

$$= \left(\left(4^x + 44\right)^{\frac{2}{5}\cdot 2} + \frac{1}{\left(2^{x-1}+7\right)^{\frac{1}{3}}}\right)^8$$

Given $T_4 = 336$

$$\Rightarrow {}^{8}C_3\left(4^x+44\right)^{\frac{8-3}{5}}\cdot\left(\frac{1}{\left(2^{x-1}+7\right)^{\frac{1}{3}}}\right)^3=336$$

$$\Rightarrow \frac{4^x+44}{2^{x-1}+7}=6$$

Let $2^x = t$

$\Rightarrow t^2 + 44 = 3t + 42$

$\Rightarrow t^2 - 3t + 2 = 0$

$\Rightarrow t = 1$ or $t = 2$

$\Rightarrow x = 0$ or $x = 1$

BASED ON FINDING COEFFICIENTS

3. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x-\frac{2}{x}\right)^{10}$ is 1 : 32.

सिद्ध कीजिए $(1-x^2)^{10}$ के प्रसार में x^{10} का गुणांक तथा $\left(x-\frac{2}{x}\right)^{10}$ के प्रसार में x रहित पद के गुणांक का अनुपात 1 : 32 है।

Ans.

Sol. $\left(1-x^2\right)^{10}=\sum_{r=0}^{10}{}^{10}C_r\left(-x^2\right)^r$

Coefficient of x^{10} is ${}^{10}C_5(-1)^5$

$$\left(x-\frac{2}{x}\right)^{10}=\sum_{r=0}^{10}{}^{10}C_r x^{10-r}\cdot(-2)^r\cdot x^{-r}$$

Term independent of x if $r = 5$

$\Rightarrow$ Coefficient of term independent of $x = {}^{10}C_5(-2)^5$

$$\Rightarrow \frac{{}^{10}C_5(-1)^5}{{}^{10}C_5(-2)^5}=\frac{1}{32}$$

4. Let a and b be the coefficient of x^3 in $(1+x+2x^2+3x^3)^4$ and $(1+x+2x^2+3x^3+4x^4)^4$ respectively. Find the value of $(a-b)$.

यदि $(1 + x + 2x^2 + 3x^3)^4$ तथा $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ के प्रसार में x^3 के गुणांक क्रमशः a तथा b है तो (a – b) का मान होगा ।

Ans. 0

Sol. Coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$

$$= \frac{\lfloor 4}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4} \cdot 1^{P_1} \cdot x^{P_2} \cdot 2^{P_3} \cdot x^{2P_3} \cdot 3^{P_4} \cdot x^{3P_4}$$

$$= \frac{\lfloor 4}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4} \cdot 1^{P_1} \cdot 2^{P_3} \cdot 3^{P_4} \cdot x^{P_2+2P_3+3P_4}$$

$P_1 + P_2 + P_3 + P_4 = 4$

$P_2 + 2P_3 + 3P_4 = 3$

$\Rightarrow a = 40$

$$(1 + x + 2x^2 + 3x^3 + 4x^4)^4 = \frac{\lfloor 4 \cdot 2^{P_3} \cdot 3^{P_4} \cdot 4^{P_5}}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4 \lfloor P_5} x^{P_2+2P_3+3P_4+4P_5}$$

For x^3

$P_1 + P_2 + P_3 + P_4 + P_5 = 4$

$P_2 + 2P_3 + 3P_4 + 4P_5 = 3$

$\Rightarrow b = 40$

$\Rightarrow b - a = 0$

5. Find the coefficient of x^r in the expression of

$(x + 3)^{n-1} + (x + 3)^{n-2}(x + 2) + (x + 3)^{n-3}(x + 2)^2 + ... + (x + 2)^{n-1}$.

$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + ... + (x+2)^{n-1}$ में x^r का गुणांक ज्ञात कीजिए।

Ans. ${}^nC_r(3^{n-r} - 2^{n-r})$

Sol. $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + ... + (x+2)^{n-1}(x+3)^{n-1}$

G.P. with $r = \dfrac{(x+2)}{x+3}$

$$= (x+3)^{n-1} \dfrac{\left[\left[\dfrac{(x+2)}{x+3}\right]^n - 1\right]}{\dfrac{x+2}{x+3} - 1}$$

$$= (x+3)^{n-1} \cdot \dfrac{(x+3)}{-1} \cdot \left[\left(\dfrac{x+2}{x+3}\right)^n - 1\right]$$

$$= -(x+3)^n \left[\left(\dfrac{x+2}{x+3}\right)^n - 1\right]$$

$= (x+3)^n - (x+2)^n$

Coefficient of $x^r = {}^nC_r(3^{n-r} - 2^{n-r})$

6. Find the term independent of x in the expansion of $(1 + x + 2x^3)\left(\dfrac{3x^2}{2} - \dfrac{1}{3x}\right)^9$.

$(1 + x + 2x^3)\left(\dfrac{3x^2}{2} - \dfrac{1}{3x}\right)^9$ के प्रसार में x रहित पद ज्ञात कीजिए।

Ans. $\dfrac{17}{54}$

Sol. Co-efficient of x^0 in $(1 + x + 2x^3)\left(\dfrac{3}{2}x^2 - \dfrac{1}{3x}\right)^9$

= Co-efficient of x^0 in $\left(\dfrac{3}{2}x^2 - \dfrac{1}{3x}\right)^9$ + Co-efficient of x^{-1} in $\left(\dfrac{3}{2}x^2 - \dfrac{1}{3x}\right)^9$

+ 2 Co-efficient of x^{-3} in $\left(\dfrac{3}{2}x^2 - \dfrac{1}{3x}\right)^9$

$$= {}^9C_{r_1}\left(\frac{3}{2}x^2\right)^{9-r_1}\left(-\frac{1}{3x}\right)^{r_1} + {}^9C_{r_2}\left(\frac{3}{2}x^2\right)^{9-r_2}\left(-\frac{1}{3x}\right)^{r_2} + 2\ {}^9C_{r_3}\left(\frac{3}{2}x^2\right)^{9-r_3}\left(-\frac{1}{3x}\right)^{r_3}$$

$= {}^{9}C_6\left(\frac{3}{2}\right)^3\left(-\frac{1}{3}\right)^6 + 2\ {}^{9}C_7\left(\frac{3}{2}\right)^2\left(-\frac{1}{3}\right)^7 = \frac{17}{54}$

7. Let $(1+x^2)^2 \cdot (1+x)^n = \sum_{K=0}^{n+4} a_K \cdot x^K$. If a_1, a_2 & a_3 are in AP, find n.

माना $(1+x^2)^2 \cdot (1+x)^n = \sum_{K=0}^{n+4} a_K \cdot x^K$. यदि a_1, a_2 तथा a_3 समान्तर श्रेणी में है तो n का मान ज्ञात कीजिए।

Ans. n = 2 or 3 or 4

Sol. $\left(1+x^2\right)^2(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$

$\Rightarrow \left(1+2x^2+x^4\right)(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$

$\Rightarrow \left(1+2x^2+x^4\right)\sum_{r=0}^{n} {}^{n}C_r x^r = a_0 + a_1 x + ...$

$a_1 = {}^{n}C_1 = n$

$a_2 = {}^{n}C_2 + 2\cdot{}^{n}C_0 = \frac{n(n-1)}{2} + 2 = \frac{n^2-n+4}{2}$

$a_3 = {}^{n}C_3 + 2\cdot{}^{n}C_1 = \frac{n(n-1)(n-2)}{6} + 2\cdot n$

$\Rightarrow a_1 = n, a_2 = \frac{n^2-n+4}{2}, a_3 = \frac{n(n-1)(n-2)}{6} + 2n$

Given $2a_2 = a_1 + a_3$

$\Rightarrow n^2 - n + 4 = n + \frac{n(n-1)(n-2)}{6} + 2n$

$\Rightarrow n^2 - 2n + 4 = \frac{\left(n^2-n\right)(n-2)}{6} + 2n$

$\Rightarrow n^2 - 2n + 4 - 2n = \frac{n(n-1)(n-2)}{6}$

$\Rightarrow n(n-2) + (-2)(n-2) = \frac{n(n-1)(n-2)}{6}$

$\Rightarrow 6(n-2)^2 - (n-2)\,n\,(n-1) = 0$

$\Rightarrow (n-2)[6n - 12 - n^2 + n] = 0$

$(n-2)[7n - 12 - n^2] = 0$

$n = 2, 3, 4$

8. Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0.$

समीकरण $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$ के सभी मूलों (वास्तविक या काल्पनिक) का योगफल ज्ञात कीजिए।

Ans. 500

Sol. $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$

$$\because \left(\frac{1}{2} - x\right)^{2001} = \sum_{r=0}^{2001} {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} \cdot (-x)^r$$

$$= {}^{2001}C_0 \left(\frac{1}{2}\right)^{2001} + {}^{2001}C_1 \left(\frac{1}{2}\right)^{2000}(-x) + {}^{2001}C_2 \left(\frac{1}{2}\right)^{2002} \cdot (-x)^2 + \ldots + {}^{2001}C_{2000} \left(\frac{1}{2}\right)^1 (-x)^{2000}$$

$$+ {}^{2001}C_{2001}(-x)^{2001}$$

$$\Rightarrow \sum_{r}^{2000} {}^{2001}C_r \left(\frac{1}{2}\right)^{2001-r} (-x)^r = 0$$

Sum of the roots of this equation

$$= -\frac{\text{Coefficient of } x^{1999}}{\text{coefficient of } x^{2000}}$$

$$= +\frac{\left(\frac{1}{2}\right)^2 \cdot {}^{2001}C_{1999}}{\left(\frac{1}{2}\right)^1 \cdot {}^{2001}C_{2000}}$$

$$= \frac{1}{4} \times \frac{2}{1} \times \frac{\underline{|2001}\,\underline{|1}\,\underline{|2000}}{\underline{|2}\,\underline{|1999}\,\underline{|2001}}$$

$= \frac{1}{4} \times 2000 = 500$

NUMERICALLY GREATEST TERM

9. Find numerically greatest term in the expansion of

(i) $(2 + 3x)^9$ when $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

निम्न के प्रसार में संख्यात्मक महत्तम पद होगा

(i) $(2 + 3x)^9$ जब $x = \frac{3}{2}$ (ii) $(3 - 5x)^{15}$ जब $x = \frac{1}{5}$

Ans. (i) $T_7 = \frac{7.3^{13}}{2}$ (ii) 455×3^{12}

Sol. (i) $(2 + 3x)^9$; $x = \frac{3}{2}$

$$p = \frac{n+1}{1+\left|\frac{2}{3x}\right|} = \frac{10}{1+\left|\frac{2}{3\times\frac{3}{2}}\right|} = \frac{10}{1+\frac{4}{9}} = \frac{90}{13}$$

$p \notin Z$

$\Rightarrow$ NGT is $T_{[P]+1} = T_7 = {}^9C_6(2)^3 \cdot 3^6$

$$= \frac{9\times8\times7}{6} \times 8 \times 3^6 \times \left(\frac{3}{2}\right)^6$$

$$= \frac{3^{13}\cdot 7}{2}$$

(ii) $(3 - 5x)^{15}$ when $x = \frac{1}{5}$

$$P = \frac{15+1}{1+\left|\frac{3}{-5x}\right|} = \frac{16}{1+3} = 4 \in Z$$

$\Rightarrow T_4$ & $T_{4+1} = T_5$ are NGT

$$T_5 = {}^{15}C_4 (3)^{11} \cdot (-5x)^4 \Big|_{x=\frac{1}{5}}$$

$$= \frac{15\times14\times13\times12}{4\times3\times2}\times3^{11}$$

$$= 3^{12}\times455 = T_5$$

$$T_4 = {}^{15}C_3 3^{12}\cdot(-5x)^4$$

$$\Rightarrow T_4 = \frac{15\times14\times13}{3\times2}\times -3^{12}$$

$$\Rightarrow T_4 = -455\times3^{12}$$

$$\Rightarrow |T_4| = |T_5|$$

10. Find the index n of the binomial $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient $(n \in N)$.

n का मान ज्ञात कीजिए जिसके लिए $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ के प्रसार में 9th पद संख्यात्मक महत्तम है $(n \in N)$.

Ans. $n = 12$

Sol. For T_9 to be the numerically greatest term, $r = \left[\frac{n+1}{1+\left|\frac{x}{a}\right|}\right] = \left[\frac{n+1}{1+\left|\frac{1}{2}\right|}\right] = 8$

$$\Rightarrow \quad 8 < \frac{2(n+1)}{3} < 9 \Rightarrow 11 < n < 12.5 \Rightarrow n = 12$$

11. For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.

x के किस मान के लिए $(5 + 3x)^{10}$ के प्रसार में 4th पद संख्यात्मक महत्तम है।

Ans. $\frac{5}{8} < x < \frac{20}{21}$

Sol. 4th term is greatest in the expansion of $(5 + 3x)^{10}$

$$\Rightarrow [P] = 3$$

$$\Rightarrow 3 < P < 4$$

$$\Rightarrow 3 < \frac{10+1}{1+\left|\frac{5}{3x}\right|} < 4$$

$\Rightarrow 3+3\left|\frac{5}{3x}\right|<11$ & $11<4+4\left|\frac{5}{3x}\right|$

$\frac{15}{3|x|}<8$ & $\frac{7}{20}<\frac{1}{3|x|}$

$\Rightarrow 8|x|>5$ & $|x|<\frac{20}{21}$

$\Rightarrow |x|>\frac{5}{8}$ & $|x|<\frac{20}{21}$

$\Rightarrow x\in\left(-\infty,-\frac{5}{8}\right)\cup\left(\frac{5}{8},\infty\right)$ & $x\in\left(-\frac{20}{21},\frac{20}{21}\right)$

$\Rightarrow x\in\left(-\frac{20}{21},-\frac{5}{8}\right)\cup\left(\frac{5}{8},\frac{20}{21}\right)$

Given $x>0$

$\Rightarrow x\in\left(\frac{5}{8},\frac{20}{21}\right)$

MULTINOMIAL THEOREM

12. Find the coefficient of

(a) $x^2y^3z^4$ in the expansion of $(ax-by+cz)^9$.

(b) $a^2b^3c^4d$ in the expansion of $(a-b-c+d)^{10}$.

निम्न के गुणांक ज्ञात कीजिए।

(a) $(ax-by+cz)^9$ के प्रसार में $x^2y^3z^4$ का गुणांक

(b) $(a-b-c+d)^{10}$ के प्रसार में $a^2b^3c^4$ का गुणांक

Ans. (a) $-1260\cdot a^2b^3c^4$; (b) -12600

Sol. (a) $(ax-by+cz)^9=\sum_{P_1+P_2+P_3}\frac{\lfloor 9}{\lfloor P_1\lfloor P_2\lfloor P_3}a^{P_1}\cdot(-b)^{P_2}(+c)^{P_3}x^{P_1}y^{P_2}z^{P_3}$

For $x^2y^3z^4$

$P_1=2, P_2=3, P_3=4$

$$\Rightarrow \text{Coefficient} = -\frac{\lfloor 9}{\lfloor 2 \lfloor 3 \lfloor 4} a^2 b^3 c^4$$

$$= -\frac{9\times 8\times 7\times 6\times 5}{2\times 6} a^2 b^3 c^4$$

$$= -63 \times 20\ a^2 b^3 c^4$$

$$= -1260\ a^2 b^3 c^4$$

(b) $(a - b - c + d)^{10}$

$$= \sum_{P_1+P_2+P_3+P_4=10} \frac{\lfloor 10\ aa^{P_1} (-b)^{P_2} (-c)^{P_3} d^{P_4}}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4}$$

For $a^2b^3c^4d$

$P_1 = 2, P_2 = 3, P_3 = 4, P_4 = 1$

Coefficient of $a^2b^3c^4d$

$$= \frac{\lfloor 10}{\lfloor 2 \lfloor 3 \lfloor 4 \lfloor 1} \cdot (-1)^3 \cdot (-1)^4$$

$$= -\frac{10\times 9\times 8\times 7\times 6\times 5}{2\times 6} = -20\times 63\times 10 = -12600$$

13. Find the coefficient of

(a) x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$ (b) x^4 in the expansion of $(2 - x + 3x^2)^6$.

निम्न के गुणांक ज्ञात कीजिए।

(a) $(1 + x + x^2 + x^3)^{11}$ के प्रसार में x^4 का गुणांक

(b) $(2 - x + 3x^2)^6$ के प्रसार में x^4 का गुणांक

Ans. (a) 990 (b) 3660

Sol. $$\left(1 + x + x^2 + x^3\right)^{11} = \sum_{P_1+P_2+P_3+P4} \frac{\lfloor 11 \cdot x}{\lfloor P_1 \lfloor P_2 \lfloor P_3 \lfloor P_4}$$

For x^4

$P_4 + 2P_3 + 3P_4 = 4$

(b) $\left(2-x+3x^2\right)^6 = \sum_{r_1+r_2+r_3=6} \frac{\lfloor 6 2^{r_1}\cdot(-1)^{r_2}(3)^{r_3}}{\lfloor r_1 \lfloor r_2 \lfloor r_3} x^{r_2+2r_3}$

For x^4

$r_1 + r_2 + r_3 = 6$ & $r_2 + 2r_3 = 4$

INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

14. (a) Show that the integral part in each of the following is odd. $n \in N$

(A) $\left(5+2\sqrt{6}\right)^n$ (B) $\left(8+3\sqrt{7}\right)^n$

(b) Show that the integral part in each of the following is even. $n \in N$

(A) $\left(3\sqrt{3}+5\right)^{2n+1}$ (B) $\left(5\sqrt{5}+11\right)^{2n+1}$

(a) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग विषम है जहां $n \in N$

(A) $\left(5+2\sqrt{6}\right)^n$ (B) $\left(8+3\sqrt{7}\right)^n$

(b) सिद्ध कीजिए प्रत्येक का पूर्णांक भाग सम है जहां $n \in N$

(A) $\left(3\sqrt{3}+5\right)^{2n+1}$ (B) $\left(5\sqrt{5}+11\right)^{2n+1}$

Ans.

Sol. (a) (A) $\left(5+2\sqrt{6}\right)^n = I + f; 0 < f < 1$

$\left(5-2\sqrt{6}\right)^n = 0 + f'; 0 < f' < 1$

$\Rightarrow I + f + f' = \left(5+2\sqrt{6}\right)^n + \left(5-2\sqrt{6}\right)^n$

$= 2\left[{}^nC_0 5^n + {}^nC_2 5^{n-2}\left(2\sqrt{6}\right)^2 + ...\right]$

$\Rightarrow I + f + f' =$ even Integer

$\because\ 0 < f < 1$ & $0 < f' < 1$

$\Rightarrow 0 < f + f' < 2$

Case-I : If I is even integer then $f + f'$ must be even integer.

but $0 < f + f' < 2 \rightarrow$ There is no even integer

Case-II : If I is odd integer then $f + f'$ must be odd integer

$\Rightarrow f + f' =$ as $0 < f + f' < 2$

$\therefore$ Integral part of $\left(5+2\sqrt{6}\right)^n$ is odd.

(B) $\left(8+3\sqrt{7}\right)^n = I + f; 0 < f < 1$

$\left(8-3\sqrt{7}\right)^n = 0 + f'; 0 < f' < 1$

$\Rightarrow \left(8+3\sqrt{7}\right)^n + \left(8-3\sqrt{7}\right)^n = 2\left[{}^nC_0 8^n + {}^nC_2 8^{n-2}\left(3\sqrt{7}\right)^2 + ...\right] = I + f + f'$

$\Rightarrow I + f + f' =$ even integer

$\because\ 0 < f < 1$ & $0 < f' < 1$

$\Rightarrow 0 + f + f' < 2 \Rightarrow f + f' = 1$

$\Rightarrow$ Integral part of $\left(8+3\sqrt{7}\right)^{n}$ is odd.

(b) (A) $\left(3\sqrt{3}+5\right)^{2n+1} = I + f; 0 < f < 1$

$\left(3\sqrt{3}-5\right)^{2n+1} = f'; 0 < f' < 1$

$\downarrow$

$5\cdot19$

$\Rightarrow I - f - f' = \left(3\sqrt{3}+5\right)^{2n+1} - \left(3\sqrt{3}-5\right)^{2n+1}$

$= 2\left[{}^{2n+1}C_1\left(3\sqrt{3}\right)^{2n}(5)^1 + ...\right]$

$\Rightarrow I + f - f' =$ even integer

$-1 < -f' < 0$

$-1 < f - f' < 1$

$\Rightarrow f - f' = 0$

$\Rightarrow I$ must be even

(B) $\left(5\sqrt{5}+11\right)^{2n+1} = I + f$

$\left(5\sqrt{5}-11\right)^{2n+1} = f'$

$\Rightarrow I + f - f' = \left(5\sqrt{5}+1\right)^{2n+1} - \left(5\sqrt{5}-1\right)^{2n+1}$

$= 2\left[{}^{2n+1}C_1\left(5\sqrt{5}\right)^{2n}\cdot 1 + ...\right]$

$\because -1 < f - f' < 0$

$\Rightarrow f - f' = 0$

$\Rightarrow I$ must be even

15. If $\left(7+4\sqrt{3}\right)^n = p+\beta$ where n & p are positive integers and β is a proper fraction show that $(1-\beta)(p+\beta)=1$.

यदि $\left(7+4\sqrt{3}\right)^n = p+\beta$ जहां n & p धनात्मक पूर्णांक है तथा β भिन्नात्मक भाग है तो सिद्ध कीजिए $(1-\beta)(p+\beta)=1$.

Ans.

Sol. $\left(7+4\sqrt{3}\right)^n = p+\beta;\ n \in N \ \&\ p \in N \ \&\ \beta$ is a proper fraction.

$$\left(7-4\sqrt{3}\right)^n = \beta';\ 0<\beta'<1$$

$$\Rightarrow \beta' = \frac{(49-48)^n}{\left(7+4\sqrt{3}\right)^n} = \frac{1}{p+\beta}$$

$$\Rightarrow p+\beta+\beta' = \left(7+4\sqrt{3}\right)^n + \left(7-4\sqrt{3}\right)^n$$

$$= 2\left[{}^nC_0\,7^n + {}^nC_2\,7^{n-2}\left(4\sqrt{3}\right)^2 + ...\right]$$

$\Rightarrow \beta + \beta' < 2$

$\Rightarrow \beta + \beta' = 1$

$\Rightarrow 1 - \beta = \beta'$

$$\Rightarrow 1-\beta = \frac{1}{p+\beta}$$

$$\Rightarrow (p+\beta)(1-\beta) = 1$$

16. If $\left(6\sqrt{6}+14\right)^{2n+1} = N \ \&\ F$ be the fractional part of N, prove that $NF = 20^{2n+1}$ $(n \in N)$

यदि $\left(6\sqrt{6}+14\right)^{2n+1} = N$ तथा F, N का भिन्नात्मक भाग है तो सिद्ध कीजिए $NF = 20^{2n+1}$ $(n \in N)$

Ans.

Sol. $\left(6\sqrt{6}+14\right)^{2n+1} = I + f = N;\ 0<f<1$

$\left(6\sqrt{6}-14\right)^{2n+1} = f'$

$\Rightarrow I + f - f' = 2\left({}^{2n+1}C_1\left(6\sqrt{6}\right)^{2n} + {}^{2n+1}C_3\left(6\sqrt{6}\right)^{2n-2} + ...\right)$

$\Rightarrow f - f' = 0$

$\Rightarrow f = f'$

$\Rightarrow Nf = Nf' = \left(6\sqrt{6}+14\right)^{2n+1}\cdot\left(6\sqrt{6}-14\right)^{2n+1}$

$= \left(216-196\right)^{2n+1}$

$= 20^{2n+1}; n \in N$

17. Let I denotes the integral part and F the proper fractional part of $\left(3+\sqrt{5}\right)^n$ where $n \in N$ and if ρ denotes the rational part and σ the irrational part of the same, show that

$\rho = \frac{1}{2}(I + 1)$ and $\sigma = \frac{1}{2}(I + 2F - 1)$.

यदि I तथा F, $\left(3+\sqrt{5}\right)^n$ के प्रसार में क्रमशः पूर्णांक तथा भिन्नात्मक भाग हैं जहां $n \in N$ तथा यदि ρ परिमेय भाग तथा σ अपरिमेय भाग है तो सिद्ध कीजिए

$\rho = \frac{1}{2}(I + 1)$ and $\sigma = \frac{1}{2}(I + 2F - 1)$.

Ans.

Sol. $\left(3+\sqrt{5}\right)^n = I + F$

$\left(3-\sqrt{5}\right)^n = F'$

$\Rightarrow I + F + F' = 2\left[{}^nC_0 3^n + {}^nC_2 3^{n-2}\left(\sqrt{5}\right)^2 + ...\right]$

$\Rightarrow 0 < F + F' < 2$

$\Rightarrow F + F' = 1$

$\Rightarrow F' = 1 - F$

$$\because (I+F)=\left(3+\sqrt{5}\right)^n=\left[{}^nC_0 3^n\left(\sqrt{5}\right)^0+{}^nC_1 3^{n-1}\left(\sqrt{5}\right)^1+{}^nC_2 3^{n-2}\left(\sqrt{5}\right)^2+....\right]$$

$$(F')=\left(3-\sqrt{5}\right)^n=\left[{}^nC_0 3^n\left(\sqrt{5}\right)^0-{}^nC_1 3^{n-1}\left(\sqrt{5}\right)^1+....\right]$$

$$\Rightarrow 2\rho=I+F+F'\Rightarrow \rho=\frac{I+1}{2}$$

$$\&\ 2\sigma=I+F-F'\Rightarrow \sigma=\frac{I+2F-1}{2}$$

SUMMATION OF SERIES

18. Given $s_n=1+q+q^2+.....+q^n$ & $S_n=1+\frac{q+1}{2}+\left(\frac{q+1}{2}\right)^2+....+\left(\frac{q+1}{2}\right)^n$, $q\neq 1$, prove that ${}^{n+1}C_1+{}^{n+1}C_2.s_1+{}^{n+1}C_3.s_2+....+{}^{n+1}C_{n+1}.s_n=2^n\,.\,S_n$.

दिया गया है $s_n=1+q+q^2+.....+q^n$ और $S_n=1+\frac{q+1}{2}+\left(\frac{q+1}{2}\right)^2+....+\left(\frac{q+1}{2}\right)^n$, $q\neq 1$, तो सिद्ध कीजिए ${}^{n+1}C_1+{}^{n+1}C_2.s_1+{}^{n+1}C_3.s_2+....+{}^{n+1}C_{n+1}.s_n=2^n\,.\,S_n$.

Ans.

Sol. $s_n=\frac{q^{n+1}-1}{q-1}$ and $S_n=\frac{\left(\frac{q+1}{2}\right)^{n+1}-1}{\frac{q+1}{2}-1}$

${}^{n+1}C_1+{}^{n+1}C_2 s_1+.....+{}^{n+1}C_{n+1}\,s_n$

$={}^{n+1}C_1+{}^{n+1}C_2\left(\frac{q^2-1}{q-1}\right)+.....+{}^{n+1}C_{n+1}\,\frac{q^{n+1}-1}{q-1}$

$=\frac{1}{q-1}[{}^{n+1}C_1 q+{}^{n+1}C_2 q^2+....+{}^{n+1}C_{n+1}q^{n+1}-{}^{n+1}C_1-{}^{n+1}C_2-....-{}^{n+1}C_{n+1}]$

$=\frac{1}{q-1}[(1+q)^{n+1}-1-2^{n+1}+1]=\frac{1}{q-1}[(1+q)^{n+1}-2^{n+1}]=\left(\frac{\left(\frac{q+1}{2}\right)^{n+1}-1}{\frac{q-1}{2}}\right).2^n=2^n\,S_n$

19. Find the sum of the series $\sum_{r=0}^{n}(-1)^r.{}^nC_r\left[\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{3r}}+\frac{15^r}{2^{4r}}+.....\text{up to m terms}\right]$

श्रेणी $\sum_{r=0}^{n}(-1)^r.{}^nC_r\left[\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{3r}}+\frac{15^r}{2^{4r}}+.....\text{up to m terms}\right]$ का योगफल ज्ञात कीजिए।

Ans. $\frac{\left(2^{mn}-1\right)}{\left(2^{n}-1\right)\left(2^{mn}\right)}$

Sol. $\sum_{r=0}^{n}(-1)^{r}.\ {}^{n}C_{r}\left[\frac{1}{2^{r}}+\left(\frac{3}{4}\right)^{r}+\left(\frac{7}{8}\right)^{r}+\text{..... .m terms}\right]$

$=\sum_{r=0}^{n}\left[(-1)^{r}\ {}^{n}C_{r}\frac{1}{2^{r}}+(-1)^{r}\ {}^{n}C_{r}\left(\frac{3}{4}\right)^{r}+\text{..... .m terms}\right]$

$=[\ {}^{n}C_{0}-{}^{n}C_{1}\left(\frac{1}{2}\right)+{}^{n}C_{2}\left(\frac{1}{2}\right)^{2}+\text{.....}]+$

$[\ {}^{n}C_{0}-{}^{n}C_{1}\left(\frac{3}{4}\right)+{}^{n}C_{2}\left(\frac{3}{4}\right)^{2}+\text{.....}]+\text{.....m terms}$

$=\left(1-\frac{1}{2}\right)^{n}+\left(1-\frac{3}{4}\right)^{n}+\left(1-\frac{7}{8}\right)^{n}+\text{.....m terms}$

$=\frac{1}{2^{n}}\left(\frac{1-\frac{1}{2^{mn}}}{1-\frac{1}{2^{n}}}\right)=\frac{1}{2^{n}-1}\left(1-\frac{1}{2^{mn}}\right)$

20. Using the theory of permutation or otherwise prove that

$${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}={}^{105}C_{90}$$

क्रमचय सिद्धान्त या अन्यथा प्रयोग कर सिद्ध कीजिये।

$${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}={}^{105}C_{90}.$$

Ans.

Sol. ${}^{100}C_{10}+5\cdot{}^{100}C_{11}+10\cdot{}^{100}C_{12}+10\cdot{}^{100}C_{13}+5\cdot{}^{100}C_{14}+{}^{100}C_{15}$

$={}^{5}C_{0}\cdot{}^{100}C_{10}+{}^{5}C_{1}\cdot{}^{100}C_{11}+{}^{5}C_{2}\cdot{}^{100}C_{12}+{}^{5}C_{3}\cdot{}^{100}C_{13}+{}^{5}C_{4}\cdot{}^{100}C_{14}+{}^{5}C_{5}\cdot{}^{100}C_{15}$

$|r_1-r_2|=-10=\text{Constant}$

$(x+1)^{5}={}^{5}C_{0}\,x^{5}+{}^{5}C_{1}\,x^{4}+{}^{5}C_{2}\,x^{3}+{}^{5}C_{3}\,x^{2}+{}^{5}C_{4}\,x^{1}+{}^{5}C_{5}$

$(1+x)^{100}={}^{100}C_{0}+{}^{100}C_{1}\,x+\text{.....}+{}^{100}C_{15}\,x^{15}+\text{....}+{}^{100}C_{100}\,x^{100}$

$$(1+x)^{105}=\left[\sum_{r=0}^{5}{}^{5}C_{r}x^{r}\right]\left[\sum_{r=0}^{100}{}^{100}C_{r}x^{r}\right]$$

Comparing coefficient of x^{15}

$^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 100 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15}$

$= {}^{105}C_{90}$

21. Let $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r}$. Find the total number of divisors of $f(11)$.

माना $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r}$. तब $f(11)$ के कुल विभाजक ज्ञात कीजिये।

Ans. 24

Sol. $f(n) = \sum_{r=0}^{n} \sum_{k=r}^{n} \binom{k}{r} = \sum_{r=0}^{n} \left[\sum_{k=r}^{n} {}^{k}C_{r} \right]$

$= \sum_{r=0}^{n} \left[{}^{r}C_{r} + {}^{r+1}C_{r} + ... + {}^{n}C_{r} \right]$

$\sum_{r=0}^{n} \left[{}^{r+1}C_{r+1} + {}^{r+1}C_{r} + {}^{r+2}C_{r} + ... + {}^{n}C_{r} \right]$

$= \sum_{r=0}^{n} \left[{}^{r+2}C_{r+1} + {}^{r+2}C_{r} + ... + {}^{n-1}C_{r} + {}^{n}C_{r} \right]$

$= \sum_{r=0}^{n} {}^{n+1}C_{r+1} = {}^{n+1}C_{1} + {}^{n+1}C_{2} + ... + {}^{n+1}C_{n+1}$

$= 2^{n+1} - 1$

$\Rightarrow f(11) = 2^{12} - 1 = 4095 = 5 \times 9 \times 7 \times 13 = 5 \times 7 \times 13 \times 3^2$

$\Rightarrow$ Total no. of divisor $= 2 \times 2 \times 2 \times 3 = 24$

22. Given that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, find the values of :

(i) $a_0 + a_1 + a_2 + ... + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 ..... + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2$

दिया गया है $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, तो निम्न के मान ज्ञात कीजिए –

(i) $a_0 + a_1 + a_2 + ... + a_{2n}$; (ii) $a_0 - a_1 + a_2 - a_3 ..... + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2$

Ans. (i) 3^n (ii) 1, (iii) a_n

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2 x^2 + ... + a_{2n}x^{2n}$

(i) $a_0 + a_1 + a_2 + ... + a_{2n} = (1 + 1 + 1)^n = 3^n$

(ii) $a_0 - a_1 + a_2 - a_3 + ... + a_{2n} = (1 - 1 + (-1)^2)^n = 1$

(iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 =$

$\because \left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + ... + a_{2n} x^{2n}$ _____(i)

$x \to \frac{1}{x}$

$\left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ... + \frac{a_{2n}}{x^{2n}}$

$\left[\frac{\left(1 + x + x^2\right)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + ... + a_{2n}}{x^{2n}}$

$x \to -x$

$\left[\frac{\left(1 - x + x^2\right)^n}{x^{2n}}\right] = \frac{a_0 x^{2n} - a_1 x^{2n-1} + a_2 x^{2n-2} + ... + a_{2n}}{x^{2n}}$ _____(ii)

By (i) × (ii); we get

$\Rightarrow \left[\left(1 + x^2\right)^2 - x^2\right]^n = ... + x^{2n}\left(a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2\right)$

Comparing coefficient of x^{2n} in $[1 + x^4 + x^2]^n$

$\because\ (1 + x + x^2)^n = a_0 + a_1 x + ... + a_{2n} x^{2n}$

$x \to x^2$

$(1 + x^2 + x^4)^n = a_0 + a_1 x^2 + ... + a_n x^{2n} + ... + a_{2n} x^{4n}$

$\Rightarrow a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 = a_n$

23. Prove that $\sum_{K=0}^{n} {}^nC_K \sin Kx . \cos(n - K)x = 2^{n-1} \sin nx$.

सिद्ध कीजिए $\sum_{K=0}^{n} {}^nC_K \sin Kx . \cos(n - K)x = 2^{n-1} \sin nx$

Ans.

Sol. $S = {}^nC_0.\sin(0x).\cos nx + {}^nC_1.\sin x.\cos(n-1)x$

$\quad\quad\quad\quad ..... + {}^nC_n \sin nx.\cos(0x) ..... (1)$

$S = {}^nC_0.\sin nx.\cos(0x) + {}^nC_1.\sin(n-1)x.\cos x +$

$\quad\quad\quad\quad ..... + {}^nC_n \sin(0x)\cos nx ..... (2)$

Add (1) & (2)

$2S = ({}^nC_0 + {}^nC_1 + ... + {}^nC_n)\sin nx \Rightarrow S = 2^{n-1}\sin nx$

24. Find the number of divisors of the number

$N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + ... + 2000 \cdot {}^{2000}C_{2000}$.

$N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + ... + 2000 \cdot {}^{2000}C_{2000}$ के विभाजकों की संख्या ज्ञात कीजिए।

Ans. 8016

Sol. $N = \sum_{r=0}^{2000} {}^{2000}C_r \cdot r$

$= \sum_{r=1}^{2000} \frac{2000}{r} \cdot r \cdot {}^{1999}C_{r-1}$

$= 2000 \sum_{r=1}^{2000} {}^{1999}C_{r-1}$

$= 2 \times 10^3 \times (2)^{1999} = 2^{2000} \cdot 2^3 \cdot 5^3 = 2^{2003} \cdot 5^3$

$\Rightarrow$ No. of divisor $= 2004 \times 4 = 8016$

25. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, ..., C_n$ are the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in N$, then prove the following:

क्रमचय सिद्धान्त का प्रयोग कर सर्वसमिकाओं को सिद्ध कीजिये। यदि $C_0, C_1, C_2, ..., C_n$, $(1 + x)^n$, $n \in N$ के प्रसार में द्विपद गुणांक है। तब निम्न को सिद्ध कीजिए –

(a) $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2 = \frac{(2n)!}{n!\,n!}$

(b) $C_0 C_1 + C_1 C_2 + C_2 C_3 + ... + C_{n-1} C_n = \frac{(2n)!}{(n+1)!\,(n-1)!}$

(c) $C_o C_r + C_1 C_{r+1} + C_2 C_{r+2} + ... + C_{n-r} C_n = \frac{2n!}{(n-r)!\,(n+r)!}$

(d) $\sum_{r=0}^{n-2} \left({}^nC_r \cdot {}^nC_{r+2} \right) = \frac{(2n)!}{(n-2)!\,(n+2)!}$

Ans.

Sol. (a) $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2$

$$(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$$

$$(x+1)^n = C_0x^n + C_1x^{n-1} + ... + C_n$$

$$\Rightarrow (1+x)^{2n} = \left[\sum_{r=0}^{n} C_r x^r\right]\left[\sum_{r=0}^{n} C_r x^{n-r}\right]$$

Comparing coefficient of x^n in $(1+x)^{2n}$

$$\Rightarrow C_0^2 + C_1^2 + C_2^2 + ...C_n^2 = {}^{2n}C_n = \frac{\lfloor 2n}{\lfloor n \lfloor n}$$

(b) $C_0C_1 + C_1C_2 + C_2C_3 + ... + C_{n-1}C_n$

Comparing coefficient of x^{n-1} in $(1+x)^{2n}$

$$\Rightarrow C_0C_1 + C_1C_2 + ... + C_{n-1}C_n = {}^{2n}C_{n-1} = \frac{\lfloor 2n}{\lfloor n-1 \lfloor n+1}$$

(c) We know that

$(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_2x^r + ... + C_nx^n$ _____(i)

$$\left(1+\frac{1}{x}\right)^n = C_0 + C_1\frac{1}{x} + C_2\frac{1}{x^2} + ...$$

$$+ C_r\frac{1}{x^r} + ... + C_n\frac{1}{x^n}$$ _____(ii)

Multiply (i) & (ii) we get

$$\frac{(1+x)^{2n}}{x^n} = (C_0 + C_1x + ... C_rx^r + ... + C_nx^n)$$

$$\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{C_n}{x^n}\right)$$ _____(iii)

Now coefficient of $\frac{1}{x^r}$ in RHS

$= C_0C_r + C_1C_{r+1} + ... C_{n-r}C_n$

$\therefore$ Coefficient of $\frac{1}{x^r}$ in LHS

= coefficient of x^{n-r} in $(1+x)^{2n} = {}^{2n}C_{n-r}$

$$= \frac{2n!}{(n-r)!(n+r)!}$$

But (iii) is an identity

$\therefore$ Coefficient of $\frac{1}{x^r}$ in RHS

= coefficient of $\frac{1}{x^r}$ in LHS

$$\Rightarrow \quad C_0C_r+C_1C_{r+1}+....+C_{n-r}C_n=\frac{2n!}{(n-r)!(n+r)!}$$

(d) $\sum_{r=0}^{n-2} {}^nC_r\,{}^nC_{r+2} = {}^2C_0\,{}^nC_2 + {}^nC_1\,{}^nC_3 + {}^nC_2\,{}^nC_4 + ... + {}^nC_{n-2}\,{}^nC_n$

Comparing coefficient of x^{n-2} in

$$\sum_{r=0}^{n-2} {}^nC_r\,{}^nC_{r+2} = \frac{\lfloor 2n}{\lfloor n-2 \rfloor n+2}$$

26. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, ..., C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$, then prove the following:

क्रमचय सिद्धान्त का प्रयोग कर सर्वसमिकाओं को सिद्ध कीजिये। यदि $C_0, C_1, C_2, ..., C_n$, $(1+x)^n$, $n \in N$ के प्रसार में द्विपद गुणांक है। तब निम्न को सिद्ध कीजिए।

(a) $C_1 + 2C_2 + 3C_3 + ...+ n \cdot C_n = n \cdot 2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + ...+ (n+1)C_n = (n+2)2^{n-1}$

(c) $C_0 + 3C_1 + 5C_2 + ...+ (2n+1)C_n = (n+1)\,2^n$

(d) $(C_0+C_1)(C_1+C_2)(C_2+C_3) ..... (C_{n-1}+C_n) = \frac{C_0 . C_1 . C_2 .... C_{n-1}\,(n+1)^n}{n!}$

(e) $1 \cdot C_o^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + ... + (2n+1)\,C_n^2 = \frac{(n+1)\,(2n)!}{n!\,n!}$

Ans.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1x + ... + {}^nC_nx^n$

(a) $\sum_{r=1}^{n} r \cdot {}^nC_r = \sum_{r=1}^{n} r \cdot \frac{n}{r}\,{}^{n-1}C_{r-1}$

$= n\left[{}^{n-1}C_0 + {}^{n-1}C_1 + ... + {}^{n-1}C_{n-1}\right]$

$= n \cdot 2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + ... + (n+1)C_n$

$$= \sum_{r=0}^{n} (r+1)^{n} C_r = \sum_{r=0}^{n} r \cdot {}^{n}C_r + \sum_{r=0}^{n} {}^{n}C_r$$

$$= n \cdot 2^{n-1} + 2^n = 2^{n-1}(n+2)$$

(c) $C_0 + 3C_1 + 5C_2 + ... + (2n+1)C_n$

$$= \sum_{r=0}^{n} (2r+1)\, {}^{n}C_r$$

$$= 2\sum_{r=0}^{n} r \cdot {}^{n}C_r + \sum_{r=0}^{n} {}^{n}C_r$$

$$= 2^n \cdot 2^{n-1} + 2^n$$

$$= 2^n (n+1)$$

(d) $(C_0 + C_1)(C_1 + C_2).....(C_{n-1} + C_n)$

$$= ({}^{n+1}C_1)({}^{n+1}C_2).....({}^{n+1}C_n) = ((n+1).C_0)\left(\frac{n+1}{2}.C_1\right)..... \left(\frac{n+1}{n}.C_{n-1}\right)$$

$$= \frac{(n+1)^n}{n!} C_0 C_1 ..... C_{n-1}$$

(e) $1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + ... + (2n+1)C_n^2$

$$= \sum_{r=0}^{n} (2r+1) C_r^2$$

$$= 2\sum_{r=0}^{n} r \cdot C_r^2 + \sum_{r=0}^{n} C_r^2$$

$$= 2\sum_{r=0}^{n} r \cdot \frac{n}{r}\, {}^{n-1}C_{r-1} \cdot {}^{n}C_r + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n \cdot \sum_{r=1}^{n} {}^{n-1}C_{r-1} \cdot {}^{n}C_r + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n\left[\underbrace{{}^{n-1}C_0}_{\text{Coefficient of } x^{n-1}} \underbrace{{}^{n}C_1}_{\text{Coefficient of } x} + {}^{n-1}C_1\, {}^{n}C_2 + ... + {}^{n-1}C_{n-1} \cdot {}^{n}C_n\right] + \sum_{r=0}^{n} \left({}^{n}C_r\right)^2$$

$$= 2n \cdot \text{coefficient of } x^n \text{ in } (1+x)^{2n-1} + \frac{\lfloor 2n}{\lfloor n \rfloor n}$$

$$= 2n\cdot\frac{\lfloor 2n-1}{\lfloor n-1\lfloor n}+\frac{\lfloor 2n}{\lfloor n\lfloor n}$$

$$= 2n\left[\frac{\lfloor 2n-1}{\lfloor n-1\lfloor n}+\frac{\lfloor 2n-1}{\lfloor n\lfloor n}\right]=\frac{\lfloor 2n}{\lfloor n\lfloor n-1}\left[1+\frac{1}{n}\right]$$

$$=\frac{\lfloor 2n}{\lfloor n\lfloor n}\cdot(n+1)$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 27 to 29

A path of length n is a sequence of points $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ with integer coordinates such that for all i between 1 and $n-1$ both inclusive,

either $x_{i+1} = x_i + 1$ and $y_{i+1} = y_i$ (in which case we say the ith step is rightward)

or $x_{i+1} = x_i$ and $y_{i+1} = y_i+1$ (in which case we say that the ith step is upward).

This path is said to start at (x_1, y_1) and end at (x_n, y_n). Let P(a, b), for a and b non-negative integers, denotes the number of paths that start at (0, 0) and end at (a, b).

n लम्बाई का पाथ एक अनुक्रम है जिसके बिन्दु पूर्णांक निर्देशांकों $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ है। इस प्रकार की 1 और $n-1$ के मध्य, जहां 1 और $n-1$ भी शामिल है, सभी i के लिए

या तो $x_{i+1} = x_i + 1$ और $y_{i+1} = y_i$ (में i^{th} स्टेप दांयी ओर है)

या $x_{i+1} = x_i$ और $y_{i+1} = y_i+1$ (में i^{th} स्टेप ऊपर की ओर है)

इस पथ का प्रारम्भिक बिन्दु (x_1, y_1) और अन्त बिन्दु (x_n, y_n) है। माना P(a, b), जहां a, b अऋणात्मक पूर्णांक है। उन पाथों की संख्या दर्शाते हैं जिसका प्रारम्भिक बिन्दु (0, 0) और अन्त बिन्दु (a, b) है।

27. The value of $\sum_{i=0}^{10} P(i, 10-i)$, is :

$\sum_{i=0}^{10} P(i, 10-i)$ का मान होगा:

(A) 1024 (B) 512 (C) 256 (D) 128

Ans. A

Sol. $$\sum_{r=0}^{10} P(i,10-i)=\sum_{i=0}^{10}\frac{\lfloor 10}{\lfloor i\lfloor 10-i}$$

$$=\sum_{i=0}^{10} {}^{10}C_i = 2^{10} = 1024$$

28. Number of ordered pairs (i, j) where $i \neq j$ for which $P(i, 100-i) = P(j, 100-j)$, is :

$P(i, 100-i) = P(j, 100-j)$ के लिए क्रमित युग्मों (i, j) की संख्या (जहां $i \neq j$) है:

(A) 50 (B) 99 (C) 100 (D) 101

Ans. C

Sol. $i \neq j \quad P(i, 100-i) = P(j, 100-j)$

$\neq {}^{100}C_i = {}^{100}C_j$

$\Rightarrow i = j$ (wrong) or $\underbrace{i+j=100}_{\substack{(0,100)\\(1,99)\\(50,50)\\(49,51)\\(99,1)\\(100,0)}}$ & $i \neq j$

$\Rightarrow$ No. of ordered pair = 101 – 1 = 100

29. The sum $P(43, 4) + \sum_{j=1}^{5} P(49-j, 3)$ is equal to :

$P(43, 4) + \sum_{j=1}^{5} P(49-j, 3)$ का योग बराबर :

(A) P(4, 48) (B) P(3, 49) (C) P(4, 47) (D) P(5, 47)

Ans. A

Sol. $P(43,4) + \sum_{j=1}^{5} P(49-j,3)$

$= \frac{\lfloor 47}{\lfloor 43 \lfloor 4} + \sum_{j=1}^{5} \frac{\lfloor 52-j}{\lfloor 49-j \lfloor 3}$

$= {}^{47}C_4 + \sum_{j=1}^{5} {}^{52-j}C_3$

$= {}^{47}C_4 + {}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$

$= {}^{48}C_4 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$

$= {}^{52}C_4$

$= \frac{\lfloor 52}{\lfloor 4 \lfloor 48}$

= P(4, 48)

Paragraph for Question Nos. 30 to 32

Let $S_k = 3^k \binom{100}{0}\binom{100}{k} - 3^{k-1}\binom{100}{1}\binom{99}{k-1} + 3^{k-2}\binom{100}{2}\binom{98}{k-2} - \ldots\ldots\ldots + (-1)^k \binom{100}{k}\binom{100-k}{0}$

where $\binom{n}{r}$ represents the coefficient of x^r in the binomial expansion of $(1+x)^n \ \forall\ n \in N$

and $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

माना $S_k = 3^k \binom{100}{0}\binom{100}{k} - 3^{k-1}\binom{100}{1}\binom{99}{k-1} + 3^{k-2}\binom{100}{2}\binom{98}{k-2} - \ldots\ldots\ldots + (-1)^k \binom{100}{k}\binom{100-k}{0}$

जहां $\binom{n}{r}$ $(1+x)^n \ \forall\ n \in N$ के प्रसार में x^r का गुणांक है और $V_k = \left(\frac{1}{2}\right)^k S_k = M(100, k)$.

30. The value of $\sum_{k=0}^{100} S_k\, S_{100-k}$ equals :

$\sum_{k=0}^{100} S_k\, S_{100-k}$ का मान है :

(A) $\binom{200}{100}$ (B) $2^{100}\binom{200}{100}$ (C) $2^{200}\binom{200}{100}$ (D) $2^{99}\binom{200}{99}$

Ans. B

Sol. $S_K = 3^K\ {}^{100}C_0\ {}^{100}C_K - 3^{K-1}\ {}^{100}C_1\ {}^{99}C_{K-1} + \ldots. + (-1)^K \cdot {}^{100}C_K\ {}^{100-K}C_0$

$(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1\, x + \ldots. + {}^{100}C_K\, x^K + \ldots. + {}^{100}C_{100}\, x^{100}$

$$S_K = \sum_{r=0}^{K} 3^{K-r}\ {}^{100}C_r\ {}^{100-r}C_{K-r} \cdot (-1)^r$$

$$= \sum_{r=0}^{K} 3^{K-r} \frac{\lfloor 100}{\lfloor 100-r \rfloor r} \frac{\lfloor 100-r}{\lfloor 100-K \rfloor K-r} (-1)^r$$

$$= \sum_{r=0}^{K} 3^{K-r} \cdot (-1)^r\ {}^{100}C_K \cdot {}^{K}C_r$$

$= {}^{100}C_K\ (3-1)^K = 2^K \cdot {}^{100}C_K$

$$\Rightarrow V_K = \frac{1}{2^K} S_K = {}^{100}C_K = M(100, K)$$

$$\sum_{K=0}^{100} S_K S_{100-K} = \sum_{K=0}^{100} {}^{100}C_K \cdot {}^{100}C_{100-K} \cdot 2^K \cdot 2^{100-K}$$

$$= 2^{100} \sum_{K=0}^{100} \left({}^{100}C_K\right)^2 \rightarrow \text{Coefficient of } x^{100} \text{ in } (1+x)^{200}$$

$$= 2^{100} \cdot {}^{200}C_{100}$$

$$= 2^{100} \cdot \binom{200}{100}$$

31. The value of the $M(100, 49) + M(100, 50)$ is equal to :

$M(100, 49) + M(100, 50)$ का मान है :

(A) $M(99, 50)$ (B) $M(100, 51)$ (C) $M(101, 49)$ (D) $M(101, 51)$

Ans. D

Sol. $\because M(100, K) = {}^{100}C_K$

$\Rightarrow M(100, 49) + M(100, 50)$

$= {}^{100}C_{49} + {}^{100}C_{50}$

$= {}^{100}C_{50} = M(101, 50) = M(101, 51)$

32. In an examination, there are 100 true-false type questions. The number of ways in which one student can make his choice to attempt or not to attempt the questions is given by :

किसी परीक्षा में 100 सत्य असत्य प्रकार के प्रश्न है। तब कोई विद्यार्थी कुल कितने प्रकार से प्रश्न का उत्तर दे या नहीं दे सकता है :

(A) $\sum_{k=0}^{100} S_k$ (B) $\sum_{k=0}^{100} V_k$ (C) $\sum_{k=1}^{100} S_k$ (D) $\sum_{k=1}^{100} V_k$

Ans. A

Sol. $100 \rightarrow$ question (True/False)

$= {}^{100}C_0 \cdot 2^0 + {}^{100}C_1 \cdot 2 + ... + {}^{100}C_{100} \cdot 2^{100}$

$$= \sum_{K=0}^{100} {}^{100}C_K \cdot 2^K = \sum_{K=0}^{100} S_K$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

33. In the expansion of $\left(x-\frac{\alpha}{x}\right)^n$ and $\left(x+\frac{\beta}{x^2}\right)^n$ in powers of x have one term independent of x, then n is divisible by :

यदि $\left(x-\frac{\alpha}{x}\right)^n$ तथा $\left(x+\frac{\beta}{x^2}\right)^n$ के प्रसार में x की घातों के रूप में केवल एक पद ही x से स्वतन्त्र हो तब n विभाजित होगा:

(A) 2 (B) 3 (C) 4 (D) 6

Ans. ABD

Sol. $\left(x-\frac{\alpha}{x}\right)^n=\sum_{r=0}^{n}{}^nC_r x^{n-2r}\cdot(-\alpha)^r \rightarrow$ Term independent of x if $n=2r$

$\left(x+\frac{\beta}{x^2}\right)^n=\sum_{r=0}^{n}{}^nC_r x^{n-3r}\cdot(\beta)^r \rightarrow$ Term independent of x if $n=3r$

$\Rightarrow$ n is divisible by 2, 3 or 6.

34. In the expansion of $(a+b+c)^{10}$:

(A) total number of terms is 66 (B) coefficient of a^8bc is 90

(C) coefficient of $a^4b^5c^3$ is 0 (D) none of these

$(a+b+c)^{10}$ के प्रसार में :

(A) कुल पदों की संख्या 66 है। (B) a^8bc का गुणांक 90 है।

(C) $a^4b^5c^3$ का गुणांक 0 है। (D) इनमकें से कोई नहीं

Ans. ABC

Sol. $(a+b+c)^{10}$

Total terms are ${}^{10+3-1}C_2={}^{12}C_2=\frac{12\times11}{2}=66$

$(a+b+c)^{10}=\sum_{P_1+P_2+P_3=10}\frac{\lfloor 10}{\lfloor P_1\lfloor P_2\lfloor P_3}a^{P_1}b^{P_2}c^{P_3}$

Coefficient of a^8bc is $\frac{\lfloor 10}{\lfloor 8\lfloor 1\lfloor 1}=90$

Coefficient of $a^4b^5c^3$ is 0 as $P_1+P_2+P_3=12\neq10$

35. In the expansion of $\left(x+\frac{\alpha}{x^2}\right)^n$, $a \neq 0$, if no term is independent of x, then n is :

$\left(x+\frac{a}{x^2}\right)^n$, $a \neq 0$ के प्रसार में कोई भी पद x से स्वतन्त्र नहीं है तब n =

(A) 10 (B) 12 (C) 16 (D) 20

Ans. ACD

Sol. $\left(x+\frac{a}{x^2}\right)^n ; a \neq 0 = \sum_{r=0}^{n} x^{n-3r} \cdot {}^nC_r \cdot a^r$

If no term is independent of x then

$n - 3r \neq 0$

$\Rightarrow n \neq 3r$

$\Rightarrow n \neq 0, 3, 6, 9, 12, 15$ .....

$\Rightarrow n = 10, 16 \& 20$

36. In the expansion of $\left(x+\frac{1}{x}+2\right)^{2009}$

(A) Middle term is ${}^{4018}C_{2009}$
(B) The term independent of x is ${}^{4018}C_{2009}$
(C) Coefficient of x in the expansion is ${}^{4018}C_{2010}$
(D) Coefficient of 1/x in the expansion is ${}^{4018}C_{2008}$

$\left(x+\frac{1}{x}+2\right)^{2009}$ के प्रसार में –

(A) मध्य पद ${}^{4018}C_{2009}$ है। (B) x से स्वतंत्र पद ${}^{4018}C_{2009}$ है।
(C) x का गुणांक ${}^{4018}C_{2010}$ है। (D) 1/x का गुणांक ${}^{4018}C_{2008}$ है।

Ans. ABCD

Sol. $\left(x+\frac{1}{x}+2\right)^{2009}$

$=\left(\frac{x^2+2x+1}{x}\right)^{2009} = \frac{1}{x^{2009}}(x+1)^{4018}$

$=1\left[\sum_{r=0}^{4018} {}^{4018}C_r x^{r-2009}\right]$

Term independent of x if r = 2009

$\Rightarrow$ Coefficient of this is ${}^{4018}C_{2009}$.

Coefficient of x = ${}^{4018}C_{2010}$

Coefficient of $\frac{1}{x}$ = ${}^{4018}C_{2008}$

Coefficient of middle term is ${}^{4018}C_{2009}$

37. Which of the following statement(s) is/are correct?

(A) $1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+.....+\infty=4$

(B) Integral part of $\left(9+4\sqrt{5}\right)^n$, is even, $n \in N$.

(C) $\left({}^nC_0+{}^nC_1+{}^nC_2+.....+{}^nC_n\right)^2=1+{}^{2n}C_1+{}^{2n}C_2+.....+{}^{2n}C_{2n}$

(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x|<\frac{2}{3}$

निम्न में से कौनसा/से कथन सत्य है ?

(A) $1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+.....+\infty=4$

(B) $\left(9+4\sqrt{5}\right)^n$ का पूर्णांक मान सम है, जहाँ $n \in N$.

(C) $\left({}^nC_0+{}^nC_1+{}^nC_2+.....+{}^nC_n\right)^2=1+{}^{2n}C_1+{}^{2n}C_2+.....+{}^{2n}C_{2n}$

(D) यदि $|x|<\frac{2}{3}$ तो $\frac{1}{(3+2x)^2}$ को x के बढ़ते घात में अनन्त पदों तक लिखा जा सकता है

Ans. AC

Sol. A) $\left(1-\frac{1}{2}\right)^{-2}=\left(\frac{1}{2}\right)^{-2}=\frac{1}{2^{-2}}=4 \quad \Rightarrow$ **(A)** is correct

(B) it should be odd $\Rightarrow$ **(B)** is incorrect

(C) L.H.S. = $(2^n)^2$; R.H.S. = $2^{2n} \quad \Rightarrow$ **(C)** is correct

(D) $|x|<\frac{3}{2} \quad \Rightarrow$ **(D)** is incorrect

38. If $\left(9+\sqrt{80}\right)^n = 1+f$ where I, n are integers and $0<f<1$, then

(A) I is an odd integer (B) I is an even integer

(C) $(I+f)(1-f)=1$ (D) $1-f=\left(9-\sqrt{80}\right)^n$

यदि $\left(9+\sqrt{80}\right)^n = I + f$ जहाँ I तथा n पूर्णांक है जहाँ $0 < f < 1$ तो

(A) I विषम पूर्णांक (B) I सम पूर्णांक है

(C) $(I + f)(1 - f) = 1$ (D) $1 - f = \left(9-\sqrt{80}\right)^n$

Ans. ACD

Sol. $\left(9+4\sqrt{5}\right)^n = 3 + f$

let $\left(9-4\sqrt{5}\right)^n = f'$

$I + f + f' = \left(9+4\sqrt{5}\right)^n + \left(9-4\sqrt{5}\right)^n$ = even integer

$0 < f + f' < 2 \Rightarrow f + f' = 1$

$\therefore$ I is an odd integer $\Rightarrow$ (A)

39. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ If A_0, A_1, A_2 are in A.P. then the value of n is :

माना $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ यदि A_0, A_1, A_2 समान्तर श्रेणी में हो, तो n बराबर होगा :

(A) 2 (B) 3 (C) 5 (D) 7

Ans. AB

Sol. $(1 + x^2)^2 (1 + x)^n = (1 + 2x^2 + x^4)\left({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ...\right)$

$= {}^nC_0 + {}^nC_1 x + ({}^nC_2 + 2 \cdot {}^nC_0) x^2 + ...$

Hence $A_0 = 1$; $A_1 = {}^nC_1$; $A_2 = {}^nC_2 + 2$ which are in A.P.

40. If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to ${}^xC_y$; then the possible value(s) of x + y can be :

यदि ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ का मान ${}^xC_y$ के बराबर है ; तो x + y के संभव मान हो सकते है :

(A) 112 (B) 114 (C) 196 (D) 198

Ans. BD

Sol. ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$

$= ({}^{100}C_7 + {}^{100}C_6) + 3\,({}^{100}C_7 + {}^{100}C_8) + 3\,({}^{100}C_8 + {}^{100}C_9) + {}^{100}C_9 + {}^{100}C_{10}$

$= {}^{101}C_7 + 3 \cdot {}^{101}C_8 + 3 \cdot {}^{101}C_9 + {}^{101}C_{10}$

$= ({}^{101}C_7 + {}^{101}C_8) + 2\,[{}^{101}C_8 + {}^{101}C_9] + [{}^{101}C_{10} + {}^{101}C_9]$

$= {}^{102}C_8 + 2 \cdot {}^{102}C_9 + {}^{102}C_{10}$

$= {}^{103}C_9 + {}^{103}C_{10}$

$= {}^{104}C_{10} = {}^{104}C_{94}$

$\Rightarrow x + y = 114$ or $x + y = 198$

MATCH THE COLUMN TYPE QUESTION(S)

41. **Match the column**

	Column I	**Column II**
(A)	If the coefficients of x^7 and x^8 in $\left(2+\frac{x}{3}\right)^n$ are equal, then n is	(P) 2
(B)	If the coefficient of x in the expansion of $\left(x-\frac{1}{ax^2}\right)^{10}$ is –15, then the value of a, is	(Q) 01
(C)	The remainder when 2^{2003} in divided by 17 is	(R) 55
(D)	The last two digits of 3^{400}, are	(S) 8

	Column I	**Column II**
(A)	यदि $\left(2+\frac{x}{3}\right)^n$ के प्रसार में x^7 व x^8 के गुणांक बराबर है तब n =	(P) 2
(B)	$\left(x-\frac{1}{ax^2}\right)^{10}$ के प्रसार में x का गुणांक –15 है तब a =	(Q) 01
(C)	2^{2003} को 17 से विभाजित करने पर शेषफल	(R) 55
(D)	3^{400} के अन्तिम दो अंक	(S) 8

Ans. (A) R (B) P (C) S (D) Q

Sol. (A) Coefficient of x^7 & x^8 are equal in $\left(2+\frac{x}{3}\right)^n$

$$\Rightarrow {}^nC_7 2^{n-7}.\frac{x^7}{3^7} = {}^nC_8 2^{n-8}.\frac{x^8}{3^8}$$

$$\Rightarrow \frac{\lfloor n}{\lfloor n-7 \rfloor 7}\frac{2^n}{2^7}.\frac{1}{3^7} = \frac{\lfloor n}{\lfloor n-8 \rfloor 8}\frac{2^n}{2^8}.\frac{1}{3^8}$$

$$\Rightarrow \frac{1}{(n-7)} = \frac{1}{6\cdot 8} \qquad \Rightarrow n-7 = 48$$

$$\Rightarrow n = 55$$

$$(B)\ \left(x-\frac{1}{ax^2}\right)^{10} = \sum_{r=0}^{10} {}^{10}C_r x^{10-r}.\frac{(-1)^r}{a^r \cdot x^{2r}}$$

$$= \sum_{r=0}^{10} {}^{10}C_r \cdot x^{10-3r}.\left(\frac{-1}{a}\right)^r$$

For x^1 ; $10 - 3r = 1$

$3r = 9$

r = 3

$$\Rightarrow {}^{10}C_3\left(\frac{-1}{a}\right)^3 = -15$$

$$\Rightarrow \frac{10\times 9\times 8}{6} = 15\cdot a^3$$

$\Rightarrow 9 \times 8 \times 2 = 18a^3$

$\Rightarrow a^3 = 8$

$\Rightarrow a = 2$

(C) $2^{2003} = 2 \cdot 2^{2002} = 2^3 \cdot (2)^{2000} = 8 \cdot (16)^{500}$

$= 8 \cdot [17 - 1]^{500}$

$$= 8\sum_{r=0}^{500} {}^{500}C_r 17^{500-r}\cdot(-1)^r$$

$= 17\lambda + {}^{8500}C_{500}$

$\therefore$ Remainder = 8

(D) $3^{400} = (81)^{100} = (80 + 1)^{100}$

$$= \sum_{r=0}^{100} {}^{100}C_r (80)^r$$

$= {}^{100}C_0 + {}^{100}C_1 \cdot 80 + 100\lambda$

$= {}^{100}C_0 + 100\lambda' \; ; \lambda \in Z$

$= 100\,\lambda' + 1$

$\because$ Last digit will be 01.

Exercise J-M

1. The value of $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ is : **[JEE(Main)Jan.2023]**

$\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r$ का मान है : **[JEE(Main)Jan.2023]**

(A) ${}^{44}C_{22}$ (B) ${}^{45}C_{24}$ (C) ${}^{45}C_{23}$ (D) ${}^{44}C_{23}$

Ans. C

Sol. $\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_r = \sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_{23-r}$

$= {}^{22}C_0 \, {}^{23}C_{23} + {}^{22}C_1 \, {}^{23}C_{22} + \cdots + {}^{22}C_{21} \, {}^{23}C_2 + {}^{22}C_{22} \, {}^{23}C_1$

$(1+x)^{22} = {}^{22}C_0 + {}^{22}C_1 x + \cdots + {}^{22}C_{21} x^{21} + {}^{22}C_{22} x^{22}$

$(1+x)^{23} = {}^{23}C_0 + {}^{23}C_1 x + \cdots + {}^{23}C_{22} x^{22} + {}^{23}C_{23} x^{23}$

coefficient of x^{23} in $(1+x)^{22}(1+x)^{23} = \sum_{r=0}^{22} {}^{22}C_r \, {}^{23}\sum_{r=0}^{22} {}^{22}C_r \, {}^{23}C_{23-r}$

$= {}^{45}C_{23}$

2. If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots\ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ then α is equal to : **[JEE(Main)Jan.2023]**

यदि $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \ldots\ldots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ है, तो α बराबर है : **[JEE(Main)Jan.2023]**

(A) 15 (B) 60 (C) 10 (D) 30

Ans. A

Sol. $S = 0 \cdot \left({}^{30}C_0\right)^2 + 1 \cdot \left({}^{30}C_1\right)^2 + 2 \cdot \left({}^{30}C_2\right)^2 + \ldots\ldots + 30 \cdot \left({}^{30}C_{30}\right)^2$

$S = 30 \cdot \left({}^{30}C_0\right)^2 + 29 \cdot \left({}^{30}C_1\right)^2 + 28 \cdot \left({}^{30}C_2\right)^2 + \ldots + \cdot 0 \cdot \left({}^{30}C_0\right)^2$

$2S = 30 \cdot \left({}^{30}C_0{}^2 + {}^{30}C_1{}^2 + \ldots\ldots \cdot + {}^{30}C_{30}{}^2\right)$

$S = 15 \cdot {}^{60}C_{30} = 15 \cdot \frac{60!}{(30!)^2}$

$\frac{15 \cdot 10!}{(30!)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$

$\Rightarrow \alpha = 15$

3. If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to :

[JEE(Main)Jan.2023]

यदि $(1+x)^{10}$ के द्विपद प्रसार में x^{10-r} का गुणांक a_r है, तो $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4895 (B) 5445 (C) 3025 (D) 1210

Ans. D

Sol. Coeff. of x^{10-r} in $(1+x)^{10}$

$a_r = {}^{10}C_r$

$$\frac{a_r}{a_{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}} = \frac{10-r+1}{r} = \frac{11-r}{r}$$

$$\sum_{r=0}^{10} r^3\left(\frac{11-r}{r}\right)^2 = \sum_{r=0}^{10} r(11-r)^2$$

$$= \sum_{r=0}^{10}\left(r^3 - 22r^2 + 121r\right)$$

$$= \sum_{r=0}^{10} r^3 - 22\sum_{r=0}^{10} r^2 + 121\sum_{r=0}^{10} r = \left(\frac{10(10+1)}{2}\right)^2 - 22\times\frac{10(10+1)(20+1)}{6} + 121\times\frac{10\cdot 11}{2}$$

$$= 55^2 - 11\times 11\times 70 + 5\times 11^3 = 11^2[25-70+55] = 11^2\times 10 = 1210$$

4. $\sum_{k=0}^{6} {}^{51-k}C_3$ is equal to : **[JEE(Main)Jan.2023]**

$\sum_{k=0}^{6} {}^{51-k}C_3$ बराबर है : **[JEE(Main)Jan.2023]**

(A) ${}^{51}C_4 - {}^{45}C_4$ (B) ${}^{52}C_4 - {}^{45}C_4$ (C) ${}^{51}C_3 - {}^{45}C_3$ (D) ${}^{52}C_3 - {}^{45}C_3$

Ans. B

Sol. ${}^{51}C_3 + {}^{50}C_3 + - - - - - - - - - + {}^{45}C_3$

$\Rightarrow {}^{45}C_4 + {}^{45}C_3 + {}^{46}C_3 + - - - - - - - - - + {}^{51}C_3 - {}^{45}C_4$

$\Rightarrow {}^{46}C_4 + {}^{46}C_3 + - - - - - - - -$

$= {}^{51}C_4 + {}^{51}C_3 - {}^{45}C_4 = {}^{52}C_4 - {}^{45}C_4$

5. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}\,K}{a} = \frac{2^{\ell}\,m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to : **[JEE(Main)Jan.2023]**

माना $(1+x)^{99}$ के प्रसार में x की विषम घातों के गुणांको का योग K है। माना $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$ के प्रसार में मध्य पद a है। यदि $\frac{{}^{200}C_{99}\,K}{a}=\frac{2^{\ell}\,m}{n}$ है, जहाँ m तथा n विषम संख्याऐं है, तो क्रमित युग्म (l, n) बराबर है। **[JEE(Main)Jan.2023]**

(A) (50, 51) (B) (51,101) (C) (50,101) (D) (51,99)

Ans. C

Sol. Sum of odd coeff. of $x=\frac{2^{99}-0}{2}=k$

$k=2^{98}$

Middle term $\alpha = T101 = {}^{200}C_{100}\frac{(2)^{100}}{2^{50}}$

$= {}^{200}C_{100}\,2^{50}$

Now $\frac{{}^{200}C^{99}\times k}{\alpha}=\frac{200!}{100!99!}\times\frac{2^{98}}{{}^{200}C_{100}\times 2^{50}}$

$=\frac{100}{101}\times 2^{48}$

$\frac{25}{101}\times 2^{50}$

6. If the coefficient of x^{15} in the expansion of $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3}-\frac{1}{bx^{1/3}}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

[JEE(Main)Jan.2023]

यदि $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$ के प्रसार में x^{15} का गुणांक, $\left(ax^{1/3}-\frac{1}{bx^{1/3}}\right)^{15}$, के प्रसार में x^{-15} के गुणांक के बराबर है, जहाँ a तथा b घनात्मक संख्याएँ है, तो ऐसे प्रत्येक क्रमित युग्म (a, b) के लिए : **[JEE(Main)Jan.2023]**

(A) a = b (B) ab = 1 (C) a = 3b (D) ab = 3

Ans. B

Sol. Coefficient of x^{15} in $\left(ax^3+\frac{1}{bx^{1/3}}\right)^{15}$

$T_{r+1} = {}^{15}C_r\left(ax^3\right)^{15-r}\left(\frac{1}{bx^{1/3}}\right)^r$

$45 - 3r - \frac{r}{3} = 15$

$30 = \frac{10r}{3}$

$r = 9$

Coefficient of $x^{15} = {}^{15}C_9\, a^6\, b^{-9}$

Coefficient of x^{-15} in $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$

$T_{r+1} = {}^{15}C_r \left(ax^{1/3}\right)^{15-r} \left(-\frac{1}{bx^3}\right)^r$

$5 - \frac{r}{3} - 3r = -15$

$\frac{10r}{3} = 20$

$r = 6$

Coefficient $= {}^{15}C_6\, a^9 \times b^{-6}$

$\Rightarrow \frac{a^9}{b^6} = \frac{a^6}{b^9}$

$\Rightarrow a^3b^3 = 1 \Rightarrow ab = 1$

7. The coefficient of x^{301} in $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots\ldots + x^{500}$ is :

[JEE(Main)Jan.2023]

$(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots\ldots + x^{500}$ में x^{301} का गुणांक है : **[JEE(Main)Jan.2023]**

(A) ${}^{501}C_{302}$ (B) ${}^{500}C_{300}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{200}$

Ans. D

Sol. $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots + x^{500}$

$$= (1+x)^{500} \left\{ \frac{1 - \left(\frac{x}{1+x}\right)^{501}}{1 - \frac{x}{1+x}} \right\}$$

$$= (1+x)^{500} \frac{\left((1+x)^{501} - x^{501}\right)}{(1+x)^{501}} \cdot (1+x)$$

$= (1+x)^{501} - x^{501}$

Coefficient of x^{301} in $(1+x)^{501} - x^{501}$ is given by ${}^{501}C_{301} = {}^{501}C_{200}$

8. Let $x=\left(8\sqrt{3}+13\right)^{13}$ and $y=\left(7\sqrt{2}+9\right)^{9}$. If [t] denotes the greatest integer $\le t$, then :

[JEE(Main)Jan.2023]

(A) [x] is even but [y] is odd (B) [x] + [y] is even

(C) [x] and [y] are both odd (D) [x] is odd but [y] is even

माना $x=\left(8\sqrt{3}+13\right)^{13}$ और $y=\left(7\sqrt{2}+9\right)^{9}$ है। यदि [t] महत्तम पूर्णांक $\le t$ है, तब : **[JEE(Main)Jan.2023]**

(A) [x] सम है परन्तु [y] विषम है (B) [x] + [y] सम है

(C) [x] एवं [y] दोनों विषम है (D) [x] विषम है परन्तु [y] सम है

Ans. B

Sol. $x=\left(8\sqrt{3}+13\right)={}^{13}C_0\cdot\left(8\sqrt{3}\right)^{13}+{}^{13}C_1\left(8\sqrt{3}\right)^{12}+....$

$x'=\left(8\sqrt{3}-13\right)^{13}={}^{13}C_0\left(8\sqrt{3}\right)^{13}-{}^{13}C_1\left(8\sqrt{3}\right)^{12}(13)^1+...$

$x-x'=2\left[{}^{13}C_1\cdot\left(8\sqrt{3}\right)^{12}(13)^1+{}^{13}C_3\left(8\sqrt{3}\right)^{10}\cdot(13)^3....\right]$

therefore, $x-x'$ is even integer, hence [x] is evem

Now, $y=\left(7\sqrt{2}+9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}+{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y'=\left(7\sqrt{2}-9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}-{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y-y'=2\left[{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_3\left(7\sqrt{2}\right)^{6}(9)^3+...\right]$

$y-y'=$ Even integer, hence [y] is even

9. The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ is : **[JEE(Main)Jan.2023]**

$\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{2^{51}}{51!}$ (B) $\frac{2^{50}}{51!}$ (C) $\frac{2^{51}}{50!}$ (D) $\frac{2^{50}}{50!}$

Ans. B

Sol. $\sum_{r=1}^{26}\frac{1}{(2r-1)!(51-(2r-1))!}=\sum_{r=1}^{26}{}^{51}C_{(2r-1)}\frac{1}{51!}$

$=\frac{1}{51!}\left\{{}^{51}C_1+{}^{51}C_3+...+{}^{51}C_{51}\right\}=\frac{1}{51!}\left(2^{50}\right)$

10. The sum $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ is equal to : **[JEE(Main)Jan.2023]**

योगफल $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{13e}{4}+\frac{5}{4e}$ (B) $\frac{13e}{4}+\frac{5}{4e}-4$

(C) $\frac{11e}{2}+\frac{7}{2e}$ (D) $\frac{11e}{2}+\frac{7}{2e}-4$

Ans. B

Sol. $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\frac{2n(2n-1)+8n+8}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\left\{\frac{1}{(2n-2)!}+\frac{4}{(2n-1)!}+\frac{8}{2n!}\right\}$

$=\frac{1}{2}\left\{\left(\frac{e+e^{-1}}{2}\right)+4\left(\frac{e-e^{-1}}{2}\right)+8\left(\frac{e+e^{-1}}{2}-1\right)\right\}$

$=\frac{1}{4}\left(13e+5e^{-1}-16\right)$

$=\frac{13}{4}e+\frac{5}{4e}-4$

11. The coefficient of x^{-6}, in the expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is _____ . **[JEE(Main)Jan.2023]**

$\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$ के प्रसार में x^{-6} का गुणांक है। **[JEE(Main)Jan.2023]**

Ans. 5040

Sol. $T_{r+1}={}^9C_r\left(\frac{5}{2x^2}\right)^r.\left(\frac{4x}{5}\right)^{9-r}={}^9C_r\frac{2^{18-3r}}{5^{9-2r}}x^{9-3r}$

for coefficient of $x^{-6}\Rightarrow 9-3r=-6\Rightarrow r=5$

coff. of $x^{-6}=\frac{{}^9C_5.2^3}{5^{-1}}=\frac{9.8.7.5}{4.3.2.1}.(8\times 5)$

= 5040

12. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ is –84 and the coefficient of x^{-3l} is $2^{\alpha}\beta$, where $\beta<0$ is an odd number, then $|\alpha\ell-\beta|$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ के द्विपद प्रसार में अचर पद –84 है तथा x^{-3l} का गुणांक $2^{\alpha}\beta$ है, जहाँ $\beta<0$ एक विषम संख्या है, तो $|\alpha\ell-\beta|$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 98

Sol. In, $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$

$$T_{r+1}={}^9C_r\frac{\left(x^{5/2}\right)^{9-r}}{2^{9-r}}\left(\frac{-4}{x^{\ell}}\right)^r$$

$$=(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r x^{\frac{45}{2}-\frac{5r}{2}-lr}$$

$$=45-5r-21r=0$$

$$r=\frac{45}{5+21} \qquad \text{_____(1)}$$

Now, according to the question, $(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r=-84$

$$=(-1)^r\ {}^9C_r\ 2^{3r-9}=21\times4$$

Only natural value of r possible if $3r-9=0$

$r=3$ and ${}^9C_3=84$

$\therefore l=5$ from equation (1)

Now, coefficient of $x^{-31}=x^{\frac{45}{2}-\frac{5r}{2}-lr}$ at $l=5$, gives

$r=5$

$$\therefore {}^9C_5(-1)\frac{4^5}{2^4}=2^{\alpha}\times\beta$$

$$=-63\times2^7$$

$\Rightarrow \alpha = 7, \beta = -63$

$\therefore$ value of $|\alpha\ell - \beta| = 98$

13. Suppose $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is _____. **[JEE(Main)Jan.2023]**

माना $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$ है। तो α का मान है _____ **[JEE(Main)Jan.2023]**

Ans. 1012

Sol. $\sum r \cdot {}^{2023}C_r = \sum r \cdot (2023)^{2022} C_{r-1}$

$2023 \sum r^{2022} C_{r-1}$

$2023\left(\sum_{1}^{2023} (r-1)^{2022} C_{r-1} + {}^{2022}C_{r-1}\right)$

$= 2023\left(\left(\sum_{r=2}^{2023} (2022)^{2021} C_{r-2}\right) + \sum_{r=1}^{2023} {}^{2022}C_{r-1}\right) = 2023\left(2022 \cdot 2^{2021} + 2^{2022}\right)$

$= 2^{2022} \cdot 2023(1011+1) = 2^{2022}(1012) \cdot (2023)$

$\because\ \alpha = 1012$

14. Let the sum of the coefficients of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$, be 376. Then the coefficients of x^4 is _____. **[JEE(Main)Jan.2023]**

माना $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$ के प्रसार में प्रथम तीन पदों के गुणांकों का योग 376 है। तो x^4 का गुणांक _____है।

[JEE(Main)Jan.2023]

Ans. 406

Sol. ${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 \cdot 3^2 = 376$

$\Rightarrow n = 10$

Now $T_{r+1} = {}^{10}C_r \cdot \left(\frac{-3}{x^2}\right)^r \cdot (x)^{10-r}$

$= {}^{10}C_r \cdot (-3)^r \cdot x^{10-3r}$

For $x^4 \Rightarrow 10 - 3r = 4 \Rightarrow r = 2$

Coff. of $x^4 = {}^{10}C_2 \cdot (-3)^2 = 405$

15. The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is _____. **[JEE(Main)Jan.2023]**

$\left(2x+\frac{1}{x^7}+3x^2\right)^5$ के प्रसार में अचर पद है। **[JEE(Main)Jan.2023]**

Ans. 1080

Sol. General term is $\Sigma\frac{5!(2x)^{n_1}\left(x^{-7}\right)^{n_2}\left(3x^2\right)^{n_3}}{n_1!n_2!n_3!}$

For constant term,

$n_1 + 2n_3 = 7n_2$

& $n_1 + n_2 + n_3 = 5$

Only possibility $n_1 = 1, n_2 = 1, n_3 = 3$

$\Rightarrow$ constant term = 1080

16. The remainder when $(2023)^{2023}$ is divided by 35 is _____. **[JEE(Main)Jan.2023]**

$(2023)^{2023}$ को 35 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 7

Sol. $-72023 = -7 \times 7^{2022}$

$$=\frac{-7^{2022}}{5}$$

$$=\frac{-[50-1]^{1011}}{5}=\frac{[5\lambda-1]}{5}$$

remainder = + 1

when divided by 5 now remainder is 7 when divided 35

17. If the co-efficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to _____ . **[JEE(Main)Jan.2023]**

यदि $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11}$ के प्रसार में x^9 का गुणांक एवं $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11}$ के प्रसार में x^{-9} का गुणांक बराबर हैं, तब $(\alpha\beta)^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. Coefficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right) = {}^{11}C_6\cdot\frac{\alpha^5}{\beta^6}$

$\because$ Both are equal

$$\therefore \frac{11}{C_6}\cdot\frac{\alpha^5}{\beta^6} = -\frac{11}{C_5}\cdot\frac{\alpha^6}{\beta^5}$$

$$\Rightarrow \frac{1}{\beta} = -\alpha$$

$$\Rightarrow \alpha\beta = -1$$

$$\Rightarrow (\alpha\beta)^2 = 1$$

18. Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of these three terms, is _____ .

[JEE(Main)Jan.2023]

माना $(1 + 2x)^n$ द्विपद प्रसार में तीन क्रमागत पदों के गुणंकों का अनुपात 2 : 5 : 8 है। इन तीन पदों में मध्य पद का गुणांक है।

[JEE(Main)Jan.2023]

Ans. 1120

Sol. $t_{r+1} = {}^nC_r (2x)^r$

$$\Rightarrow \frac{{}^nC_r(2)^{r-1}}{{}^nC_r(2)^r} = \frac{2}{5}$$

$$\Rightarrow \frac{\frac{n!}{(r-1)!(n-r+1)!}}{\frac{n!(2)}{r!(n-r)!}} = \frac{2}{5}$$

$$\Rightarrow \frac{r}{n-r+1} = \frac{4}{5} \Rightarrow 5r = 4n - 4r + 4$$

$$\Rightarrow 9r = 4\,(n + 1) \qquad ....(1)$$

$$\Rightarrow \frac{{}^nC_r(2)^r}{{}^nC_{r+1}(2)^{r+1}} = \frac{5}{8}$$

$$\Rightarrow \frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r+1)!(n-r-1)!}} = \frac{5}{4} \Rightarrow \frac{r+1}{n-r} = \frac{5}{4}$$

$\Rightarrow 4r + 4 = 5n - 5r \Rightarrow 5n - 4 = 9r$ ....(2)

From (1) and (2)

$\Rightarrow 4n + 4 = 5n - 4 \Rightarrow n = 8$

(A) $\Rightarrow r = 4$

so, coefficient of middle term is

$${}^8C_4\,2^4 = 16\times\frac{8\times7\times6\times5}{4\times3\times2\times1} = 16\times70 = 1120$$

19. Let $\sum_{n=0}^{\infty}\frac{n^3\left((2n)!\right)+(2n-1)(n!)}{(n!)\left((2n!)\right)} = ae+\frac{b}{e}+c,$ where $a,b,c\in Z$ and $e=\sum_{n=0}^{\infty}\frac{1}{n!}$. Then a^2-b+c is equal to :

[JEE(Main)Jan.2023]

माना $\sum_{n=0}^{\infty}\frac{n^3\left((2n)!\right)+(2n-1)(n!)}{(n!)\left((2n!)\right)} = ae+\frac{b}{e}+c,$ है जहाँ $a,b,c\in Z$ तथा $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ है। तो a^2-b+c बराबर है।

[JEE(Main)Jan.2023]

Ans. 26

Sol. $$\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}+\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{(2n)\times n!}{n!\times(2n)!}-\frac{n!}{n!(2n)!}$$

$$\sum_{n=0}^{\infty}\frac{1}{(2n-1)!}-\sum_{n=0}^{\infty}\frac{1}{(2n)!}$$

$$\frac{e^x-e^{-x}}{2}\bigg|_{x=1}-\frac{e^x+e^{-x}}{2}\bigg|_{x=1}$$

$$=\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}=\sum_{n=0}^{\infty}\frac{n^3}{n!}$$

$n^3 = an(n-1)(n-2)+bn(n-1)+cn+d$

$n=1\Rightarrow c=1$

$n=2\Rightarrow 8=2b+2\Rightarrow b=3$

$$=\sum_{n=0}^{\infty}\frac{n(n-1)(n-2)+3n(n-1)+n}{n!}$$

$$\sum_{n=3}^{\infty}\frac{1}{(n-3)!}+3\sum_{n=2}^{\infty}\frac{1}{(n-2)!}+\sum_{n=1}^{\infty}\frac{1}{(n-1)!}$$

$$=\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}+\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$$

$$=5e+\frac{e}{2}-\frac{1}{e}-\frac{e}{2}+\frac{1}{e}$$

$= 5e - \frac{1}{e}$

$a = 5, b = -1, c = 0$

Now $a^2 - b + c = 26$

20. 50th root of a number x is 12 and 50th root of another number y is 18. Then the remainder obtained on dividing (x + y) by 25 is _____ . **[JEE(Main)Jan.2023]**

एक संख्या x का 50वाँ मूल 12 है एवं दूसरी संख्या y का 50वाँ मूल 18 है। तब (x + y) को 25 से विभाजित करने पर प्राप्त शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 23

Sol. $(x)^{\frac{1}{50}} = 12$ and $(y)^{\frac{1}{50}} = 18$

$\Rightarrow x = 12^{50} \Rightarrow y = 18^{50}$

$x + y = 12^{50} + 18^{50}$

$(2^2 \times 3^1)^{50} + (3^2 \times 2)^{50}$

$= 6^{50}(2^{50} + 3^{50})$

$= (5 + 1)^{50}(4^{25} + 9^{25})$

$= (25\lambda + 1)[(5 - 1)^{25} + (10 - 1)^{25}]$

$= (25\lambda + 1)(25k - 1 + 25\mu - 1)$

$= (25\lambda + 1)(25t + 23)$

$= 25\lambda + 23$

so remiander is 23, when $x + y$ is divided by 25.

21. Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ has term $\beta x^{-\alpha}$, $\beta \in N$. Then α is equal to _____ . **[JEE(Main)Jan.2023]**

माना $\alpha > 0$ न्यूनतम संख्या है, जिसके लिए $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ के प्रसार के एक पद में $\beta x^{-\alpha}$ तो α बराबर है _____ , $\beta \in N$

[JEE(Main)Jan.2023]

Ans. 2

Sol. $\alpha > 0, \left(x^{2/3} + \frac{2}{x^3}\right)^{30}$

$T_{r+1} = {}^{30}C_r \left(x^{2/3}\right)^{30-r} \left(\frac{2}{x^3}\right)^r$

$= {}^{30}C_r x^{20-\frac{2r}{3}-3r}$

$= {}^{30}C_r (2)^r x^{\frac{60-11r}{3}}$ $\quad 0 \le r \le 30$

For $\beta x^{-\alpha}$, $\alpha > 0$ the smallest value of α is 2 (for r = 6)

$T7 = {}^{30}C_6 (2)^6 x^{-2}$

$\alpha = 2$

22. The remainder on dividing 5^{99} by 11 is _____ . **[JEE(Main)Jan.2023]**

5^{99} को 11 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $5^{99} = 5^4.5^{95}$

$= 625\ [5^5]^{19}$

$= 625\ [3125]^{19}$

$= 625\ [2124 + 1]^{19}$

$= 625\ [11k \times 19 + 1]$

$= 625 \times 11k \times 19 + 625$

$= 11k_1 + 616 + 9$

$= 11(k_2) + 9$

Remainder = 9

23. The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____. **[JEE(Main)Jan.2023]**

$19^{200} + 23^{200}$ को 49 से विभाजित करने पर शेषफल है। **[JEE(Main)Jan.2023]**

Ans. 29

Sol. $(21 - 2)^{200} + (21 + 2)^{200} = 49\lambda + 2^{201}$

$2^{201} = 8^{67} = (7 + 1)^{67} = 49\lambda + 7 \times 67 + 1$

$= 49\lambda + 470$

$= 49\ (\lambda + 9) + 29$

So, remainder = 29

24. Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2\left(10-3^x\right)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____ . **[JEE(Main)Jan.2023]**

माना $2^{(x-2)\log_2 3}$ की बढ़ती घातों में $\left(\sqrt{2^{\log_2(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ के द्विपद प्रसार में छठा पद 21 है। यदि इस प्रसार में दूसरा, तीसरा तथा चौथा द्विपद गुणांक एक A.P. के क्रमशः पहला, तीसरा तथा पाँचवा पद है, तो x के सभी संभव मानों के वर्गो का योग है। **[JEE(Main)Jan.2023]**

Ans. 4

Sol. $T_6 = {}^mC_5\left(10-3^x\right)^{\frac{m-5}{2}}\cdot\left(3^{x-2}\right)=21$ ...(1)

${}^mC_1, {}^mC_2, {}^mC_3$ are in A.P.

$2.{}^mC_2 = {}^mC_1 + {}^mC_3$

Solving for m, we get

m = 2 (rejected), 7

Put in equation (1)

$21.\left(10-3^x\right)\frac{3^x}{9}=21$

$3^x = 3^0, 3^2$

x = 0, 2

Sum of all the squares of all possible values of x = 4

25. If the term without x in the expansion of $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to _____.

[JEE(Main)Jan.2023]

यदि $\left(x^{\frac{2}{3}}+\frac{\alpha}{x^3}\right)^{22}$ के प्रसार में x से स्वतंत्र पद 7315 है, तो $|\alpha|$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $T_{r+1} = {}^{22}C_r\cdot\left(x^{\frac{2}{3}}\right)^{22-r}.(\alpha)^r, x^{-3r}$

$= {}^{22}C_r.x^{\frac{44}{3}-\frac{2r}{3}-3r}(\alpha)^r$

$\frac{44}{3}=\frac{11r}{3}$

r = 4

${}^{22}C_4.ga^4 = 7315$

$\frac{22\times21\times20\times19}{24}.\alpha^4 = 7315$

$\alpha = 1$

26. If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6}:1$, then the third term from the beginning is : **[JEE(Main)April,2023]**

यदि $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ के विस्तार में आरंभ से पाँचवें पद का अंत से पाँचवे पद से अनुपात $\sqrt{6}:1$ है, तब आरंभ से तीसरा पद है : **[JEE(Main)April,2023]**

(A) $30\sqrt{3}$ (B) $60\sqrt{3}$ (C) $60\sqrt{2}$ (D) $30\sqrt{2}$

Ans. B

Sol. $$\frac{{}^nC_4 2^{\frac{n-4}{4}}\cdot\left(3^{\frac{-1}{4}}\right)^4}{{}^nC_4 3^{-\left(\frac{n-4}{4}\right)}\cdot\left(2^{\frac{1}{4}}\right)^4}=\frac{\sqrt{6}}{1}$$

$\Rightarrow n=10$

So $T_3 = {}^{10}C_2 2^{\frac{1}{4}\cdot 8}\cdot 3^{-\frac{1}{4}\cdot 2} = \frac{45.4}{\sqrt{3}} = 60\sqrt{3}$

27. If the coefficients of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$ are equal, then : **[JEE(Main)April,2023]**

यदि $\left(ax^2+\frac{1}{2bx}\right)^{11}$ में x^7 तथा $\left(ax-\frac{1}{3bx^2}\right)^{11}$ में x^{-7} के गुणांक बराबर है, तो : **[JEE(Main)April,2023]**

(A) $64ab = 243$ (B) $243ab = 64$ (C) $729ab = 32$ (D) $32ab = 729$

Ans. C

Sol. Coefficient of x^7 in $\left(ax^2+\frac{1}{2bx}\right)^{11}$

$$T_{r+1} = {}^{11}C_r\left(ax^2\right)^{11-r}\left(\frac{1}{2bx}\right)^r$$

$$= {}^{11}C_r (a)^{11-r}\left(\frac{1}{2b}\right)^r x^{22-3r}$$

$22-3r=7 \Rightarrow r=5$

Coefficient of x^{-7} in $\left(ax-\frac{1}{3bx^2}\right)^{11}$

$T_{r+1}={}^{11}C_r(ax)^{11-r}\left(-\frac{1}{3bx^2}\right)^r$

$={}^{11}C_r a^{11-r}\left(-\frac{1}{3b}\right)^r x^{11-3r}$

$\therefore \quad {}^{11-3r}=-7 \Rightarrow r=6$

$\Rightarrow \quad {}^{11}C_5(ab)^6\left(\frac{1}{2b}\right)^5=32$

$\Rightarrow \quad 729ab=32$

28. Among the statements :

(S1) : $2023^{2022}-1999^{2022}$ is divisible by 8

(S2) : $13(13)^n-11n-13$ is divisible by 144 for infinitely many $n\in\mathbb{N}$: **[JEE(Main)April,2023]**

(A) both (S1) and (S2) are correct (B) both (S1) and (S2) are incorrect

(C) only (S2) is correct (D) only (S1) is correct

कथनों :

(S1) : $2023^{2022}-1999^{2022}$, 8 से विभाज्य है

(S2) : $13(13)^n-11n-13$ अनंत $n\in\mathbb{N}$ के लिए 144 से विभाज्य है

में से **[JEE(Main)April,2023]**

(A) दोनों (S1) तथा (S2) सही है (B) दोनों (S1) तथा (S2) गलत है

(C) केवल (S2) सही है (D) केवल (S1) सही है

Ans. D

Sol. Statement – 1 : $x^{2022}-y^{2022}$ is divisible by $(x-y)$

Hence $(2023)^{2022}-(1999)^{2022}$ is divisible by $2023-1999=24$

$\Rightarrow$ statement : 1 is correct

Statement – 2 : $13\,(13)^n-11n-13$

$=13(12+1)^n-11n-13$

$=13\left[12^n+n(12)^{n-1}+\frac{n(n-1)}{2}(12)^{n-2}+\ldots\ldots\ldots\ldots+{}^nC_n\right]-11n-13$

$=13\left[12^n+n(12)^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)^2\right]-11n-13$

$=13\left[12^n+n(12)^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)+1\right]+156n-11n$

$=13\left[12^n+n\cdot12^{n-1}+{}^nC_2(12)^{n-2}+\ldots\ldots\ldots+{}^nC_{n-2}(12)^2\right]+145n$

Which is not divisible by 144 for all $n \in N$ but divisible by 144 for infinitely many values of n if $n = 144m$ where $m \in N$.

29. If the coefficients of three consecutive terms in the expansion of $(1+x)^n$ are in the ratio 1 : 5 : 20, then the coefficient of the fourth term is : **[JEE(Main)April,2023]**

यदि $(1+x)^n$ के प्रसार में तीन क्रमागत पदों के गुणांकों का अनुपात 1 : 5 : 20 है, तो चौथे पद का गुणांक है : **[JEE(Main)April,2023]**

(A) 1827 (B) 2436 (C) 5481 (D) 3654

Ans. D

Sol. $\frac{{}^nC_{r-1}}{1} = \frac{{}^nC_r}{5} = \frac{{}^nC_{r+1}}{20}$

$$\because \quad \frac{{}^nC_r}{{}^nC_{r-1}} = 5 \Rightarrow \frac{\frac{\lfloor n}{\lfloor n-r}\lfloor r}{\frac{\lfloor n}{\lfloor n-r+1}\lfloor r-1}} = 5$$

$$\Rightarrow \frac{n-r+1}{r} = 5 \Rightarrow n = 6r-1 \ldots \text{ (i)}$$

$$\because \quad \frac{{}^nC_{r+1}}{{}^nC_r} = 4 \Rightarrow \frac{n-r}{r+1} = 4 \Rightarrow n = 5r+4.$$

from (i) and (ii), $r = 5, n = 29$

$\Rightarrow$ Coefficient of fourth term $= {}^{29}C_3 = 3654$

30. The absolute difference of the coefficients of x^{10} and x^7 in the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{11}$ is equal to : **[JEE(Main)April,2023]**

$\left(2x^2 + \frac{1}{2x}\right)^{11}$ के प्रसार में x^{10} तथा x^7 के गुणांकों का निरपेक्ष अंतर बराबर है : **[JEE(Main)April,2023]**

(A) $11^3 - 11$ (B) $12^3 - 12$ (C) $13^3 - 13$ (D) $10^3 - 10$

Ans. B

Sol. $T_{r+1} = {}^{11}C_r \left(2x^2\right)^{11-r} (2x)^{-r}$

For coefficient of x^7

$\therefore \quad 22 - 2r - r = 7$

$\therefore \quad r = 5$

For coefficient of x^{10}

$22 - 3r = 10$

$\Rightarrow r = 4$

Absolute difference $= \left| {}^{11}C_5 \cdot \frac{2^6}{2^5} - {}^{11}C_4 \cdot \frac{2^7}{2^4} \right|$

$= |924 - 2640| = 1716$

$= 12^3 - 12$

31. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by : **[JEE(Main)April,2023]**

(A) 14 but not by 34 (B) both 14 and 34 (C) 34 but not by 14 (D) neither 14 nor 34

$25^{190} - 19^{190} - 8^{190} + 2^{190}$ **[JEE(Main)April,2023]**

(A) 14 से विभाज्य है परन्तु 34 से नहीं (B) 14 तथा 34 दोनों से विभाज्य है

(C) 34 वे विभाज्य है परन्तु 14 से नहीं (D) न तो 14 से नही 34 से विभाज्य है

Ans. C

Sol. $(25^{190} - 19^{190}) - (8^{190} - 2^{190})$ is divisible by 6.

$(25^{190} - 8^{190}) - (19^{190} - 2^{190})$ is divisible by 17.

$25^{190} - 8^{190}$ is not divisible by 7

but $19^{190} - 2^{190}$ is divisible by 7

So, $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by 34 but not 14

32. If the coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$ and the coefficient of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$ are equal, then a^4b^4 is equal to : **[JEE(Main)April,2023]**

यदि $\left(ax - \frac{1}{bx^2}\right)^{13}$ में x^7 का गुणांक तथा $\left(ax + \frac{1}{bx^2}\right)^{13}$ में x^{-5} का गुणांक बराबर हैं, तो a^4b^4 बराबर है : **[JEE(Main)April,2023]**

(A) 22 (B) 33 (C) 44 (D) 11

Ans. A

Sol. Coefficient of x^7 in $\left(ax - \frac{1}{bx^2}\right)^{13}$

$T_{r+1} = {}^{13}C_r (9x)^{13-r} \left(-\frac{1}{bx^2}\right)^r$

$\Rightarrow r = 2$

$\therefore$ Coeff. $= {}^{13}C_2 \frac{a^{11}}{b^2}$

Similarly coeff. of x^{-5} in $\left(ax + \frac{1}{bx^2}\right)^{13}$

$\Rightarrow r = 6$

$\therefore$ Coeff. $= {}^{13}C_6 \frac{a^7}{b^6}$

Now, ${}^{13}C_2 \frac{a^{11}}{b^2} = {}^{13}C_6 \frac{a^7}{b^6}$

$\Rightarrow a^4b^4 = 22$

33. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to : **[JEE(Main)April,2023]**

माना $(22)^{2022} + (2022)^{22}$ को 3 से विभाजित करने पर शेषफल α है तथा 7 से विभाजित करने पर शेषफल β है। तो $(\alpha^2 + \beta^2)$ बराबर है : **[JEE(Main)April,2023]**

(A) 13 (B) 5 (C) 20 (D) 10

Ans. B

Sol. For $\alpha : (21+1)^{2022} + \underbrace{(2022)^{22}}_{\text{divisible by 3}}$

$= 3K_1 + 1$

For $\beta : (21+1)^{2022} + (2023-1)^{22}$

$= 7\lambda + 1 + 7\mu + 1$

$= 7K_2 + 2$

So, $\alpha = 1, \beta = 2$

$\alpha^2 + \beta^2 = 5$

34. If the coefficients of x and x^2 in $(1 + x)^p (1 - x)^q$ are 4 and –5 respectively, then $2p + 3q$ is equal to : **[JEE(Main)April,2023]**

$(1 + x)^p (1 - x)^q$ में x तथा x^2 के गुणांक क्रमशः 4 तथा –5 हैं, तो $2p + 3q$ बराबर है : **[JEE(Main)April,2023]**

(A) 63 (B) 69 (C) 60 (D) 66

Ans. A

Sol. $(1+x)^p(1-x)^q$

$= \left(1 + px + \frac{p(p-1)}{2}x^2 + \ldots.\right)\left(1 - qx + \frac{q(q-1)}{2}x^2 + \ldots.\right)$

$= 1 + (p-q)x + \left(\frac{p(p-1)}{2} + \frac{q(q-1)}{2} - pq\right)x^2 + \ldots..$

$p - q = 4 \Rightarrow p = q + 4 \quad \ldots(i)$

$\frac{p(p-1)}{2} + \frac{q(q-1)}{2} - pq = -5$

$\frac{(q+4)(q+3)}{2} + \frac{q(q-1)}{2} - (q+4)q = -5$

$\Rightarrow \quad 6-q=-5$

$\Rightarrow \quad q=11$

$\therefore \quad p=15$

$2p+3q=63$

35. If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then |x| is equal to : **[JEE(Main)April,2023]**

यदि $\left(\frac{4x}{5}-\frac{5}{2x}\right)^{2022}$ के द्विपद प्रसार में अंत से 1011वाँ पद, आरंभ से 1011वें पद का 1024 गुना है, तो |x| बराबर है :

[JEE(Main)April,2023]

(A) 8 (B) 12 (C) 10 (D) 15

Ans. 10 **(Bonus)**

Sol. T_{1011} from beginning $= T_{1010+1}$

$={}^{2022}C_{1010}\left(\frac{4x}{5}\right)^{1012}\left(\frac{-5}{2x}\right)^{1010}$

T_{1011} from end

$={}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$

Given : $={}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$

$=2^{10}\cdot{}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1010}\left(\frac{4x}{5}\right)^{1012}$

$\left(\frac{-5}{2x}\right)^{2}=2^{10}\left(\frac{4x}{5}\right)^{2}$

$x^4=\frac{5^4}{2^{16}}$

$|x|=\frac{5}{16}$

36. The sum of the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+2}$, which are in the ratio 1 : 3 : 5, is equal to : **[JEE(Main)April,2023]**

$(1+x)^{n+2}$ के द्विपद प्रसार में तीन क्रमागत पदों के गुणांकों, जो 1 : 3 : 5 के अनुपात में है, का योग बराबर है :

[JEE(Main)April,2023]

(A) 25 (B) 92 (C) 41 (D) 63

Ans. D

Sol. Coefficients are ${}^{n+2}C_{r-1}, {}^{n+2}C_r, {}^{n+2}C_{r+1}$

Now $\frac{{}^{n+2}c_{r-1}}{{}^{n+2}c_r} = \frac{1}{3} \Rightarrow n+3 = 4r$ ....(1)

$\frac{{}^{n+2}c_r}{{}^{n+2}c_{r+1}} = \frac{3}{5} \Rightarrow 3n+1 = 8r$ ....(2)

By(1) & (2) $n = 5, r = 2$

Now coefficients are ${}^{7}C_1, {}^{7}C_2, {}^{7}C_3$

sum = 63

37. If $\frac{1}{n+1}{}^{n}C_n + \frac{1}{n}{}^{n}C_{n-1} + \ldots + \frac{1}{2}{}^{n}C_1 + {}^{n}C_0 = \frac{1023}{10}$ then n is equal to : **[JEE(Main)April,2023]**

यदि $\frac{1}{n+1}{}^{n}C_n + \frac{1}{n}{}^{n}C_{n-1} + \ldots + \frac{1}{2}{}^{n}C_1 + {}^{n}C_0 = \frac{1023}{10}$ तो n बराबर है : **[JEE(Main)April,2023]**

(A) 7 (B) 6 (C) 8 (D) 9

Ans. D

Sol. $\sum_{r=0}^{n} \frac{1}{r+1}{}^{n}C_r = \frac{1023}{10} \quad \left(\because {}^{n+1}C_{r+1} = \frac{n+1}{r+1}{}^{n}C_r\right)$

$\Rightarrow \sum_{r=0}^{n} \frac{1}{n+1}{}^{n+1}C_{r+1} = \frac{1023}{10}$

$\Rightarrow \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n+1}\right] = \frac{1023}{10}$

$\Rightarrow \frac{2^{n+1}-1}{n+1} = \frac{1023}{10} = \frac{2^{10}-1}{10}$

$\Rightarrow n+1 = 10$

$\Rightarrow n = 9$

38. The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to : **[JEE(Main)April,2023]**

$(1-x)^{100}$ के द्विपद प्रसार में प्रथम 50 पदों के गुणांकों का योग बराबर है : **[JEE(Main)April,2023]**

(A) ${}^{99}C_{49}$ (B) $-{}^{99}C_{49}$ (C) $-{}^{101}C_{50}$ (D) ${}^{101}C_{50}$

Ans. B

Sol. $(1-x)^{100} = Co - C_1x + C_2x^2 - C_3x^3 + \ldots C_{99}x^{99} + C_{100}x^{100}$

$\Rightarrow Co - C_1 + C_2 - C_3 + \ldots\ldots - C_{99} + C_{100} = 0$

$2\left(Co - C_1 + C_2 + \ldots\ldots - C_9\right) + C_{50} = 0$

$$C_0 - C_1 + C_2 + \ldots C_{99} = -\frac{1}{2}^{100} C_{50}$$

$$-\frac{1}{2}\frac{100!}{50!50!} = -\frac{1}{2} \times \frac{100 \times 99!}{50!50!} = -^{99}C_{49}$$

39. Fractional part of the number $\frac{4^{2022}}{15}$ is equal to : **[JEE(Main)April,2023]**

संख्या $\frac{4^{2022}}{15}$ का भिन्न भाग है : **[JEE(Main)April,2023]**

(A) $\frac{4}{15}$ (B) $\frac{14}{15}$ (C) $\frac{1}{15}$ (D) $\frac{8}{15}$

Ans. C

Sol. $\frac{4^{2022}}{15}$

$\because 4 \equiv 4(\text{mod}15)$

$4^2 \equiv 1(\text{mod}15)$

$4^{2022} \equiv 1(\text{mod}15)$

$\therefore$ Fractional part of $\frac{4^{2022}}{15} = \frac{1}{15}$

40. The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is : **[JEE(Main)April,2023]**

$\left(2x^3 - \frac{1}{3x^2}\right)^5$ के प्रसार में x^5 का गुणांक है : **[JEE(Main)April,2023]**

(A) 9 (B) 8 (C) $\frac{26}{3}$ (D) $\frac{80}{9}$

Ans. D

Sol. General term for $\left(2x^3 - \frac{1}{3x^2}\right)^5$

$T_{r+1} = {}^5C_r \left(-\frac{1}{3x^2}\right)^r \left(2x^3\right)^{5-r}$

$= {}^5C_r(-1)^r 3^{-r} 2^{5-r} \cdot x^{15-5r}$

$15 - 5r = 5 \Rightarrow r = 2$

Coeff. of $x^5 = {}^5C_2(-1)^2 3^{-2} 2^3$

$= 10 \times \frac{1}{9} \times 8$

$= \frac{80}{9}$

41. Let $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$. If $p_1 = 20$ and $p_2 = 210$, then 2(a+b+c) is equal to :

[JEE(Main)April,2023]

यदि $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a, b, c \in \mathbb{N}$, $p_1 = 20$ तथा $p_2 = 210$ हो, तो 2(a+b+c) का मान बराबर होगा :

[JEE(Main)April,2023]

(A) 8 (B) 12 (C) 6 (D) 15

Ans. B

Sol. $\left(a + bx + cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i,$

Coefficient of $x^1 = 20$

$20 = \frac{10!}{9!1!} \times a^9 \times b^1$

$a^9 \cdot b = 2$

$a = 1, b = 2$

Coefficient of $x^2 = 210$

$210 = \frac{10!}{9!1!} \times a^9 \times c^1 + \frac{10!}{8!2!} \times a^8 b^2$

$210 = 10 \cdot c + 45 \times 4$

$10c = 30$

$c = 3$

$2 (a + b + c) = 12$

42. The coefficient of x^{18} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$ is _____. **[JEE(Main)April,2023]**

$\left(x^4 - \frac{1}{x^3}\right)^{15}$ के प्रसार में x^{18} का गुणांक है _____ **[JEE(Main)April,2023]**

Ans. 5005

Sol. $\left(x^4 - \frac{1}{x^3}\right)^{15}$

$T_{r+1} = {}^{15}C_r \left(x^4\right)^{15-r} \left(\frac{-1}{x^3}\right)^r$

$60-7r=18$

$r=6$

Hence coeff. of $x^{18} = {}^{15}C_6 = 5005$

43. Let [t] denote the greatest integer $\le t$. If the constant term in the expansion of $\left(3x^2-\frac{1}{2x^5}\right)^7$ is α, then $[\alpha]$ is equal to _____. **[JEE(Main)April,2023]**

माना [t] महत्तम पूर्णांक $\le t$ है। यदि $\left(3x^2-\frac{1}{2x^5}\right)^7$ के प्रसार में अचर पद α है, तो $[\alpha]$ बराबर है _____

[JEE(Main)April,2023]

Ans. 1275

Sol. $T_{r+1} = {}^7C_r\left(3x^2\right)^{7-r}\left(-\frac{1}{2x^5}\right)^r$

$T_{r+1} = {}^7C_r 3^{7-r}\left(-\frac{1}{2}\right)^r x^{14-7r}$

For term independent of x : $14-7r=0 \Rightarrow r=2$

Required coefficient $= {}^7C_2 3^{7-2}\left(-\frac{1}{2}\right)^2 = \frac{7.6}{2}\cdot 3^5\cdot\frac{1}{2^2}$

$=\frac{7\cdot 3^6}{4}=1275.85$

$[1275.85] = 1275$

44. The coefficient of x^7 in $\left(1-x+2x^3\right)^{10}$ is _____. **[JEE(Main)April,2023]**

$\left(1-x+2x^3\right)^{10}$ में x^7 का गुणांक है _____ **[JEE(Main)April,2023]**

Ans. 960

Sol. General term $=\frac{10!}{r_1!\cdot r_2!r_3!}(-1)^{r_2}\cdot(2)^{r_3}x^{r_2+3r_3}$

where $r_1+r_2+r_3=10$ and $r_2+3r_3=7$

r_1	r_2	r_3
3	7	0
5	4	1
7	1	2

Required coefficient

$=\frac{10!}{3!\cdot 7!}(-1)^7+\frac{10!}{5!\cdot 4!}(-1)^4(2)+\frac{10!}{7!\cdot 2!}(-1)^1(2)^2$

$= -120 + 2520 - 1440 = 960$

45. The number of integral terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to _____.

[JEE(Main)April,2023]

$\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ के प्रसार में पूर्णांक पदों की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 171

Sol. The number of integral term in the expression of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{680}$ is equal to

General term $= {}^{680}C_r \left(3^{\frac{1}{2}}\right)^{680-r} \left(5^{\frac{1}{4}}\right)^{r}$

$= {}^{680}C_r 3^{\frac{680-r}{2}} 5^{\frac{r}{4}}$

Value's of r, where $\frac{r}{4}$ goes to integer

r = 0, 4, 8, 12, ...680

All value of r are accepted for $\frac{680-r}{2}$ as well so

No of integral terms = 171.

46. Let α be the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^{n}$, $n \le 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

[JEE(Main)April,2023]

माना $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^{n}$, $n \le 15$ के द्विपद प्रसार में अचर पद α है। यदि इस प्रसार में शेष पदों के गुणांकों का योग 649 है तथा x^{-n} का गुणांक $\lambda\alpha$ है, तो λ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 36

Sol. $T_{k+1} = {}^{n}C_k (x)^{\frac{n-k}{2}} (-6)^k (x)^{\frac{-3}{2}k}$

$\frac{n-k}{2} - \frac{3}{2}k = 0$

$n - 4k = 0$

$(-5)^n - \left({}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}} \right) = 649$

By observation $(625 + 24 = 649)$, we get $n = 4$

$\because$ $n = 4 \& k = 1$

Required is coefficient of x^{-4} is $\left(\sqrt{4} - \frac{6}{x^{\frac{3}{2}}} \right)^4$

${}^4C_1 (-6)^3$

By calculating we will get $\lambda = 36$

47. The remainder, when 7^{103} is divided by 17, is _____. **[JEE(Main)April,2023]**

7^{103} को 17 से विभाजित करने पर शेषफल है _____ **[JEE(Main)April,2023]**

Ans. 12

Sol. 7^{103} divided by 17

$7 \equiv 7 \pmod{17}$

$7^2 \equiv -2 \pmod{17}$

$7^6 \equiv -8 \pmod{17}$

$7^8 \equiv -1 \pmod{17}$

$7^{16} \equiv 1 \pmod{17}$

$7^{103} \equiv 12 \pmod{17}$

$\therefore$ Remainder = 12

48. The number of elements in the set $\{n \in \mathbb{N} : 10 \le n \le 100$ and $3^n - 3$ is a multiple of $7\}$ is _____. **[JEE(Main)April,2023]**

समुच्चय में अवयवों की संख्या होगी $\{n \in \mathbb{N} : 10 \le n \le 100$ तथा $3^n - 3$, 7 का गुणक $\}$_____ **[JEE(Main)April,2023]**

Ans. 15

Sol. $n \in [10, 100]$

$3^n - 3$ is multiple of 7

$3^n = 7\lambda + 3$

$n = 1,7,13,20,.....97$

Number of possible values of $n = 15$

49. If A denotes the sum of all the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then : **[JEE(Main)Jan.2024]**

यदि $(1 - 3x + 10x^2)^n$ के प्रसार में सभी गुणांकों का योग A है तथा $(1 + x^2)^n$ के प्रसार में सभी गुणांकों का योग B है, तो : **[JEE(Main)Jan.2024]**

(A) $B = A^3$ (B) $3A = B$ (C) $A = B^3$ (D) $A = 3B$

Ans. C

Sol. $A = \text{sum of coefficient} = (1 - 3(1) + 10(1)^2)^n = 8^n$

(Putting $x = 1$)

$B = \text{Sum of coefficient} = (1 + (1)^2)^n = 2^n$

$\Rightarrow 8^n = (23)^n = (2^n)^3 = B^3$

$\Rightarrow A = B^3$

50. ${}^{n-1}C_r = (k^2 - 8)\,{}^nC_{r+1}$ if and only if : **[JEE(Main)Jan.2024]**

${}^{n-1}C_r = (k^2 - 8)\,{}^nC_{r+1}$ है यदि और केवल यदि : **[JEE(Main)Jan.2024]**

(A) $2\sqrt{2} < k \le 3$ (B) $2\sqrt{3} < k < 3\sqrt{3}$

(C) $2\sqrt{3} < k \le 3\sqrt{2}$ (D) $2\sqrt{2} < k < 2\sqrt{3}$

Ans. A

Sol. ${}^{n-1}C_r = \left(k^2 - 8\right)\frac{n}{r+1}\,{}^{n-1}C_r$

$k^2 - 8 = \frac{r+1}{n}$

$0 < k^2 - 8 \le 1$

$8 < k^2 \le 9$

$2\sqrt{2} < k \le 3$ or $-3 \le k < -2\sqrt{2}$

51. Suppose $2 - p, p, 2 - \alpha, \alpha$ are the coefficients of four consecutive terms in the expansion of $(1 + x)^n$. Then the value of $p^2 - \alpha^2 + 6\alpha + 2p$ equals : **[JEE(Main)Jan.2024]**

माना $(1 + x)^n$ के प्रसार में चार क्रमागत पदों के गुणांक $2 - p, p, 2 - \alpha, \alpha$ हैं। तो $p^2 - \alpha^2 + 6\alpha + 2p$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 8 (B) 10 (C) 6 (D) 4

Ans. Bonus

Sol. $2 - p, p, 2 - \alpha, \alpha$

Binomial coefficients are

${}^nC_r, {}^nC_{r+1}, {}^nC_{r+2}, {}^nC_{r+3}$ respectively

$\Rightarrow {}^nC_r + {}^nC_{r+1} = 2$

$\Rightarrow {}^{n+1}C_{r+1} = 2$ _____(1)

Also, ${}^nC_{r+2} + {}^nC_{r+3} = 2$

$\Rightarrow {}^{n+1}C_{r+3} = 2$ _____(2)

From (1) and (2)

${}^{n+1}C_{r+1} = {}^{n+1}C_{r+3}$

$\Rightarrow 2r + 4 = n + 1$

$n = 2r + 3$

${}^{2r+4}C_{r+1} = 2$

Data Inconsistent

52. Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$.

Then $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ is : **[JEE(Main)Jan.2024]**

माना $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$ के प्रसार में सातवें तथा तेरहवें पदों के गुणांक क्रमशः m तथा n है। तो $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{4}{9}$ (B) $\frac{1}{9}$ (C) $\frac{1}{4}$ (D) $\frac{9}{4}$

Ans. D

Sol. $m = {}^{18}C_6\left(\frac{1}{3}\right)^{12}\left(\frac{1}{2}\right)^6$

$n = {}^{18}C_{12}\left(\frac{1}{3}\right)^{6}\left(\frac{1}{2}\right)^{12}$

$\Rightarrow \frac{n}{m} = \frac{3^6}{2^6}$

$\Rightarrow \left(\frac{n}{m}\right)^{\frac{1}{3}} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

53. The coefficient of x^{2012} in the expansion of $(1-x)^{2008}(1+x+x^2)^{2007}$ is equal to _____. **[JEE(Main)Jan.2024]**

$(1-x)^{2008}(1+x+x^2)^{2007}$ के प्रसार में x^{2012} का गुणांक बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 0

Sol. $\left((1-x)^{2008}\right)\left[1+x+x^2\right]^{2007}$

$(1-x)\left[1-x^3\right]^{2007}$

$(1-x)^{2007}\, c_r \left(-x^3\right)^r$

$^{2007}c_r\left[(-x)^{3r}+(-x)^{3r+1}\right]$

$\therefore$ 3r = 2012

or 3r + 1 = 2012

no such r exists

$\therefore$ Coefficient is zero

54. If $\frac{^{11}C_1}{2}+\frac{^{11}C_2}{3}+...+\frac{^{11}C_9}{10}=\frac{n}{m}$ with gcd(n, m) = 1, then n + m is equal to _____.

[JEE(Main)Jan.2024]

यदि $\frac{^{11}C_1}{2}+\frac{^{11}C_2}{3}+...+\frac{^{11}C_9}{10}=\frac{n}{m}$ है तथा gcd(n, m) = 1 है, तो n + m बराबर है _____

[JEE(Main)Jan.2024]

Ans. 2041

Sol. $\sum_{r=1}^{9}\frac{^{11}C_r}{r+1}$

$=\frac{1}{12}\sum_{r=1}^{9}{}^{12}C_{r+1}$

$=\frac{1}{12}\left[2^{12}-26\right]=\frac{2035}{6}$

$\therefore$ m + n = 2041

55. Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____ . **[JEE(Main)Jan.2024]**

$64^{32^{32}}$ को 9 से विभाजित करने पर शेषफल है _____ . **[JEE(Main)Jan.2024]**

Ans. 1

Sol. Let $32^{32} = t$

$64^{32^{32}} = 64^t = 8^{2t} = (9-1)^{2t}$

$= 9k + 1$

Hence remainder = 1

56. Number of the integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _____.

[JEE(Main)Jan.2024]

$\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ के प्रसार में पूर्णांक पदों की संख्या है _____। **[JEE(Main)Jan.2024]**

Ans. 138

Sol. $T_{r+1} = {}^{824}C_r (7)^{\frac{824-r}{2}} (11)^{\frac{r}{6}}$

$\Rightarrow {}^{824}C_r (7)^{412-\frac{r}{2}} (11)^{\frac{r}{6}}$

$r = 0,6,12,.....,822$

Total 138

57. Let $\alpha = \sum_{k=0}^{n}\left(\frac{\left({}^{n}C_k\right)^2}{k+1}\right)$ and $\beta = \sum_{k=0}^{n-1}\left(\frac{{}^{n}C_k\,{}^{n}C_{k+1}}{k+2}\right)$. If $5\alpha = 6\beta$, then n equals _____.

[JEE(Main)Jan.2024]

माना $\alpha = \sum_{k=0}^{n}\left(\frac{\left({}^{n}C_k\right)^2}{k+1}\right)$ तथा $\beta = \sum_{k=0}^{n-1}\left(\frac{{}^{n}C_k\,{}^{n}C_{k+1}}{k+2}\right)$ हैं। यदि $5\alpha = 6\beta$ हैं, तो n बराबर है _____

[JEE(Main)Jan.2024]

Ans. 10

Sol. $\alpha = \sum_{k=0}^{n} \frac{{}^{n}C_k \cdot {}^{n}C_k}{k+1} \cdot \frac{n+1}{n+1}$

$= \frac{1}{n+1}\sum_{k=0}^{n} {}^{n+1}C_{k+1} \cdot {}^{n}C_{n-k}$

$\alpha = \frac{1}{n+1} \cdot {}^{2n+1}C_{n+1}$

$\beta = \sum_{k=0}^{n-1} {}^{n}C_k \cdot \frac{{}^{n}C_{k+1}}{k+2}\frac{n+1}{n+1}$

$\frac{1}{n+1}\sum_{k=0}^{n-1} {}^{n}C_{n-k} \cdot {}^{n+1}C_{k+2}$

$= \frac{1}{n+1} \cdot {}^{2n+1}C_{n+2}$

$\frac{\beta}{\alpha} = \frac{{}^{2n+1}C_{n+2}}{{}^{2n+1}C_{n+1}} = \frac{2n+1-(n+2)+1}{n+2}$

$\frac{\beta}{\alpha} = \frac{n}{n+2} = \frac{5}{6}$

$n = 10$

58. In the expansion of $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, the sum of the coefficients of x^3 and x^{-13} is equal to _____. **[JEE(Main)Jan.2024]**

$(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x \neq 0$, के प्रसार में x^3 तथा x^{-13} के गुणांकों का योग है _____

[JEE(Main)Jan.2024]

Ans. 118

Sol. $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$

$=(1+x)(1-x^2)\left(\left(1+\frac{1}{x}\right)^3\right)^5$

$=\frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}}$

$=\frac{(1+x)^{17}-x(1+x)^{17}}{x^{15}}$

= coeff (x^3) in the expansion = coeff(x^{18}) in $(1+x)^{17} - x(1+x)^{17}$

$= 0 - 1$

$= -1$

coeff (x^{-13}) in the expansion = coeff (x^2) in $(1+x)^{17} - x(1+x)^{17}$

$=\binom{17}{2}-\binom{17}{1}$

$= 17 \times 8 - 17$

$= 17 \times 7$

$= 119$

Hence Answer = 119 – 1 = 118

59. Let the coefficient of x^r in the expansion of

$(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r = \beta^n - \gamma^n, \beta, \gamma \in \mathbb{N}$, then the value of $\beta^2+\gamma^2$ equals _____. **[JEE(Main)Jan.2024]**

माना $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$ के प्रसार में x^r का गुणांक α_r है। यदि

$\sum_{r=0}^{n} \alpha_r = \beta^n - \gamma^n, \beta, \gamma \in \mathbb{N}$ है, तो $\beta^2 + \gamma^2$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 25

Sol. $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \ldots.(x+2)^{n-1}$

$$= (x+3)^{n-1} \frac{\left\{1 - \left(\frac{x+2}{x+3}\right)^n\right\}}{1 - \left(\frac{x+2}{x+3}\right)}$$

$$= (x+3)^n - (x+2)^n$$

$$\sum_{r=0}^{n} \alpha_r = \sum_{r=0}^{n} \left[(x+3)^n - (x+2)^n\right] = \beta^n - \gamma^n$$

$$\Rightarrow 4^n - 3^n = \beta^n - \gamma^n$$

$$\Rightarrow \beta = 4, \quad \gamma = 3$$

$$\beta^2 + \gamma^2 = 25$$

60. If the coefficient of x^{30} in the expansion $\left(1+\frac{1}{x}\right)^6 \left(1+x^2\right)^7 \left(1-x^3\right)^8$; $x \neq 0$ is α, then $|\alpha|$ equals _____.

[JEE(Main)Jan.2024]

यदि $\left(1+\frac{1}{x}\right)^6 \left(1+x^2\right)^7 \left(1-x^3\right)^8$; $x \neq 0$, के प्रसार में x^{30} का गुणांक α है, तो $|\alpha|$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 678

Sol. coefficient of x^{30} in $\frac{(x+1)^6 \left(1+x^2\right)^7 \left(1-x^3\right)^8}{x^6}$

coefficient of x^{36} in $(1+x)^6 \left(1+x^2\right)^7 \left(1-x^3\right)^8$

General term

${}^6C_{r_1} {}^7C_{r_2} {}^8C_{r_3} (-1)^{r_1} x^{r_1+2r_2+3r_3}$

$r_1 + 2r_2 + 3r_3 = 36$

Case – I :

r_1	r_2	r_3
0	6	8
2	5	8
4	4	8
4	3	8

$r_1 + 2r_2 = 12$ (taking $r_3 = 8$)

Case – II :

r_1	r_2	r_3
1	7	7
3	6	7
5	5	7

$r_1 + 2r_2 = 15$ (taking $r_3 = 7$)

Case – III :

r_1	r_2	r_3
4	7	6
6	6	6

$r_1 + 2r_2 = 18$ (taking $r_3 = 6$)

Coefficient = $7 + (15 \times 21) + (15 \times 35) + (35)$
$-(6 \times 8) - (20 \times 7 \times 8) - (6 \times 21 \times 8) + (15 \times 28)$
$+ (7 \times 28) = -678 = \alpha$
$|\alpha| = 678$

61. The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}} + 5^{\frac{1}{3}}\right)^{15}$ is equal to : **[JEE(Main)April,2024]**

$\left(2^{\frac{1}{5}} + 5^{\frac{1}{3}}\right)^{15}$ के प्रसार में सभी परिमेय पदों का योग है : **[JEE(Main)April,2024]**

(A) 6131 (B) 3133 (C) 931 (D) 633

Ans. B

Sol. $T_{r+1} = {}^{15}C_r \left(5^{\frac{1}{3}}\right)^r \left(2^{\frac{1}{5}}\right)^{15-r}$

$= {}^{15}C_r 5^{\frac{r}{3}} \cdot 2^{\frac{15-r}{5}}$

$R = 3\lambda, 15\mu$
$\Rightarrow r = 0, 15$
2 rational terms

$\Rightarrow {}^{15}C_0 2^3 + {}^{15}C_{15}(5)^5$

$= 8 + 3125 = 3133$

62. If the coefficients of x^4, x^5 and x^6 in the expansion of $(1 + x)^n$ are in the arithmetic progression, then the maximum value of n is : **[JEE(Main)April,2024]**

यदि $(1 + x)^n$ के प्रसार में x^4, x^5 तथा x^6 के गुणांक समांतर श्रेणी में हैं, तो n का उच्चतम मान है :

[JEE(Main)April,2024]

(A) 28 (B) 7 (C) 21 (D) 14

Ans. D

Sol. $2^nC_5 = {}^nC_4 + {}^nC_6$

$$\frac{2\cdot n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

$$\frac{2!}{5\times 4!(n-5)(n-6)!} = \frac{1}{4!(n-4)(n-5)(n-6)!} + \frac{1}{6\times 5\times 4!(n-6)!}$$

$$\frac{2}{5(n-5)} = \frac{1}{(n-4)(n-5)} + \frac{1}{30}$$

$$\frac{2(n-4)}{5} = 1 + \frac{(n-4)(n-5)}{30}$$

$12n - 40 = 30 + n^2 - 9n + 20$

$n^2 - 21n + 98 = 0$

$n = 14, 7$

63. If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}}\right)^{12}$, $x \neq 0$ is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to : **[JEE(Main)April,2024]**

यदि $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}}\right)^{12}$, $x \neq 0$ के प्रसार में अचर पद $\alpha \times 2^8 \times \sqrt[5]{3}$ है, तो 25α बराबर है : **[JEE(Main)April,2024]**

(A) 724 (B) 742 (C) 693 (D) 639

Ans. C

Sol. ${}^{12}C_r \left(\frac{\sqrt[5]{3}}{x}\right)^{12-r} \left(\frac{2x}{\sqrt[3]{5}}\right)^r = {}^{12}C_r \frac{3^{\frac{12-r}{5}}.2^r}{5^{r/3}} \times x^{2r-12}$

$2r - 12 = 0 \Rightarrow r = 6$

$$\alpha \times 2^8 \times \sqrt[5]{3} = {}^{12}C_6 \frac{3^{\frac{6}{5}}2^6}{5^2} = \frac{11.3.4.7.3.3^{\frac{1}{5}}2^6}{5^2}$$

$$\alpha = \frac{3^2.7.11}{5^2} \Rightarrow 25\alpha = 693$$

64. Let $0 \le r \le n$. If ${}^{n+1}C_{r+1} : {}^{n}C_r : {}^{n-1}C_{r-1} = 55 : 35 : 21$, then $2n + 5r$ is equal to : **[JEE(Main)April,2024]**

माना $0 \le r \le n$ है। यदि ${}^{n+1}C_{r+1} : {}^{n}C_r : {}^{n-1}C_{r-1} = 55 : 35 : 21$ है, तो $2n + 5r$ बराबर है : **[JEE(Main)April,2024]**

(A) 55 (B) 60 (C) 62 (D) 50

Ans. D

Sol. $\frac{n+1}{r+1} \times {}^{n}C_r : {}^{n}C_r : \frac{r}{n} {}^{n}C_r = 55 : 35 : 21$

$\Rightarrow \frac{n+1}{r+1} = \frac{55}{34}$ and $\frac{n}{r} = \frac{35}{21}$

$\Rightarrow 7n - 11r = 4$ .....(1)

$3n - 5r = 4$ .....(2)

Solving (1) and (2)

$r = 6$ and $n = 10$

$\Rightarrow 2n + 5r = 20 + 30 = 50$

65. If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal to : **[JEE(Main)April,2024]**

यदि $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ के प्रसार में x से स्वतंत्र पद 105 है, तो a^2 बराबर है : **[JEE(Main)April,2024]**

(A) 4 (B) 2 (C) 9 (D) 6

Ans. A

Sol. $T_{r+1} = {}^{10}C_r \left(\sqrt{a}x^2\right)^{10-r} \left(\frac{1}{2x^3}\right)^r$

$2(10 - r) - 3r = 0$

$r = 4$

$T_5 = {}^{10}C_4 \left(\sqrt{a}\right)^6 \left(\frac{1}{2}\right)^4 = 105$

$$= \frac{10.9.8.7}{4.3.2.1} \times \frac{a^3}{2.2.2.2} = 5 \times 7 \times 3z$$

$a^3 = 8$, $a = 2 \Rightarrow a^2 = 4$

66. The coefficient of x^{70} in $x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + ... + x^{54}(1+x)^{46}$ is ${}^{99}C_p - {}^{46}C_q$.. Then a possible value of $p+q$ is : **[JEE(Main)April,2024]**

$x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + ... + x^{54}(1+x)^{46}$ में x^{70} ${}^{99}C_p - {}^{46}C_q$ है। तो $p+q$ का एक संभव मान है : **[JEE(Main)April,2024]**

(A) 68 (B) 83 (C) 55 (D) 61

Ans. B

Sol. $x^2(1+x)^{98} + x^3(1+x^{97}) + x^4(1+x)^{96} + ...x^{54}(1+x)^{46}$

Coeff. of x^{70} : ${}^{98}C_{68}$: + ${}^{97}C_{67}$ + ${}^{96}C_{66}$ + ... ${}^{47}C_{17}$ + ${}^{46}C_{16}$

$= {}^{46}C_{30} + {}^{47}C_{30} + ...{}^{98}C_{30}$

$= ({}^{46}C_{31} + {}^{46}C_{30}) + {}^{47}C_{30} + ... {}^{98}C_{30} - {}^{46}C_{31}$

$= {}^{47}C_{31} + {}^{47}C_{30} + ...{}^{98}C_{30} - {}^{46}C_{31}$ .....

$= {}^{99}C_{31} - {}^{46}C_{31} = {}^{99}C_p - {}^{46}C_q$

Possible values of $(p+q)$ are 62, 83, 99, 46

$\Rightarrow p + q = 83$

67. The sum of the coefficient of $x^{\frac{2}{3}}$ and $x^{-\frac{2}{5}}$ in the binomial expansion $\left(x^{\frac{2}{3}} + \frac{1}{2}x^{-\frac{2}{5}}\right)^9$ is : **[JEE(Main)April,2024]**

$\left(x^{\frac{2}{3}} + \frac{1}{2}x^{-\frac{2}{5}}\right)^9$ के द्विपद प्रसार में $x^{\frac{2}{3}}$ तथा $x^{-\frac{2}{5}}$ के गुणांकों का योग है : **[JEE(Main)April,2024]**

(A) $\frac{63}{16}$ (B) $\frac{21}{4}$ (C) $\frac{19}{4}$ (D) $\frac{69}{16}$

Ans. B

Sol. $T_{r+1} = {}^9C_r\left(x^{2/3}\right)^{9-r} \cdot \left(\frac{1}{2}x^{-\frac{2}{5}}\right)^r = {}^9C_r\left(\frac{1}{2}\right)^r \cdot x^{6-\frac{2r}{3}-\frac{2r}{5}}$

$= {}^9C_r \cdot \frac{1}{2^r} \cdot x^{6-\frac{16r}{15}}$

(I) $6-\frac{16r}{15}=\frac{2}{3}\Rightarrow r=5$ and (II) $6-\frac{16r}{15}=\frac{-2}{3}\Rightarrow 6+\frac{2}{5}=\frac{16r}{15}=\frac{32}{5}\quad\Rightarrow r=6$

$\text{Sum}={}^{9}C_5\cdot\frac{1}{2^5}+{}^{9}C_6\cdot\frac{1}{2^6}=\frac{1}{2^6}\left\{2\cdot{}^{9}C_4+{}^{9}C_3\right\}=\frac{1}{2^6}\left({}^{9}C_4+{}^{10}C_4\right)$

$=\frac{21}{4}$

68. Let $a=1+\frac{{}^{2}C_2}{3!}+\frac{{}^{3}C_2}{4!}+\frac{{}^{4}C_2}{5!}+.....,$

$b=1+\frac{{}^{1}C_0+{}^{1}C_1}{1!}+\frac{{}^{2}C_0+{}^{2}C_1+{}^{2}C_2}{2!}+\frac{{}^{3}C_0+{}^{3}C_1+{}^{3}C_2+{}^{3}C_3}{3!}+.....$

Then $\frac{2b}{a^2}$ is equal to _____. **[JEE(Main)April,2024]**

माना $a=1+\frac{{}^{2}C_2}{3!}+\frac{{}^{3}C_2}{4!}+\frac{{}^{4}C_2}{5!}+.....,$

$b=1+\frac{{}^{1}C_0+{}^{1}C_1}{1!}+\frac{{}^{2}C_0+{}^{2}C_1+{}^{2}C_2}{2!}+\frac{{}^{3}C_0+{}^{3}C_1+{}^{3}C_2+{}^{3}C_3}{3!}+.....$

हैं, तो $\frac{2b}{a^2}$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 8

Sol. $a=1+\frac{2c_2}{\lfloor 3}+\frac{3c_2}{\lfloor 4}+\frac{4c_2}{\lfloor 5}+\cdots$

$a=1+\sum_{r=2}^{\infty}\frac{{}^{r}C_2}{(r+1)}$

$a=1+\sum_{r=2}^{\infty}\frac{r(r-1)}{2\lfloor r+1}$

$a=1+\frac{1}{2}\sum_{r=2}^{\infty}\frac{r(r+1)-2r}{\lfloor (r+1)}$

$a=1+\frac{1}{2}\left[\sum_{r=2}^{\infty}\frac{1}{\lfloor r-1}-\frac{2}{\lfloor r}+\frac{2}{\lfloor r+1}\right]$

$a=1+\frac{1}{2}\left[(e-1)-2(e-2)+2\left(e-2-\frac{1}{2}\right)\right]$

$$a = 1 + \frac{1}{2}[e-2] = \frac{e}{2}$$

$$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \ldots$$

$b = e^2$

Now, $= \frac{2b}{a^2} = \frac{2 \times e^2}{e^2} \times 4 = 8$

69. If the constant term in the expansion of $\left(1 + 2x - 3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is p, then 108p is equal to _____.

[JEE(Main)April,2024]

यदि $\left(1 + 2x - 3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ के प्रसार में अचर पद p है, तो 108p बराबर है _____

[JEE(Main)April,2024]

Ans. 54

Sol. General term of $\left(\frac{3}{2} \times^2 - \frac{1}{3x}\right)^9$

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r = {}^9C_r (-1)^r 3^{9-2r} 2^{r-9} x^{18-3r}$$

Constant term in expansion of $(1 + 2x - 3x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$= T_7 - 3T_8 = {}^9C_6\, 3^{-3}.\, 2^{-3} + 3\,{}^9C_7\, .\, 3^{-5}\, .\, 2^{-2}$

$$= \frac{3 \times 4 \times 7}{3^3.2^3} + \frac{3 \times 9 \times 4}{3^5 \times 2^2} =$$

$$p = \frac{42 + 12}{108} = \frac{54}{108}$$

$108p = 54$

70. If $1 + \frac{\sqrt{3} - \sqrt{2}}{2\sqrt{3}} + \frac{5 - 2\sqrt{6}}{18} + \frac{9\sqrt{3} - 11\sqrt{2}}{36\sqrt{3}} + \frac{49 - 20\sqrt{6}}{180}$ +.....upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right)\log_e\left(\frac{a}{b}\right)$, where a and b are integers with gcd (a, b) = 1, then 11a + 18b is equal to _____. **[JEE(Main)April,2024]**

यदि $1+\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}}+\frac{5-2\sqrt{6}}{18}+\frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}}+\frac{49-20\sqrt{6}}{180}+.....\text{upto}\infty=2+\left(\sqrt{\frac{b}{a}}+1\right)\log_e\left(\frac{a}{b}\right)$, है, जहाँ a तथा b पूर्णांक हैं तथा gcd(a, b) = 1 है, तो 11a + 18b बराबर है _____ **[JEE(Main)April,2024]**

Ans. 76

Sol. $S = 1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}}+\frac{5-2\sqrt{6}}{18}+\frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}}+.....\infty$

$= 1 + \frac{\left(1-\sqrt{2}/\sqrt{3}\right)}{2}+\frac{\left(1-\sqrt{2}/\sqrt{3}\right)^2}{6}+\frac{\left(1-\sqrt{2}/\sqrt{3}\right)^3}{12}+.....\infty$

let $1-\frac{\sqrt{2}}{\sqrt{3}}=a$

$S = 1 + \frac{a}{2}+\frac{a^2}{6}+\frac{a^3}{12}+.....$

$=1+\left(1-\frac{1}{2}\right)a+\left(\frac{1}{2}-\frac{1}{3}\right)a^2+\left(\frac{1}{3}-\frac{1}{4}\right)a^3+....$

$=1+\left(a+\frac{a^2}{2}+\frac{a^3}{3}....\infty\right)+\frac{1}{a}\left(\frac{-a^2}{2}-\frac{a^3}{3}-\frac{a^4}{4}....\infty\right)$

$=-\text{In}(1-a)+\frac{1}{a}\left(-a-\frac{a^2}{2}-\frac{a^3}{3}....\infty\right)+2$

$=-\text{In}(1-a)+\frac{1}{a}\text{In}(1-a)+2$

$=2+\left(\frac{1}{a}-1\right)\text{In}(1-a)$

$=2+\left(\frac{\sqrt{3}}{\sqrt{3-\sqrt{2}}}-1\right)\text{In}\left(1-1+\sqrt{\frac{2}{3}}\right)$

$=2+\frac{\sqrt{2}}{\sqrt{3}-\sqrt{2}}\ \text{In}\sqrt{\frac{2}{3}}$

$=2+\left(\frac{\sqrt{6}+2}{1}.\frac{1}{2}\text{In}\frac{2}{3}\right)$

$= 2 + \left(\sqrt{\frac{3}{2}} + 1\right) \text{In}\frac{2}{3}$

$\therefore 11a + 18b = 76$

71. If the second, third and fourth terms in the expansion of $(x + y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3 + x^2 + y\right)$ is equal to _____. **[JEE(Main)April,2024]**

यदि $(x + y)^n$ के प्रसार में द्वितीय, तृतीय व चतुर्थ पद क्रमशः 135, 30 तथा $\frac{10}{3}$ हैं, तो $6\left(n^3 + x^2 + y\right)$ बराबर है _____

[JEE(Main)April,2024]

Ans. 806

Sol. ${}^nC_1 x^{n-1} y = 135$ _____(i)

${}^nC_2 x^{n-2} y^2 = 30$ _____(ii)

${}^nC_3 x^{n-3} y^3 = \frac{10}{3}$ _____(iii)

By $\frac{(i)}{(ii)}$

$\frac{{}^nC_1}{{}^nC_2}\frac{x}{y} = \frac{9}{2}$ _____(iv)

By $\frac{(ii)}{(iii)}$

$\frac{{}^nC_2}{{}^nC_3}\frac{x}{y} = 9$ _____(v)

By $\frac{(iv)}{(v)}$

$\frac{{}^nC_1\,{}^nC_3}{{}^nC_2\,{}^nC_2} = \frac{1}{2}$

$\frac{2n^2(n-1)(n-2)}{6} = \frac{n(n-1)}{2}\frac{n(n-1)}{2}$

$4n - 8 = 3n - 3$

$\Rightarrow n = 5$

put in (v)

$\frac{x}{y} = 9$

$x = 9y$

put in (i)

$^5C_1 x^4 \left(\frac{x}{9}\right) = 135$

$x^5 = 27 \times 9$

$\Rightarrow x = 3, \qquad y = \frac{1}{3}$

$6(n^3 + x^2 + y)$

$= 6\left(125 + 9 + \frac{1}{3}\right)$

$= 806$

72. Let $\alpha = \sum_{r=0}^{n}\left(4r^2 + 2r + 1\right)\ {}^nC_r$ and $\beta = \left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____.

[JEE(Main)April,2024]

माना $\alpha = \sum_{r=0}^{n}\left(4r^2 + 2r + 1\right)\ {}^nC_r$ तथा $\beta = \left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$ है। यदि $140 < \frac{2\alpha}{\beta} < 281$, तब n का मान होगा _____

[JEE(Main)April,2024]

Ans. 5

Sol. $\alpha = \sum_{r=0}^{n} r^2\ {}^nC_r + 2\sum_{r=0}^{n} r\ {}^nC_r + {}^nC_r$

$\sum (4 \cdot r(r-1) + 6r + 1)^nC_r$

$4.n(n-1)2^{n-2} + 6 \cdot n \cdot 2^{n-1} + 2^n$

$= 2^n (n+1)^2$

$\beta = \sum_{r=0}^{n}\frac{{}^nC_r}{r+1} + \frac{1}{n+1} = \frac{1}{n+1}\cdot\left[2^{n+1} - 1\right] + \frac{1}{n+1} = \frac{2^{n+1}}{n+1}$

Now, $140 < \frac{2\alpha}{\beta} < 248 \quad 140 < (n+1)^3 < 281$

$n + 1 = 6 \quad n = 5$

73. The remainder when 428^{2024} is divided by 21 is_____. **[JEE(Main)April,2024]**

428^{2024} को 21 से विभाजित करने का शेषफल है_____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $21 \times 20 = 420 \Rightarrow (428)^{2024} = (420+8)^{2024}$

$= {}^{2024}C_0\,(420)^{2024} + ...+ {}^{2024}C_{2033}\,(420)^{2023} + 8^{2024}$

$=420$ [some integer] $+ (64)^{1012}$; $63 = 21 \times 3$

$=21 \times 20$ [some integer] $+ (63+1)^{1012}$

$=21 \times 20$ [some integer] $+ 63$[some other integer] $+ 1$

Hence remainder is 1.

Exercise J - A

1. The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14. Then n =

[JEE(Adv.)2013]

$(1 + x)^{n+5}$ के तीन क्रमागत पदों के गुणांक 5 : 10 : 14 के अनुपात में हैं। तब, n = **[JEE(Adv.)2013]**

Ans. 6

Sol. Let the three consecutive terms be

${}^{n+5}C_{r-1}, {}^{n+5}C_r, {}^{n+5}C_{r+1}$

$\therefore \quad \frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_r} = \frac{1}{2} \Rightarrow \frac{r}{n-r+6} = \frac{1}{2}$

$\Rightarrow \quad n = 3r - 6 \qquad .....(1)$

Alao, $\frac{{}^{n+5}C_r}{{}^{n+5}C_{r+1}} = \frac{5}{7} \Rightarrow \frac{r+1}{n-r+5} = \frac{5}{7}$

$\Rightarrow \quad 12r = 5n + 18 \qquad .....(2)$

Solving (1) and (2), we get n = 6

2. Coefficient of x^{11} in the expansion of $(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12}$ is : **[JEE(Adv.)2014]**

$(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12}$ के प्रसार में x^{11} का गुणांक होगा **[JEE(Adv.)2014]**

(A) 1051 (B) 1106 (C) 1113 (D) 1120

Ans. C

Sol. Coefficent of $x^{11} \equiv \frac{(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12} (1-x^2)^4}{(1-x^2)^4}$

Coefficent of $x^{11} \equiv (1 - x^8)^4 (1 + x^4)^8 (1 + x^3)^7 (1 - x^2)^{-4}$

$= (1 - 4x^8) (1 + x^4)^8 (7x^3 + 35x^9) (1 - x^2)^{-4}$

$= (7x^3 + 35x^9 - 28x^{11}) (1 + x^4)^8 (1 - x^2)^{-4}$

Coefficent of $x^8 = (7x + 35x^6 - 28x^8) (1 + 8x^4 + 28x^8) (1 - x^2)^{-4}$

$= (7 + 35x^6 - 28x^8 + 56x^4 + 196x^8) (1 - x^2)^{-4}$

Coefficent of $t^4 \equiv (7 + 56t^2 + 35t^3 + 168t^4) (1 - t)^{-4}$

$= 7 \cdot {}^7C_3 + 56 \cdot {}^5C_3 + 35 \cdot {}^4C_3 + 168$

$= 245 + 700 + 168 = 1113.$

Alterantive :

$2x + 3y + 4z = 11$

$(x, y, z) = (0, 1, 2)\ {}^4C_0 \times {}^7C_1 \times {}^{12}C_2$

$(1, 3, 0)\quad {}^4C_1 \times {}^7C_3$

$(2, 1, 1)\ {}^4C_2 \times {}^7C_1 \times {}^{12}C_1$

$(4, 1, 0)\quad {}^7C_1$

coefficient of $x^{11} = 66 \times 7 + 35 \times 4 + 42 \times 12 + 7$

$= 1113$

3. The coefficient of x^9 in the expansion of $(1 + x)(1 + x^2)(1 + x^3)\ldots.(1 + x^{100})$ is : **[JEE(Adv.)2015]**

$(1+x)(1+x^2)(1+x^3)\ldots.(1+x^{100})$ के विस्तार में x^9 के गुणांक का मान है। **[JEE(Adv.)2015]**

Ans. 8

Sol. $9 = (0, 9)\ (1, 8), (2, 7), (3, 6), (4, 5)$ # 5 cases

$9 = (1,2,6), (1,3,5), (2, 3, 4)$ # 3 cases

total = 8

4. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1 + x)^2 + (1 + x)^3 + \ldots.. + (1 + x)^{49} + (1 + mx)^{50}$ is $(3n + 1)\ {}^{51}C_3$ for some positive integer n. Then the value of n is **[JEE(Adv.)2016]**

माना कि m ऐसा न्यूनतम धनात्मक पूर्णांक है कि $(1 + x)^2 + (1 + x)^3 + \ldots.. + (1 + x)^{49} + (1 + mx)^{50}$ के विस्तार में x^2 का गुणांक $(3n + 1)\ {}^{51}C_3$ किसी धनात्मक पूर्णांक n के लिए है। तब n का मान है। **[JEE(Adv.)2016]**

Ans. 5

Sol. ${}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots.... + {}^{49}C_2 + {}^{50}C_2\ m^2 = (3n + 1)\ {}^{51}C_3$

$\Rightarrow {}^{50}C_3 + {}^{50}C_2\ m^2 = (3n + 1)\ {}^{51}C_3$

$\Rightarrow {}^{50}C_3 + {}^{50}C_2 + (m^2 - 1)\ {}^{50}C_2 = 3n\ {}^{51}C_3 + {}^{51}C_3$

$\Rightarrow n = \dfrac{m^2 - 1}{51}$

$\Rightarrow m^2 = 51n + 1$ must be a perfect squared

$\Rightarrow n = 5$ and $m = 16$

5. Let

$$X = \left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + 3\left({}^{10}C_3\right)^2 + \ldots. + 10\left({}^{10}C_{10}\right)^2,$$

where ${}^{10}C_r, r \in \{1, 2,\ldots., 10\}$ denote binomial coefficients. Then, the value of $\dfrac{1}{1430}X$ is _____.

[JEE(Adv.)2018]

माना कि

$$X = \left({}^{10}C_1\right)^2 + 2\left({}^{10}C_2\right)^2 + 3\left({}^{10}C_3\right)^2 + ... + 10\left({}^{10}C_{10}\right)^2,$$

जहाँ ${}^{10}C_r$, $r \in \{1, 2,...., 10\}$, द्विपद गुणांकों को दर्शाते हैं। तब $\frac{1}{1430}$ X का मान है। **[JEE(Adv.)2018]**

Ans. 646

Sol. $X = \sum_{r=1}^{10} r.\,{}^{10}C_r = 10.\sum_{r=1}^{10} {}^{9}C_{r-1}.\,{}^{10}C_{10-r} = 10.\,{}^{19}C_9$

$$\text{Now } \frac{X}{1430} = \frac{10.\,{}^{19}C_9}{1430} = \frac{{}^{19}C_9}{143} = \frac{{}^{19}C_9}{11\times 13} = \frac{19.17.16}{8} = 19\times 34 = 646$$

6. For nonnegative integers s and r, let $\binom{s}{r} = \begin{cases} \dfrac{s!}{r!(s-r)!} & \text{if } r \le s, \\ 0 & \text{if } r > s. \end{cases}$

For positive integers m and n, let $g(m,n) = \sum_{p=0}^{m+n} \dfrac{f(m,n,p)}{\binom{n+p}{p}}$

where for any nonnegative integers p, $f(m, n, p) = \sum_{i=0}^{p} \binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$

Then which of the following statements is / are TRUE ? **[JEE(Adv.)2020]**

(A) g (m, n) = g (n, m) for all positive integers m, n

(B) g(m, n + 1) = g(m+1, n) for all positive integers m, n

(C) g (2m, 2n)=2g (m, n) for all positive integers m, n

(D) g (2m, 2n) = (g (m, n))^2 for all positive integers m, n

माना कि ऋणेतर पूर्णांकों s एवं r के लिए $\binom{s}{r}=\begin{cases}\dfrac{s!}{r!(s-r)!} & \text{if } r\le s,\\ 0 & \text{if } r>s.\end{cases}$

है। माना कि धनात्मक पूर्णांको m एवं n के लिए $g(m,n)=\sum_{p=0}^{m+n}\dfrac{f(m,n,p)}{\binom{n+p}{p}}$

है, जहाँ किसी ऋणेतर पूर्णांक p के लिए $f(m,n,p)=\sum_{i=0}^{p}\binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$

है। तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) सभी धनात्मक पूर्णांकों m, n के लिए g (m, n) = g (n, m) है

(B) सभी धनात्मक पूर्णांकों m, n के लिए g (m, n + 1) = g (m + 1, n) है

(C) सभी धनात्मक पूर्णांकों m, n के लिए g (2m, 2n) = 2g (m, n) है

(D) सभी धनात्मक पूर्णांकों m, n के लिए g (2m, 2n) = $(g(m,n))^2$ है

Ans. ABD

Sol. $f(m,n,p)=\sum_{i=0}^{p}\left({}^{m}C_{i}\dfrac{(n+i)!}{(n+i-p)!p!}\times\dfrac{(p+n)!}{(n+i)!(p-i)!}\right)$

$=\sum_{i=0}^{p}{}^{m}C_{i}\left(\dfrac{n!}{(n+i-p)!(p-i)!}\right)\left(\dfrac{p+n!}{(n)!(p)!}\right)$

$={}^{n+p}C_{p}\sum_{i=0}^{p}{}^{m}C_{i}\,{}^{n}C_{p-i}$

$f(m,n,p)={}^{n+p}C_{p}\left({}^{m}C_{0}\,{}^{n}C_{p}+{}^{m}C_{1}\,{}^{n}C_{p-1}+.....+{}^{m}C_{p}\,{}^{n}C_{0}\right)$

$={}^{n+p}C_{p}.{}^{m+n}C_{p}$

$g(m,n)=\sum_{p=0}^{m+n}\dfrac{{}^{n+p}C_{p}\;{}^{m+n}C_{p}}{{}^{n+p}C_{p}}=2^{m+n}$

(A) g(m, n) =

$(2)^{m+n}=(2)^{n+m}=g(n,m)$

(B) $g(m, n+1) = (2)m+n+1 = (2)^{m+n+1} = (2)^{m+1+n}$

(C) $g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$

(D) $g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$

$= (2^{m+n})^2 = (g(m, n))^2$

Hence, option (A), (B), (D) are correct answer.

7. Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of 2b is _____. **[JEE(Adv.)2023]**

माना कि a एवं b दो शून्येतर (nonzero) वास्तविक संख्याएं है। यदि $\left(ax^2 + \frac{70}{27bx}\right)^4$ के प्रसार (expansion) में x^5 का गुणांक (coefficient), $\left(ax - \frac{1}{bx^2}\right)^7$ के प्रसार में x^{-5} के गुणांक के बराबर है, तब 2b का मान है _____ **[JEE(Adv.)2023]**

Ans. 3

Sol. $T_{r+1} = {}^4C_r \left(a \cdot x^2\right)^{4-r} \cdot \left(\frac{70}{27bx}\right)^r$

$= {}^4C_r \cdot a^{4-r} \cdot \frac{70^r}{(27b)^r} \cdot x^{8-3r}$

here $8 - 3r = 5$

$8 - 5 = 3r \Rightarrow r = 1$

$\therefore$ coeff. $= 4 \cdot a^3 \cdot \frac{70}{27b}$

$T_{r+1} = {}^7C_r (ax)^{7-r} \left(\frac{-1}{bx^2}\right)^r$

$= {}^7C_r \cdot a^{7-r} \left(\frac{-1}{b}\right)^r \cdot x^{7-3r}$

$7 - 3r = -5 \Rightarrow 12 = 3r \Rightarrow r = 4$

Coeff. : ${}^7C_4 \cdot a^3 \cdot \left(\frac{-1}{b}\right)^4 = \frac{35a^3}{b^4}$

Now $\frac{35a^3}{b^4} = \frac{280a^3}{27b}$

$b^3 = \frac{35 \times 27}{280} = b = \frac{3}{2} \Rightarrow 2b = 3$

8. Let $a_0, a_1, \ldots, a_{23}$ be real numbers such that $\left(1+\frac{2}{5}x\right)^{23} = \sum_{i=0}^{23} a_i x^i$ for every real number x Let a_r be the largest among the numbers a_j for $0 \leq j \leq 23$. Then the value of r is _____. **[JEE(Adv.)2025]**

मान लीजिए कि $a_0, a_1, \ldots, a_{23}$ इस प्रकार की वास्तविक संख्याएँ (real numbers) हैं कि सभी वास्तविक संख्याओं x के लिए, $\left(1+\frac{2}{5}x\right)^{23} = \sum_{i=0}^{23} a_i x^i$ है। मान लीजिए कि संख्याओं a_j, $0 \leq j \leq 23$, में सबसे बड़ी संख्या a_r है। तब r का मान _____है।

[JEE(Adv.)2025]

Ans. 6

Sol. $x = 1$

$\left(1+\frac{2}{5}\right)^{23} = a_0 + a_1 + a_2 + a_3 \ldots\ldots\ldots a_{23}$

For N.G.T.

$\frac{n+1}{1+\left|\frac{x}{y}\right|} \qquad \Rightarrow \frac{23+1}{1+\frac{5}{2}} = \frac{48}{7}$

$\left[\frac{48}{7}\right] = 6 = r$

So T_7 is the numerically Greater term.

$r = 6$

CHALLENGER EXERCISES
Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The coefficient of x^{53} in the expansion $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is :

$\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ के प्रसार में x^{53} का गुणांक होगा :

(A) ${}^{100}C_{47}$ (B) ${}^{100}C_{53}$ (C) $-{}^{100}C_{53}$ (D) $-{}^{100}C_{100}$

Ans. C

Sol. $S = \sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$

$S = {}^{100}C_0 (x-3)^{100} + {}^{100}C_1 (x-3)^{99} \cdot 2 + ... + {}^{100}C_{100} \cdot 2^{100}$

$S = (2 + (x-3))^{100} = (x-1)^{100}$

Co-efficient of $x^{53} = -{}^{100}C_{53} = -{}^{100}C_{47}$

2. If the sum of the coefficients in the expansion of $\left(x^2 + \frac{2}{x^3}\right)^n$ is 243, then the term independent of x, is equal to :

$\left(x^2 + \frac{2}{x^3}\right)^n$ के प्रसार में गुणांको का योगफल 243 है तब x से स्वतन्त्र पद होगा :

(A) 11 (B) $\frac{7}{2}$ (C) 15 (D) 40

Ans. D

Sol. $\left(x^2 + \frac{2}{x^3}\right)^n = \sum_{r=0}^{n} {}^nC_r x^{2(n-r)} \left(\frac{2}{x^3}\right)^r$

$= \sum_{r=0}^{n} {}^nC_r x^{2n-5r} \cdot 2^r$

For independent term

$2n = 5r$

$\Rightarrow r = 2$

Given

$\left(1 + \frac{2}{1}\right)^n = \sum_{r=0}^{n} {}^nC_r 2^r$

$= 243$

$\Rightarrow 3^n = 243$

$\Rightarrow n = 5$

$\Rightarrow$ Coefficient

$= {}^5C_2 \cdot 2^2 = 10 \cdot 4 = 40$

3. The value of $4\ \{{}^nC_1 + 4\ .\ {}^nC_2 + 4^2\ .\ {}^nC_3 + ... + 4^{n-1}\}$ is :

$4\ \{{}^nC_1 + 4\ .\ {}^nC_2 + 4^2\ .\ {}^nC_3 + ... + 4^{n-1}\}$ का मान होगा :

(A) 0 (B) $5^n + 1$ (C) 5^n (D) $5^n - 1$

Ans. D

Sol. $4\left\{{}^nC_1 + 4.{}^nC_2 + ... + 4^{n-1}.{}^nC_n\right\}$

$= \sum_{r=1}^{n} {}^nC_r \cdot 4^r$

$= -{}^nC_0 + \sum_{r=0}^{n} {}^nC_r 4^r$

$= -1 + (1+4)^n$

$= 5^n - 1$

4. The value of $\sum_{r=0}^{n} \frac{1}{(2r)!(2n-2r)!}$, is equal to :

$\sum_{r=0}^{n} \frac{1}{(2r)!(2n-2r)!}$, का मान होगा :

(A) $\frac{2^{2n}}{(2n)!}$ (B) $\frac{2^{2n-1}}{(2n)!}$ (C) $\frac{2^{n-1}}{n!}$ (D) $\frac{2^n}{n!}$

Ans. B

Sol. $\sum_{r=0}^{n} \frac{1}{\lfloor 2r \rfloor 2n-2r} = \frac{1}{\lfloor 2n} \sum_{r=0}^{n} \frac{\lfloor 2n}{\lfloor 2r \rfloor 2n-2r}$

$= \frac{1}{\lfloor 2n} \sum_{r=0}^{n} {}^{2n}C_{2r}$

$= \frac{1}{\lfloor 2n} \left[{}^{2n}C_0 + {}^{2n}C_2 + {}^{2n}C_4 + ... + {}^{2n}C_{2n}\right]$

$$= \frac{1}{\lfloor 2n} \cdot \frac{2^{2n}}{2} = \frac{2^{2n-1}}{\lfloor 2n}$$

5. In the expansion of $(1+x)^n (1+y)^n (1+z)^n$, the sum of the co-efficients of the terms of degree 'r' is :

$(1+x)^n (1+y)^n (1+z)^n$ के प्रसार में पदों की घात 'r' के रूप में गुणांको का योगफल होगा :

(A) ${}^{n^3}C_r$ (B) ${}^{n}C_{r^3}$ (C) ${}^{3n}C_r$ (D) $3 . {}^{2n}C_r$

Ans. C

Sol. Put $x = y = z$, we get $(1+x)^{3n}$

$\therefore$ coeff. of terms of degree $r = {}^{3n}C_r$

6. The sum of the last n coefficients in the expansion of $(1+x)^{2n-1}$ when expanding in ascending power of x, is equal to :

$(1+x)^{2n-1}$ के प्रसार में अन्त से n पदों के गुणांको का योगफल क्या होगा जबकि प्रसार x की बढ़ती हुई घात के रूप में किया जाए:

(A) 2^{2n-1} (B) 2^{2n-2} (C) 2^{2n} (D) $2^{2n} - 2^n$

Ans. B

Sol. $(1+x)^{2n-1} = {}^{2n-1}C_0 + {}^{2n-1}C_1 x + ... + \underbrace{{}^{2n-1}C_n x^n + ... + {}^{2n-1}C_{2n-1} x^{2n-1}}_{\text{last n terms}}$

$$\Rightarrow {}^{2n-1}C_n + + {}^{2n-1}C_{n+1} + ... + {}^{2n-1}C_{2n-1} = \frac{2^{2n-1}}{2} = 2^{2n-2} \qquad \left\{\because {}^{n}C_r = {}^{n}C_{n-r}\right\}$$

7. The fractional part of $\frac{3^{2n}}{8}$, is equal to :

(A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) None of these

$\frac{3^{2n}}{8}$ का भिन्नात्मक भाग होगा :

(A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{3^{2n}}{8} = \frac{9^n}{8} = \frac{(8+1)^n}{8}$

$$= \frac{\sum_{r=0}^{n} {}^{n}C_r 8^r}{8} = \frac{{}^{n}C_0 + 8\lambda}{8}; \lambda \in I$$

$$= \frac{1}{8} + \lambda; \lambda \in I$$

Fractional part of $\frac{3^{2n}}{8}$ is $\frac{1}{8}$.

8. Given positive integers r > 1, n > 2, and the coefficients of the $(3r)^{th}$ and $(r + 2)^{th}$ terms in the binomial expansion of $(1 + x)^{2n}$ are equal, which, if any, of the following statements is true ?

दिया गया है धनात्मक पूर्णांक r > 1 के लिए $(r + x)^{2n}$ के प्रसार में $(3r)^{th}$ व $(r + 2)$ पदों के गुणांक बराबर है जहां n > 2 तब ?

(A) n = 2r (B) n = 3r (C) n = 2r + 1 (D) None of these

Ans. A

Sol. r > 1, n > 2

$$(1+x)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r x^r$$

Given

Coefficient of $3r^{th}$ term = Coefficient of $(r + 2)^{th}$ term

$\Rightarrow {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$

$\Rightarrow 3r - 1 = r + 1$ or $3r - 1 + r + 1 = 2n$

$\Rightarrow 2r = 2$ or $4r = 2n$

$\Rightarrow r = 1$ or $n = 2r$

as r > 1

$\Rightarrow n = 2r$

9. If in the expansion of $(1 + x)^m (1 - x)^n$, coefficients of x and x^2 are 3 and - 6 respectively, then m is :

$(1 + x)^m (1 - x)^n$ के प्रसार में x तथा x^2 के गुणांक 3 तथा - 6 है तब m बराबर होगा :

(A) 6 (B) 9 (C) 12 (D) 24

Ans. C

Sol. Consider $(1+x)^m(1-x)^n$

$\Rightarrow [{}^mC_0 + {}^mC_1x + {}^mC_2x^2 + ... + {}^mC_mx^m][{}^nC_0 + {}^nC_1(-x) + {}^nC_2(-x)^2 + ... {}^nC_n(-x)^n]$

$[\because (1+x)^n = {}^nC_0 + {}^nC_1x + ... {}^nC_nx^n]$

On multiplying and comparing

Now coefficient of $x = {}^mC_0 . {}^nC_1(-1) + {}^mC_1 . {}^nC_0 = m - n = 3$ ...(i)

Coeff. of $x^2 = {}^mC_0 . {}^nC_2(-1)^2 + {}^mC_2 . {}^nC_0 + {}^mC_1 . {}^nC_1(-1)$

$= \frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn = -6$

$= \frac{n(n-1)}{2} + \frac{(3+n)(3+n-1)}{2} - (3+n)n = -6$ From (i)

$= \frac{n^2-n}{2} + \frac{n^2+5n+6}{2} - 3n - n^2 = -6$

$= \frac{n^2-n+n^2+5n+6-6n-2n^2}{2} = -6$

$= -2n + 6 = -12$

$n! = 9$ Now $m = 3 + n = 12$

10. ${}^{n-1}C_r = (K^2-3).{}^nC_{r+1}$, if $K \in$

(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$

Ans. D

Sol. ${}^{n-1}C_r = (k^2-3) \cdot {}^nC_{r+1}$

$\frac{(n-1)!}{r!(n-r-1)!} = (k^2-3) \cdot \frac{n!}{(r+1)!(n-r-1)!}$

$\Rightarrow \frac{r+1}{n} + 3 = k^2$

$\Rightarrow \because 0 \le r \le n-1$

$1 \le r+1 \le n$

$\Rightarrow \frac{1}{n} \le \frac{r+1}{n} \le 1 \Rightarrow \frac{1}{n} \le k^2 - 3 \le 1$

$\Rightarrow 3 + \frac{1}{n} \le k^2 \le 4 \Rightarrow \sqrt{3+\frac{1}{n}} \le k \le 2$

as $n \to \infty \Rightarrow \sqrt{3} < k \le 2$

$\Rightarrow k \in \left(\sqrt{3}, 2\right]$

11. The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} ..... + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$:

$\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} ..... + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$. का मान है:

(A) $\binom{30}{10}$ (B) $\binom{30}{15}$ (C) $\binom{60}{30}$ (D) $\binom{31}{10}$

Ans. A

Sol. ${}^{30}C_0\ {}^{30}C_{10} - {}^{30}C_1\ {}^{30}C_{11} + {}^{30}C_2\ {}^{30}C_{12} + ... + {}^{30}C_{20}\ {}^{30}C_{30}$

= Coefficient of x^{20} in $(1 - x^2)^{30}$

as

$(1 + x)^{30} = {}^{30}C_0 + {}^{30}C_1\ x + {}^{30}C_2\ x^2 + ... + {}^{30}C_{20}\ x^{20} + ... + {}^{30}C_{30}\ x^{30}$

$(1 - x)^{30} = {}^{30}C_0 - {}^{30}C_1\ x + ... + {}^{30}C_{30}\ x^{30}$

$= {}^{30}C_{30} - {}^{30}C_{29}\ x + {}^{30}C_{28}\ x^2 + ... + {}^{30}C_0\ x^{30}$

$= {}^{30}C_{30} - {}^{30}C_{29}\ x + {}^{30}C_{28}\ x^2 + ... - {}^{30}C_{11}\ x^{19} + ... + {}^{30}C_{30}\ x^{30}$

$= {}^{30}C_{10}$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

12. The polynomial $1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$ may be written in the form $\sum_{i=0}^{17} a_i y^i$ where $y = x + 1$ and a_i's are constants. Then a_2 equals –

(A) The coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) The coefficient of t^3 in the expansion of $1 - (1 - t)^{17}$

यदि बहुपद $1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$ को $\sum_{i=0}^{17} a_i y^i$ के रूप में लिखा जा सकता हो, जहाँ $y = x + 1$ तथा

a_i अचर है, तो a_2 बराबर होगा –

(A) $1 - (1 - t)^{18}$ के प्रसार में t^3 का गुणांक

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) $1 - (1 - t)^{17}$ के प्रसार में t^3 का गुणांक

Ans. ABC

Sol. $1 - x + x^2 - x^3 + ...+ x^{16} - x^{17}$

$= (1 - x)\,[1 + x^2 + ... + x^{16}]$

$y = x + 1$

$\Rightarrow x = y - 1$

$$\frac{(1-x)(1-x^{18})}{(1-x^2)}$$

$$= \frac{1-x^{18}}{(1+x)}$$

$$= \frac{1-(y-1)^{18}}{y} = \frac{1-\sum_{r=0}^{18} {}^{18}C_r y^r (-1)^r}{y}$$

$$= \frac{1-\left[{}^{18}C_0 + {}^{18}C_1 y(-1) + {}^{18}C_2 y^2 (-1)^2 + ... + {}^{18}C_{18} y^{18}\right]}{y}$$

$$= -\frac{1}{y}\left[-{}^{18}C_1 y + {}^{18}C_2 y^2 + ... + {}^{18}C_{18} y^{18}\right]$$

$$= \left[{}^{18}C_1 - {}^{18}C_2 y + {}^{18}C_3 y^2 + ... - {}^{18}C_{18} y^{17}\right]$$

$$a_2 = {}^{18}C_3 = \frac{18 \times 17 \times 16}{6} = 816$$

= Coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$

$= \sum_{n=2}^{17} {}^{n}C_2$

13. In the expansion of $\left(x^3 + 3.2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$

(A) there appears a term with the power x^2

(B) there does not appear a term with the power x^2

(C) there appears a term with the power x^{-3}

(D) the ratio of the co-efficient of x^3 to that of x^{-3} is 1/3

$\left(x^3 + 3.2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$ के प्रसार में

(A) x^2 का पद आयेगा

(B) x^2 का पद नहीं आयेगा

(C) x^{-3} का पद आयेगा

(D) x^3 एवं x^{-3} के गुणांको का अनुपात 1/3 होगा।

Ans. BCD

Sol. $\left(x^3 + 3\cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$

$\left\{\log_{\sqrt{2}} \sqrt{x^3} = \frac{1}{2}\log_{\sqrt{2}} x^3 = \frac{3}{2}\log_{\sqrt{2}} x = 3\log_2 x\right\}$

$= \left(x^3 + 3\cdot 2^{\log_2 x^{-3}}\right)^{11}$

$= \left(x^3 + \frac{3}{x^3}\right)^{11}$

$= \sum_{r=0}^{11} {}^{11}C_r x^{33-3r} \cdot \frac{3^r}{x^{3r}}$

$= \sum_{r=0}^{11} {}^{11}C_r \cdot 3^r \cdot x^{33-6r}$

For x^2 $\quad 33 - 6r = -3$ $\Rightarrow 36 = 6r$

For x^3 $\quad 33 - 6r = 3$ $\Rightarrow r = 5$

$$\frac{\text{Coeff. of } x^3}{\text{Coeff. of } x^{-3}} = \frac{{}^{11}C_5 \cdot 3^5}{{}^{11}C_6 \cdot 3^6} = \frac{6}{11-6+1} \cdot \frac{}{3} = \frac{2}{6} = \frac{1}{3}$$

14. If it is known that the third term of the binomial expansion $\left(x + x^{\log_{10} x}\right)^5$ is 10^6 then x is equal to :

यदि $\left(x + x^{\log_{10} x}\right)^5$ के प्रसार में तीसरा पद 10^6 हो, तो x का मान होगा :

(A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5

Ans. AB

Sol. $\left(x + x^{\log_{10} x}\right)^5 = \sum_{r=0}^{5} {}^5C_r x^{5-r} \left(x^{\log_{10} x}\right)^r$

Given

$${}^5C_2 x^3 \left(x^{\log_{10} x}\right)^2 = 10^6$$

$$\Rightarrow x^3 \left(x^{\log_{10} x}\right)^2 = 10^5$$

Take log both sides

$$3\log_{10} x + 2\log_{10}\left(x^{\log_{10} x}\right) = 5$$

$$\Rightarrow 3\log_{10} x + 2\left(\log_{10} x\right)^2 = 5$$

Let $\log_{10} x = t$

$2t^2 + 3t - 5 = 0$

$$\Rightarrow t = \frac{-3 \pm \sqrt{9+40}}{4}$$

$$t = \frac{-3 \pm 7}{4}$$

$\Rightarrow t = 1$ or $t = -\frac{10}{4}$

$\Rightarrow x = 10$ or $\log_{10} x = -\frac{5}{2}$

$\Rightarrow x = 10$ or $x = 10^{-\frac{5}{2}}$

15. The expansion of $\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n$ is arranged in decreasing power of x. If coefficient of first three terms form an A.P. then in expansion, the integral powers of x are :

$\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n$ के प्रसार को x की अवरोही घातों के क्रम में लिखा जाता है। यदि प्रथम पदों के गुणांक एक A.P. बनाते हो, तो प्रसार में, x की पूर्णांक घातें है :

(A) 0 (B) 2 (C) 4 (D) 8

Ans. B

Sol. $\left(\sqrt{x}+\frac{1}{2\sqrt[4]{x}}\right)^n=\sum_{r=0}^{n}{}^nC_r\left(\sqrt{x}\right)^{n-r}\cdot\frac{1}{2^r}\cdot\left(\frac{1}{x}\right)^{\frac{r}{4}}$

$=\sum_{r=0}^{n}{}^nC_r\left(x\right)^{\frac{n-r}{2}}\cdot\frac{1}{2^r}\cdot x^{-\frac{r}{4}}$

$=\sum_{r=0}^{n}{}^nC_r x^{\frac{2n-2r-r}{4}}\cdot\frac{1}{2^r}$

${}^nC_0\cdot\frac{1}{2^0},{}^nC_1\cdot\frac{1}{2},{}^nC_2\cdot\frac{1}{2^2}\rightarrow$ A.P.

$1+\frac{n(n-2)}{8}=\frac{1}{2}\cdot n\cdot 2$

$\Rightarrow 8+n^2-n=8n$

$\Rightarrow n^2-9n+8=0$

$\Rightarrow n^2-8n-n+8=0$

$\Rightarrow (n-1)(n-8)=0$

$\Rightarrow n=1$ (wrong) or $n=8$

[For A.P. we need three terms in this question]

For integral power of x

$\frac{2n-3r}{4}\in I$

$\Rightarrow 2n-3r\in 4I$

$n = 8 ; r = 0, ..., 7, 8$

$16 - 3r = 4k; k \in Z$

$\Rightarrow r = \dfrac{16-4k}{3}$

$\Rightarrow r = \underset{\downarrow}{0}, \underset{\downarrow}{8}, \underset{\downarrow}{4}$
$\quad\quad\quad

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Let $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$,

find the value of a_2.

माना $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1 (1 + x) + a_2 (1 + x)^2 + ... + a_{17} (1 + x)^{17}$ तो a_2 का मान ज्ञात कीजिए।

Ans. 816

Sol. $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$

$$= \frac{1\left(1-(-x)^{18}\right)}{1-(-x)}$$

$$= \frac{1-x^{18}}{1+x}$$

$$\frac{1-x^{18}}{1+x} = a_0 + a_1(1+x) + a_2(1+x)^2 + ... + a_{17}(1+x)^{17}$$

$1 - x^{18} = a_0 (1 + x) + a_1 (1 + x)^2 + a_2 (1 + x)^3 + ... + a_{17} (1 + x)^{18}$

Let $1 + x = t$

$x = t - 1$

$1 - (t - 1)^{18} = a_0 t + a_1 t^2 + a_2 t^3 + ... + a_{17} t^{18}$

So, we need to find coefficient of t^3

$1 - (1 - t)^{18}$

$1 - \{{}^{18}C_0 - {}^{18}C_1 t_1 + {}^{18}C_2 t^2 - {}^{18}C_3 t^3 + ...\}$

So ${}^{18}C_3 \Rightarrow \frac{18}{3} \times \frac{17}{2} \times \frac{16}{1} = 816$

2. Let $a = \left(4^{1/401} - 1\right)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$.

Find the value of $(b_{2006} - b_{2005})$.

माना $a = \left(4^{1/401} - 1\right)$ और माना $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + ... + {}^nC_n \cdot a^{n-1}$ है।

तो $(b_{2006} - b_{2005})$ का मान ज्ञात कीजिए।

Ans. 2^{10}

Sol. $a = 4^{\frac{1}{401}} - 1$

$b_n = {}^nC_1 + {}^nC_2\, a + {}^nC_3\, a^2 + ... + {}^nC_n\, a^{n-1}$

$ab_n = {}^nC_1\, a^1 + {}^nC_2\, a^2 + {}^nC_3\, a^3 + ... + {}^nC_n\, a_n$

$+ \qquad +$

${}^nC_0 \qquad {}^nC_0$

$ab_n + {}^nC_0 = [1 + a]^n$

$ab_n = (1 + a)^n - 1$

$n = 2006$

$$ab_{2006} = \frac{(1+a)^{2006}}{a} - \frac{1}{a}$$

$$b_{2005} = \frac{(1+a)^{2005}}{a} - \frac{1}{a}$$

$$b_{2006} - b_{2005} = \frac{(1+a)^{2006}}{a} - \frac{1}{a} - \frac{(1+a)^{2005}}{a} + \frac{1}{a}$$

$$= \frac{(1+a)^{2006} - (1+a)^{2005}}{a}$$

$$= \frac{[1+a]^{2005}\{1+a-1\}}{a}$$

$\Rightarrow (1 + a)^{2005}$

$$1 + a = 4^{\frac{1}{401}}$$

$$(1+a)^{2005} = 4^{\frac{2005}{401}} = 4^5 = 2^{10}$$

3. Let $a = 3^{\frac{1}{223}} + 1$ and let $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - .....+ (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

If the value of $f(2007) + f(2008) = 9k$ where $k \in N$, then find k.

माना $a = 3^{\frac{1}{223}} + 1$ और माना $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - \ldots\ldots + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$.

यदि $f(2007) + f(2008) = 9k$ जहां $k \in N$ तो k का मान ज्ञात कीजिए।

Ans. 2187

Sol. $a = 3^{\frac{1}{223}} + 1$

$f(n) = {}^nC_0\, a^{n-1} - {}^nC_1\, a^{n-2} + {}^nC_2\, a^{n-3} - \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a_0$

$af(n) = {}^nC_0\, a^n - {}^nC_1\, a^{n-1} + {}^nC_2\, a^{n-2} - {}^nC_n\, a^0 \ldots\ldots + (-1)^{n-1}\, {}^nC_{n-1}\, a + (-1)^n\, {}^nC_n\, a^0$

$+$

$(-1)^n \cdot {}^nC_n\, a^0$

$= [a-1]^n$

$af(n) + (-1)^n = (a-1)^n$

$af(2007) - 1 = (a-1)^{2007}$

$af(2008) + 1 = (a-1)^{2008}$

$a\{f(2007) + f(2008)\} = (a-1)^{2007}\{1 + (a-1)\}$

$f(2007) + f(2008) = (a-1)^{2007}$

$\Rightarrow 3^{\frac{2007}{223}} \Rightarrow 3^9 = 9k$, $k = 2187$

4. Let $P = \left(2+\sqrt{3}\right)^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function.

Find the value of $\left(\frac{f^2}{1-f}\right)$.

माना $P = \left(2+\sqrt{3}\right)^5$ तथा $f = P - [P]$, जहां $[P]$ महत्तम पूर्णांक फलन है। तो $\left(\frac{f^2}{1-f}\right)$ का मान ज्ञात कीजिए।

Ans. 722

Sol. $P = \left(2+\sqrt{3}\right)^5$

$f = P - [P]$

$\frac{f^2}{1-f}$

$I + f = \left(2+\sqrt{3}\right)^5 = {}^5C_0 2^5 + {}^5C_1 2^4\sqrt{3}^1 + {}^5C_2 2^3\sqrt{3}^2 + ... + {}^5C_5\left(\sqrt{3}\right)^5$

$F = \left(2-\sqrt{3}\right)^5 = {}^5C_0 2^5 - {}^5C_1 2^4\sqrt{3} + ... - {}^5C_5\left(\sqrt{3}\right)^5$

$I + f + F = 2[{}^5C_0\, 2^5 + {}^5C_2\, 2^3\, 3 + {}^5C_4\, 2\cdot 3^2]$

$= 2\,[32 + 10\cdot 24 + 5\cdot 18]$

$= 724$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$\Rightarrow f + F = 1$

$I = 723$

$F = 1 - f$

$([P] + f)\, F = 1$

$$[P] = \frac{1}{F} - f$$

$$= \frac{1}{1-f} - f$$

$$= \frac{1-f+f^2}{1-f}$$

$$723 = 1 + \frac{f^2}{1-f}$$

$$\frac{f^2}{1-f} = 722$$

5. Prove that :
सिद्ध कीजिये

(a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + ... + \frac{n.C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

(b) $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

(c) $2 \cdot C_o + \frac{2^2 \cdot C_1}{2} + \frac{2^3 \cdot C_2}{3} + \frac{2^4 \cdot C_3}{4} + ... \frac{2^{n+1} \cdot C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$

(d) $C_o - \frac{C_1}{2} + \frac{C_2}{3} - ..... + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$

Ans.

Sol. (a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + ... + \frac{nC_n}{C_{n-1}}$

$$\sum_{r=1}^{n} \frac{r \ {}^nC_r}{{}^nC_{r-1}}$$

$$\sum_{r=1}^{n} n \frac{{}^{n-1}C_{r-1}}{{}^nC_{r-1}}$$

$$n\sum_{r=1}^{n} \frac{\frac{(n-1)!}{(n-1-r+1)!(r-1)!}}{\frac{n!}{(n-r+1)!(r-1)!}}$$

$$n\sum \frac{(n-1)!}{n!} \times \frac{(n-r+1)!}{(n-r)!}$$

$$n\sum \frac{1}{n} \times (n-r+1)$$

$$\sum_{r=1}^{n} (n-r+1)$$

$$n^2 - \frac{n(n+1)}{2} + n$$

$$\frac{2n^2 - n^2 - n + 2n}{2}$$

$$\frac{n^2+n}{2} \Rightarrow \frac{n(n+1)}{2}$$

(b) ${}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + ... + \frac{{}^nC_n}{n+1}$

$$\sum_{r=0}^{n}\frac{{}^{n}C_{r}}{r+1}$$

$$\frac{1}{n+1}\sum_{r=0}^{n}\frac{n+1}{r+1}\cdot{}^{n}C_{r}$$

$$\frac{1}{n+1}\sum_{r=0}^{n}{}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{{}^{n+1}C_{1}+{}^{n+1}C_{2}+....+{}^{n+1}C_{n+1}\right\}$$

$$+\qquad\qquad -{}^{n+1}C_{0}$$

$${}^{n+1}C_{0}$$

$$\frac{1}{n+1}\cdot\left\{2^{n+1}-1\right\}$$

(c) $2\cdot{}^{n}C_{0}+2^{2}\cdot\frac{{}^{n}C_{1}}{2}+2^{3}\cdot\frac{{}^{n}C_{2}}{3}+....+\frac{2^{n+1}\ {}^{n}C_{n}}{n+1}$

$$\sum_{r=0}^{n}\frac{2^{r+1}\ {}^{n}C_{r}}{r+1}$$

$$\frac{1}{n+1}\sum 2^{r+1}\cdot\frac{n+1}{r+1}\cdot\ {}^{n}C_{r}$$

$$\frac{1}{n+1}\sum_{r=0}^{n}2^{r+1}\ {}^{n+1}C_{r+1}$$

$$\frac{1}{n+1}\left\{2\cdot{}^{n+1}C_{1}+2^{2}\ {}^{n+1}C_{2}+...+2^{n+1}\ {}^{n+1}C_{n+1}\right\}$$

$$+\ {}^{n+1}C_{0}\qquad\qquad -{}^{n+1}C_{0}$$

$$\frac{1}{n+1}\left\{(1+2)^{n+1}-1\right\}$$

$$\frac{1}{n+1}\left\{3^{n+1}-1\right\}$$

(d) $^{n}C_0 - \frac{^{n}C_1}{2} + \frac{^{n}C_2}{3} - .... + (-1)^n \frac{^{n}C_n}{n+1}$

$\sum_{r=0}^{n} (-1)^r \frac{^{n}C_r}{r+1}$

$\frac{1}{n+1} \sum_{r=0}^{n} (-1)^r \ {}^{n+1}C_{r+1}$

$\frac{1}{n+1} \left\{ {}^{n+1}C_1 - {}^{n+1}C_2 + {}^{n+1}C_3 - ..... (-1)^n \ {}^{n+1}C_{n+1} \right\}$

$\frac{1}{n+1} \{1\}$

6. Let $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ where m, n and p are even natural numbers and C_r represents the coefficient of x^r in the expansion of $(1 + x)^{100}$. Find the value of $(m + n + p)$.

माना $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = m \left({}^{2n}C_n \right) + 2^p$ जहां m, n और p सम प्राकृत संख्याऐं हैं तथा $C_r (1+x)^{100}$ के प्रसार में x^r के गुणांक है। तब $(m + n + p)$ का मान ज्ञात कीजिए।

Ans. 502

Sol. $\sum_{r=0}^{100} \sum_{s=0}^{100} (C_r^2 + C_s^2 + C_r C_s) = \sum_{r=0}^{100} \left(\sum_{s=0}^{100} \left[\left({}^{n}C_r \right)^2 + \left({}^{n}C_s \right)^2 + {}^{n}C_r . {}^{n}C_s \right] \right)$

$$= \sum_{r=0}^{100} \left(101. \left({}^{n}C_r \right)^2 + \left[\underbrace{\left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... + \left({}^{n}C_{100} \right)^2}_{p} \right] + {}^{n}C_r \left[\underbrace{\left({}^{n}C_0 \right) + \left({}^{n}C_1 \right) + ... + \left({}^{n}C_{100} \right)}_{q} \right] \right) .....(A)$$

Now, for P : -

$\because (1+n)^n = {}^{n}C_0 + {}^{n}C_1 . x + ... {}^{n}C_n . x^n$ .....(1)

$(x+1)^n = {}^{n}C_0 . x^n + {}^{n}C_1 . x^{n-1} + ... + {}^{n}C_n$ .....(2)

Multiply equation (1) & (2) and compare coefficient of x^n both side, we get .....

$${}^{2n}C_n = \left({}^{n}C_0 \right)^2 + \left({}^{n}C_1 \right)^2 + ... C_0 = p$$

$p = {}^{200}C_{100}$

Put x = 1 in equation (1) we get

$2^n = 2^{100} = {}^nC_0 + {}^nC_1 + ...+ {}^nC_n = q$

Now, from (A)

$$\sum_{r=0}^{100}\left[101\times\left({}^nC_r\right)^2 + {}^{200}C_{100} + 2^{100}.\ {}^nC_r\right]$$

$$= 101\sum_{r=0}^{100}\left({}^nC_r\right)^2 + {}^{200}C_{100}\sum_{r=0}^{100}1 + 2^{100}\sum_{r=0}^{100}{}^nC_r$$

$= 101 \times {}^{200}C_{100} + {}^{200}C_{100} \times 101 + 2^{100} \times 2^{100}$

$= 202 \times {}^{200}C_{100} + 2^{200} = m({}^{2n}C_n) + 2^p$ (Given)

by comapre both side we get

m = 202, n = 100, p = 200

$\therefore$ m + n + p = 502

7. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, then show that the sum of the products of the C_i's taken two at a time, represented by $\underset{0 \le i < j \le n}{\Sigma\Sigma} C_i\, C_j$ is equal to $2^{2n-1} - \frac{2n!}{2(n!)^2}$.

यदि $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_n x^n$, तब सिद्ध कीजिए कि दो C_i's को एक साथ लेकर गुणन फलों का योग यदि $\underset{0 \le i < j \le n}{\Sigma\Sigma} C_i\, C_j$ से परिभाषित है, का मान $2^{2n-1} - \frac{2n!}{2(n!)^2}$ होगा।

Ans.

Sol. $(1+x)^n = C_0 + C_1x + ... + C_n x^n$

$$S = \Sigma\Sigma C_iC_j = \frac{1}{2}[(C_0 + C_1 + C_2 + ...+ C_n)^2 - (C_0^2 + C_1^2 + C_2^2 + ...+ C_n^2)]$$

$$\Rightarrow 2S = 2^{2n} - {}^{2n}C_n \Rightarrow S = 2^{2n-1} - \frac{{}^{2n}C_n}{2}$$

8. Let $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ and $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

where C_r represents coefficient of x^r in the binomial expansion of $(1 + x)^{100}$.

If $S_1 + S_2 + S_3 = a^b$ where $a, b \in N$ then find the least value of $(a + b)$.

माना $S_1 = \sum\sum_{0\le i<j\le 100} C_i\, C_j$, $S_2 = \sum\sum_{0\le j<i\le 100} C_i\, C_j$ तथा $S_3 = \sum\sum_{0\le i=j\le 100} C_i\, C_j$

जहां C_r, $(1 + x)^{100}$ में x^r के गुणांक हैं, यदि $S_1 + S_2 + S_3 = a^b$ जहां $a, b \in N$ तब $(a + b)$ का न्यूनतम मान ज्ञात कीजिए।

Ans. 66

Sol. Given that $S_1 + S_2 + S_3 = \sum_{i=0}^{100}\sum_{j=0}^{100} C_i C_j$

$= \left({}^{100}C_0 + {}^{100}C_1 + {}^{100}C_2 + ... + {}^{100}C_{100}\right)^2$

$= \left(2^{100}\right)^2 = 2^{200} = a^b$

$\Rightarrow a^b = 2^{200} = 4^{100} = 16^{50} = 32^{40} = ...$

So, least value of a + b = 16 + 50 = 66

9. The expressions $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 +..... + x^n$ are multiplied together and the terms of the product thus obtained are arranged in increasing powers of x in the form of $a_0 + a_1x + a_2x^2 +.....$, then,

(a) how many terms are there in the product.

(b) show that the coefficients of the terms in the product, equidistant from the beginning and end are equal.

(c) show that the sum of the odd coefficients = the sum of the even coefficients = $\frac{(n+1)!}{2}$

यदि व्यंजक $1 + x$, $1+x + x^2$, $1 + x + x^2 + x^3$,..... $1 + x + x^2 +..... + x^n$ को गुणा किया जाता है तथा सभी पदों को x की बढ़ती घातों में निम्न $a_0 + a_1x + a_2x^2 +.....$, प्रकार से लिखा जाता है तो –

(a) गुणनफल में पदों की संख्या कितनी होगी।

(b) सिद्ध कीजिए कि गुणनफल में प्रारम्भ तथा अन्तः से समान दूरी पर स्थित पद समान है।

(c) सिद्ध कीजिए कि विषम गुणांकों का योग = सम गुणांकों का योग = $\frac{(n+1)!}{2}$

Ans. (a) $\frac{n^2+n+2}{2}$

Sol. $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) ..... (1 + x + x^2 + ... + x^n)$

$= a_0 + a_1x + a_2x^2 + ... +$

(a) As the largest power of 'x' present in the expansion will be

$1 + 2 + 3 + ... + n$

$\Rightarrow \frac{n(n+1)}{2}$

So number of terms will be $1 + \frac{n(n+1)}{2}$

$\Rightarrow \frac{n^2+n+2}{2}$

(b) $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \ldots\ldots (1 + x + x^2 + \ldots\ldots + x^n)$

$= a_0 + a_1x + a_2x^2 + \ldots. + a_nx^{\frac{n(n+1)}{2}}$ ...(A)

Replace $x \to \frac{1}{x}$

$$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right).....\left(1+\frac{1}{x}+\frac{1}{x^2}+....+\frac{1}{x^n}\right)$$

$$= a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ...\frac{a_n}{x^{\frac{n(n+1)}{2}}}$$

$$\frac{(x+1)}{x}\left(\frac{x^2+x+1}{x^2}\right)\left(\frac{x^3+x^2+x+1}{x^3}\right).....\left(\frac{x^n+x^{n-1}+....+1}{x^n}\right)$$

$$= \frac{a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n(n+1)}{2}-1} + ... + a_n}{x^{\frac{n(n+1)}{2}}}$$

$(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \ldots\ldots (1 + x + x^2 + \ldots. + x^n) = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + ... + a_n$ ...(B)

from (A) & (B)

$$a_0 + a_1x + a_2x^2 + \ldots. + a_nx^{\frac{n(n+1)}{2}} = a_0x^{\frac{n(n+1)}{2}} + a_1x^{\frac{n^2+n-2}{2}} + \ldots + a_n$$

Comparing coefficients

$$\left.\begin{array}{l} a_0 = a_n \\ a_1 = a_{n-1} \\ \vdots \\ a_n = a_0 \end{array}\right\} \Rightarrow a_r = a_{n-r}$$

(c) Put x = 1 in (A)

$2 \cdot 3 \cdot 4 \cdot \ldots\ldots (n + 1) = a_0 + a_1 + a_2 + \ldots\ldots + a_n$

Put $x = -1$ in (A)

$0 = a_0 - a_1 + a_2 - ..... + a_n$

$a_0 + a_2 + ... + a_n = \frac{(n+1)!}{2}$

10. Let $x = \left(3\sqrt{6}+7\right)^{89}$. If $\{x\}$ denotes the fractional part of 'x' then find the remainder when $x\{x\} + (x\{x\})^2 + (x\{x\})^3$ is divided by 31.

माना $x = \left(3\sqrt{6}+7\right)^{89}$ है। यदि $\{x\}$, 'x' के भिन्नात्मक भाग को निरूपित करता है तो शेषफल ज्ञात कीजिए जब $x\{x\} + (x\{x\})^2 + (x\{x\})^3$, 31 से विभाजित किया जाये।

Ans. 0

Sol. $x = \left(3\sqrt{6}+7\right)^{89}$

Find $x\{x\} + \left(x\{x\}\right)^2 + \left(x\{x\}\right)^3$

$1 + y + y^2 + y^3 - 1$

$(1 + y) + y^2 (1 + y) - 1$

$(1 + y) (1 + y^2) - 1$

$I + f = \left(7 + 3\sqrt{6}\right)^{89} = {}^{89}C_0 7^{89} + {}^{89}C_1 7^{88}\left(3\sqrt{6}\right) + ...$

$F = \left(7 - 3\sqrt{6}\right)^{89} = ...$

$2\left[{}^{89}C_0 7^{89} + {}^{89}C_2 7^{87}\left(3\sqrt{6}\right)^1 + ...{}^{89}C_{88}(7)^1\left(3\sqrt{6}\right)^{88}\right]$

$I + f + F =$ Even Integer

$0 < f < 1$

$-1 < F < 0$

$-1 < f + F < 1$

$I =$ Even Integer

$\Rightarrow f + F = 0$

$\{x\} = -F$

$= -\left(7 - 3\sqrt{6}\right)^{89}$

$\{x\} = \left(3\sqrt{6} - 7\right)^{89}$

$x = \left(3\sqrt{6} + 7\right)^{89}$

$x\{x\} = \left\{\left(3\sqrt{6} - 7\right)\left(3\sqrt{6} + 7\right)\right\}^{89}$

$= \{54 - 49\}^{89}$

$= 5^{89}$

Now 5^{89} {5 + 25 + 125}

$5^{89} \times 5 \times 31$

Divide by 31

So remainder = 0

11. Find the sum of possible real values of x for which the sixth term of $\left(3^{\log_3 \sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7 (3^{|x-2|}-9}\right)^7$ equals 567.

x के संभव वास्तविक मानों का योग ज्ञात कीजिए, जिनके लिए $\left(3^{\log_3 \sqrt{9^{|x-2|}}} + 7^{\left(\frac{1}{5}\right)\log_7 (3^{|x-2|}-9}\right)^7$ का छठा पद 567 के बराबर है।

Ans. 4

Sol. $\left\{3^{\log_3\left(9^{|x-2|}\right)^{\frac{1}{2}}} + 7^{\frac{1}{5}\log_7 3^{|x-2|}-9}\right\}^7$

$\left\{\left(9^{|x-2|}\right)^{\frac{1}{2}} + \left\{3^{|x-2|}-9\right\}^{\frac{1}{5}}\right\}^7$

$\left\{3^{|x-2|} + 3^{\frac{|x-2|-9}{5}}\right\}^7$

$$^{7}C_{5}\left(3^{|x-2|}\right)^{2}\left(3^{\frac{|x-2|-9}{5}}\right)^{5}=567$$

$$3^{2|x-2|}.3^{|x-2|-9}=\frac{567}{21}$$

$3^{|x-2|}=t$

$$t^{2}.\frac{t}{3^{9}}=27$$

$t^{3}=3^{12}$

$t=3^{9}$

$3^{|x-2|}=3^{9}$

$|x-2|=9$

$x-2=\pm 9$

$x=2\pm 9$

$x=11,-7$

$11+(-7)$

$\Rightarrow 4$

12. Let $1+\sum_{r=1}^{10}\left(3^{r}.^{10}C_{r}+r.^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ where $\alpha, \beta \in N$ and $f(x)=x^{2}-2x-k^{2}+1$. If α, β lies between the roots of $f(x)=0$. Then find the smallest positive integral value of k.

माना $1+\sum_{r=1}^{10}\left(3^{r}.^{10}C_{r}+r.^{10}C_{r}\right)=2^{10}(\alpha\,.\,4^{5}+\beta)$ है, जहाँ $\alpha, \beta \in N$ तथा $f(x)=x^{2}-2x-k^{2}+1$ है। यदि α, β, $f(x)=0$ के मूलों के बीच स्थित है, तो k का न्यूनतम संभव पूर्णांक मान ज्ञात कीजिए।

Ans. 5

Sol. $1+\sum_{r=1}^{10}3^{r}\ {}^{10}C_{r}+\sum_{r=1}^{10}r\ {}^{10}C_{r}$

$$=1+3^{0}\ {}^{10}C_{0}+3^{1}\ {}_{10}C_{1}+3^{2}\ {}^{10}C_{2}+....+3^{10}\ {}^{10}C_{10}-3^{0}.^{10}C_{0}+\sum_{r=1}^{10}r.^{10}C_{r}$$

$$=(1+3)^{10}+\sum_{r=1}^{10}r.\frac{10}{r}.^{9}C_{r-1}$$

$$=4^{10}+10\left[^{9}C_{0}+^{9}C_{1}+^{9}C_{2}+.....+^{9}C_{9}\right]$$

$$=4^{10}+10\left(2^{9}\right)$$

$= 4^{10} + 10 \cdot 2^9$

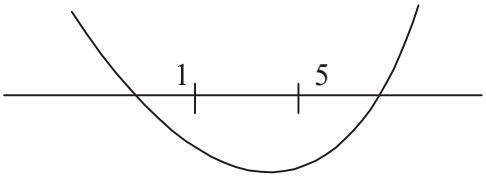
$= 4^{10} + 5 \cdot 2^{10}$

$\Rightarrow 2^{10}\left[4^5 + 5\right] = 2^{10}\left[\alpha \cdot 4^5 + \beta\right]$

$\alpha = 1, \beta = 5$

Now,

$f(x) = x^2 - 2x - k^2 + 1$



$f(1) < 0$	$f(5) < 0$
$1 - 2 - k^2 + 1 < 0$	$25 - 10 - k^2 + 1 < 0$
$-k^2 < 0$	$16 - k^2 < 0$
$k^2 > 0$	$k \in (-\infty, -4) \cup (4, \infty)$

$k \neq 0$

Smallest +ve integral value = 5

13. Let $n \in N$; $S_n = \sum_{r=0}^{3n} \left({}^{3n}C_r\right)$ and $T_n = \sum_{r=0}^{n} \left({}^{3n}C_{3r}\right)$. Find $|S_n - 3T_n|$.

माना $n \in N$; $S_n = \sum_{r=0}^{3n} \left({}^{3n}C_r\right)$ तथा $T_n = \sum_{r=0}^{n} \left({}^{3n}C_{3r}\right)$ है, तो $|S_n - 3T_n|$ का मान ज्ञात कीजिए।

Ans. 2

Sol. $S_n = \sum_{r=0}^{3n} {}^{3n}C_r$

$= {}^{3n}C_0 + {}^{3n}C_1 + {}^{3n}C_2 + ... + {}^{3n}C_{3n}$ ...(A)

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x^1 + ... + {}^{3n}C_{3n}\, x^{3n}$

Put $x = 1$

$S_n = 2^{3n}$

$T_n = \sum_{r=0}^{n} {}^{3n}C_{3r}$

$= {}^{3n}C_0 + {}^{3n}C_3 + {}^{3n}C_6 + ... + {}^{3n}C_{3n}$

$(1 + x)^{3n} = {}^{3n}C_0\, x^0 + {}^{3n}C_1\, x + {}^{3n}C_2\, x^2 + {}^{3n}C_3\, x^3 + {}^{3n}C_4\, x^4 + ... + {}^{3n}C_{3n}\, x^{3n}$

$x = w$

$(1 + w)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w + {}^{3n}C_2\, w^2 + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$x = w^2$

$(1 + w^2)^{3n} = {}^{3n}C_0 + {}^{3n}C_1\, w^2 + {}^{3n}C_2\, w + {}^{3n}C_3 + ... + {}^{3n}C_{3n}$

$(1 + w)^{3n} + (1 + w^2)^{3n} = 2\ {}^{3n}C_0 + {}^{3n}C_1\,(-1) + {}^{3n}C_2\,(-1) + 2\ {}^{3n}C_3 + ...$

Now Add (A) in it

$2^{3n} + (1 + w)^{3n} + (1 + w^2)^{3n} = 3[T_n]$

$2^{3n} + (-w^2)^{3n} + (-w)^{3n} = 3T_n$

$2^{3n} + (-1)^{3n} + (-1)^{3n} = 3T_n$

$|S_n - 3T_n|$

$= |(-1)^{3n} + (-1)^{3n}|$

$\Rightarrow 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 14 to 16

Consider the binomial expansion $R = (1 + 2x)^n = I + f$ where I is the integral part of R and f is the fractional part of R $n \in N$. Also the sum of the coefficients of R is 6561.

माना द्विपद प्रसार $R = (1 + 2x)^n = I + f$ है, जहाँ I तथा f, क्रमशः R के पूर्णांक भाग तथा भिन्नात्मक भाग है $n \in N$. R के गुणांकों का योग 6561 है।

14. The value of $(n + R - Rf)$ for $x = \frac{1}{\sqrt{2}}$ equals :

$x = \frac{1}{\sqrt{2}}$ के लिए $(n + R - Rf)$ का मान होगा:

(A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol. $I + f = (1 + 2x)^n = R$

Sum of coefficient 6561

Put $x = 1$

$3^n = 6561 = 3^8$

$n = 8$

$R = I + f = \left(1+\sqrt{2}\right)^8$

$F = \left(1-\sqrt{2}\right)^8$

$I + f = {}^8C_0 + {}^8C_1\sqrt{2} + ... + {}^8C_8\left(\sqrt{2}\right)^8$

$F = {}^8C_0 - {}^8C_1\sqrt{2} ..... + {}^8C_8\left(\sqrt{2}\right)^8$

$I + f + F = 2[{}^8C_0 + {}^8C_2 \cdot 2 + {}^8C_4\, 2^2 + {}^8C_6\, 2^3 + {}^8C_8\, 2^4]$

$I + f + F = 2 \times 577 = 1154$

$0 < f < 1$

$0 < F < 1$

$0 < f + F < 2$

$f + F = 1$

$I = 1153$

Now $n + R(1 - f)$

$8 + \left(1+\sqrt{2}\right)^8\left(1-\sqrt{2}\right)^8$

$8 + \{1 - 2\}^8$

9

15. If i^{th} terms is the greatest term for $x = 1/2$, then 'i' equals :

यदि $x = 1/2$ के लिए iवाँ पद महतम पद है, तो 'i' बराबर होगा :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \le i \le \frac{n+1}{1+\left|\frac{a}{b}\right|}$

Now $\frac{9}{1+\left|\frac{1}{2x}\right|} \Rightarrow \frac{9}{1+1} \Rightarrow \frac{9}{2} \Rightarrow 4.5$

$3.5 \le i \le 4.5$

$\Rightarrow i = 4$

So term will be 5th term.

16. If k^{th} terms is having greatest coefficient then sum of all possible value(s) of k is :

यदि kवें पद का गुणांक महतम है, तो k के सभी संभव मानों का योग होगा :

(A) 6 (B) 7 (C) 11 (D) 13

Ans. D

Sol. $n = 8$

So, greatest coefficient will be middle term.

Paragraph for Question Nos. 17 to 19

Let us consider the expansion $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$, then

माना प्रसार $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + ... + a_nx^n$ है, तो

17. $a_1 + a_3 + a_5 + a_7 + ...$ is equal to :

$a_1 + a_3 + a_5 + a_7 + ...$ बराबर है :

(A) $\frac{7^n + (-1)^n}{2}$ (B) $\frac{7^n - (-1)^n}{2}$ (C) $\frac{7^{n-1} - (-1)^{n-1}}{2}$ (D) $\frac{7^{n-1} + (-1)^{n-1}}{2}$

Ans. B

Sol. $(3 + 4x)^n = a_0 + a_1x + a_2x^2 + ... + a_nx^n$

$x = 1$

$7^n = a_0 + a_1 + a_2 + a_3 + ... + a_n$

$x = -1$

$(-1)^n = a_0 - a_1 + a_2 - a_3 - .....$

$$\frac{7^n + (-1)^n}{2} = a_0 + a_2 + a_4 + ...$$

$$\frac{7^n - (-1)^n}{2} = a_1 + a_3 + a_5 + ...$$

18. $a_0 - a_2 + a_4 - a_6 + ...$ is equal to :

$a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots$ बराबर है :

(A) $5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (B) $7^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$ (C) $7^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$ (D) $5^n \cos\left(n \tan^{-1}\frac{3}{4}\right)$

Ans. A

Sol. $(3 + 4i)^n = a_0 + a_1 i - a_2 - a_3 i + a_4 + a_5 i - a_6 \ldots\ldots\ldots\ldots$

$(3 + 4i)^n = (a_0 - a_2 + a_4 - a_6 + \ldots\ldots) + i(a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

$$\left\{5e^{i\left(\tan^{-1}\frac{4}{3}\right)}\right\}^n$$

$$5^n e^{in \tan^{-1}\frac{4}{3}}$$

$$5^n \left\{\cos\left(n \tan^{-1}\frac{4}{3}\right) + i \sin\left(n \tan^{-1}\frac{4}{3}\right)\right\}$$

$= (a_0 - a_2 + a_4 - a_6 + \ldots\ldots) + i(a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

Now $a_0 - a_2 + a_4 - a_6 + \ldots\ldots = 5^n \cos\left(n \tan^{-1}\frac{4}{3}\right)$

19. $(a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots)^2 + (a_1 - a_3 + a_5 - \ldots\ldots\ldots)^2$ equals :

$(a_0 - a_2 + a_4 - a_6 + \ldots\ldots\ldots)^2 + (a_1 - a_3 + a_5 - \ldots\ldots\ldots)^2$ बराबर है :

(A) 7^{2n} (B) 5^n (C) 5^{2n} (D) 7^n

Ans. C

Sol. $(a_0 - a_2 + a_4 - a_6 + \ldots\ldots)^2 + (a_1 - a_3 + a_5 - a_7 + \ldots\ldots)$

$= 5^{2n}$ [from above $\sin^2\theta + \cos^2\theta = 1$]

MATCH THE COLUMN TYPE QUESTION(S)

20. If $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, then :

यदि $\left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{2n} x^{2n}$, तो:

Column -I	**Column-II**
(A) $a_0 + a_1 + \ldots\ldots a_{2n}$	(P) n+1
(B) $a_0 - a_1 + a_2 - a_3 + \ldots\ldots + a_{2n}$	(Q) 1

(C) $a_0 - 2a_1 + 3a_2 + ... + (2n+1)a_{2n}$ (R) $3n$

(D) $a_0 - 2a_1 + 4a_2 - 8a_3 + ... + 2^{2n}.a_{2n}$ (S) 3^n

Ans. A – S, B – Q, C – P, D – S

Sol. $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$

[A] $\rightarrow$ S

$x = 1$

$3^n = a_0 + a_1 + a_2 + ... + a_{2n}$

[B] $\rightarrow$ Q

$x = -1$

$(1 - 1 + 1)^n = a_0 - a_1 + a_2 - a_3 + ... + a_{2n}$

$= 1$

[C] $\rightarrow$ Q

$(1 + x + x^2)^n \cdot x = a_0x + a_1x^2 + a_2x^3 + ... + a_2nx^{2n+1} = (x^n + x^{n+1} + x^{n+2})^n$

Now differentiating

$n(x^n + x^{n+1} + x^{n+2})^{n-1}\ \{nx^{n-1} + (n+1)x^n + (n+2)x^{n+1}\} = a_0 + 2a_1x + 3a_2x^2 + ... + (2n+1)a_{2n}x^{2n}$

Put $x = -1$

$n((-1)^n + (-1)^{n+1} + (-1)^n)^{n-1}\ \{n(-1)^{n-1} + (n+1)(-1)^n + (n+2)(-1)^{n+1}\}$

$= a_0 - 2a_1 + 3a_2 - ..... + (2n+1)a_{2n}$

$n(-1 + 1 - 1)^{even}\ \{+n - n - 1 + n + 2\}$

$n(n+1)$